

注意力机制

《神经网络与深度学习》第8周

注意力机制与外部记忆

注意力机制

注意力机制

人工神经网络的注意力机制

注意力机制的应用

自注意力模型

Transformer

记忆增强网络

外部记忆

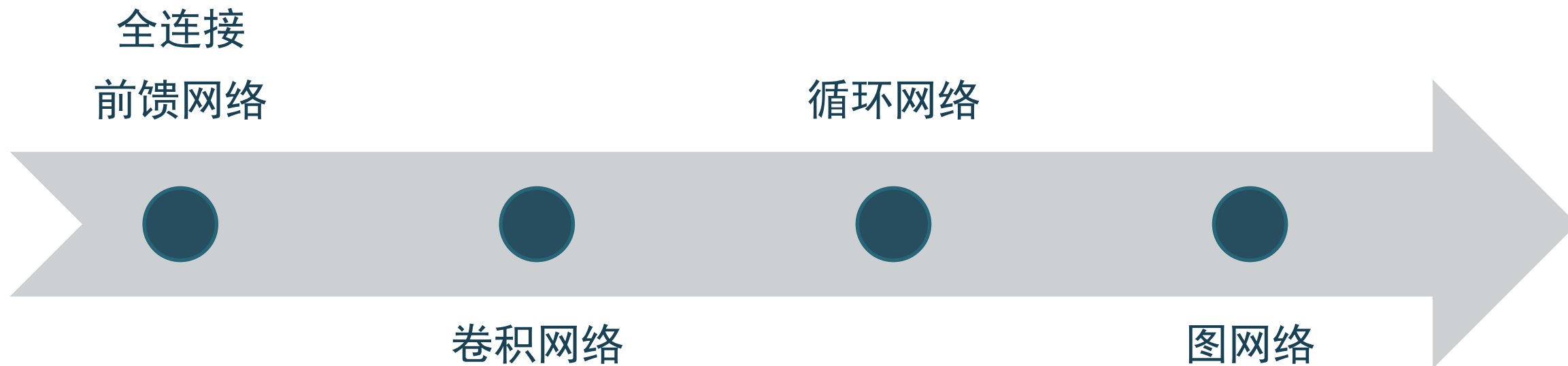
结构化的外部记忆

基于神经动力学的联想记忆

注意力机制与外部记忆

注意力机制

网络能力



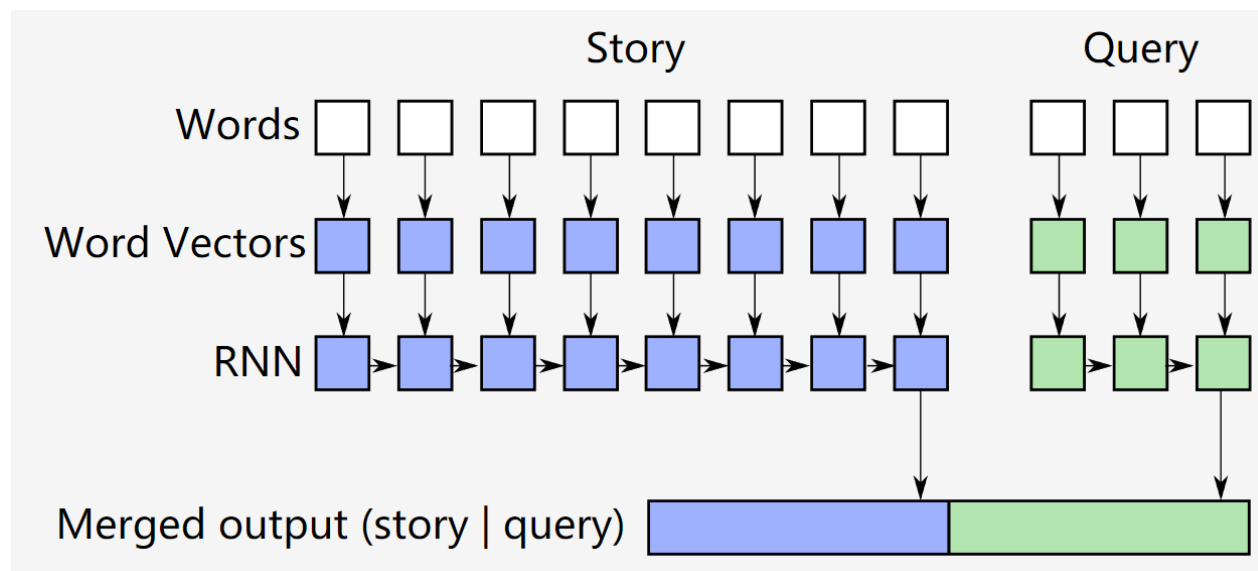
增加网络能力的另一种思路：

- 注意力机制
- 外部记忆

例子：阅读理解

Facebook bAbi tasks

- 1 John moved to the bedroom.
- 2 Mary grabbed the football there.
- 3 Sandra journeyed to the bedroom.
- 4 Sandra went back to the hallway.
- 5 Mary moved to the garden.
- 6 Mary journeyed to the office.
- 7 Where is the football? office



例子：阅读理解 SQuAD

The first recorded travels by Europeans to China and back date from this time. The most famous traveler of the period was the Venetian Marco Polo, whose account of his trip to "Cambaluc," the capital of the Great Khan, and of life there astounded the people of Europe. The account of his travels, *Il milione* (or, *The Million*, known in English as the *Travels of Marco Polo*), appeared about the year 1299. Some argue over the accuracy of Marco Polo's accounts due to the lack of mentioning the Great Wall of China, tea houses, which would have been a prominent sight since Europeans had yet to adopt a tea culture, as well the practice of foot binding by the women in capital of the Great Khan. Some suggest that Marco Polo acquired much of his knowledge **through contact with Persian traders** since many of the places he named were in Persian.

How did some suspect that Polo learned about China instead of by actually visiting it?

Answer: **through contact with Persian traders**

大脑中的注意力

人脑每个时刻接收的外界输入信息非常多，包括来源于视觉、听觉、触觉的各种各样的信息。

但就视觉来说，眼睛每秒钟都会发送千万比特的信息给视觉神经系统。

如何解决信息超载问题？

Attention Is All You Need

注意力示例



注意力实验



所有黑色牌的数字之和为多少？

如何实现？

自下而上

汇聚（pooling）

自上而下

会聚（focus）

如何实现？

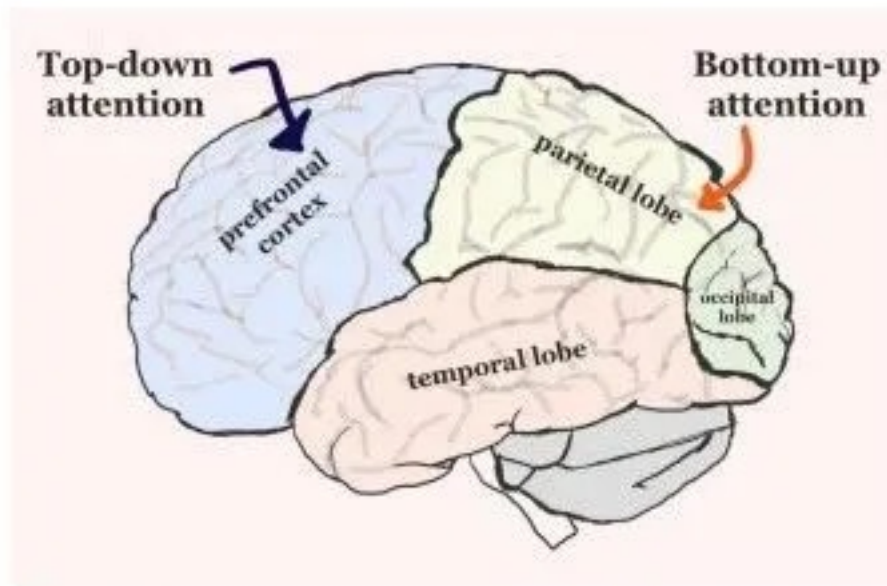
FROM ATTENTION TO CONSCIOUSNESS

C-word not taboo anymore in cognitive neuroscience

Global Workspace Theory

(Baars 1988++, Dehaene 2003++)

- Bottleneck of conscious processing
- Selected item is broadcast, stored in short-term memory, conditions perception and action
- System 2-like sequential processing, conscious reasoning & planning & imagination



如何实现？

当一个人在吵闹的鸡尾酒会上和朋友聊天时，尽管周围噪音干扰很多，他还是可以听到朋友的谈话内容，而忽略其他人的声音

同时，如果未注意到的背景声中有重要的词（比如他的名字），他会马上注意到

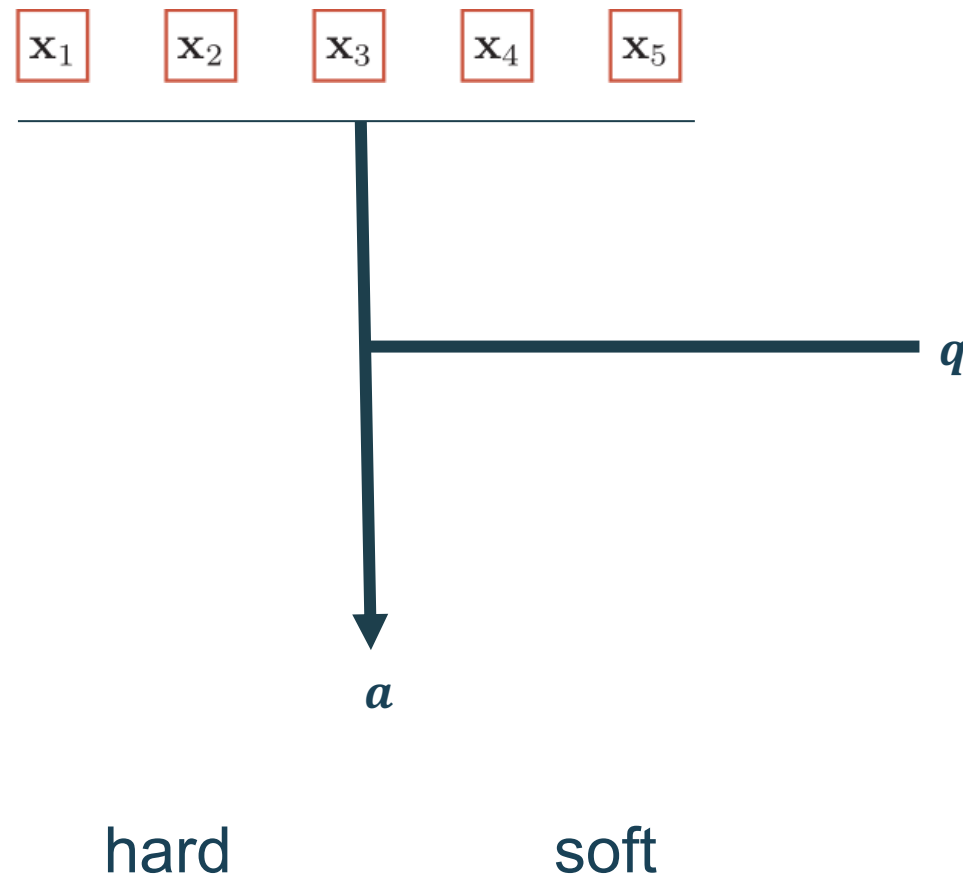
鸡尾酒会效应

人工神经网络中的注意力机制

问题



8秒内找到8



软性注意力机制 (soft attention mechanism)

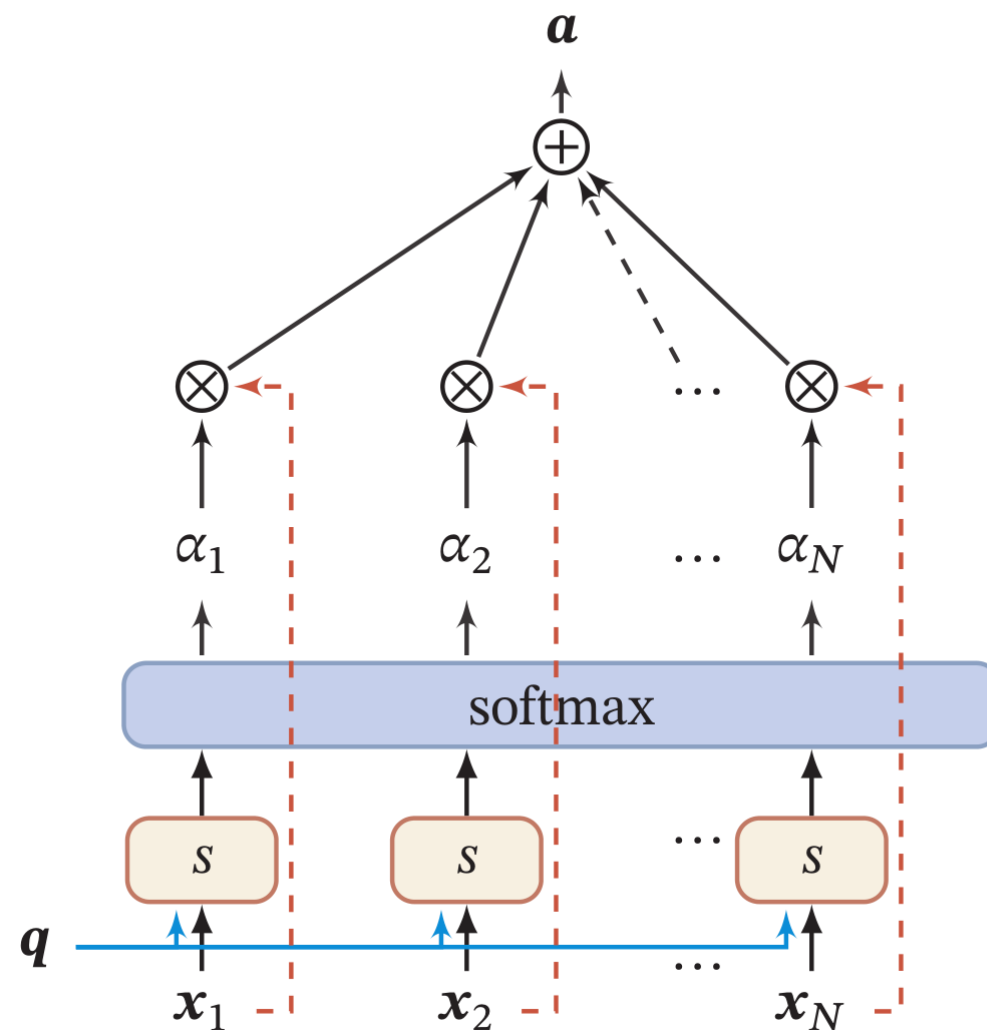
注意力机制可以分为两步

计算注意力分布 α $s(\mathbf{x}_n, \mathbf{q})$ 打分函数

$$\begin{aligned}\alpha_n &= p(z = n | \mathbf{X}, \mathbf{q}) \\ &= \text{softmax}(s(\mathbf{x}_n, \mathbf{q})) \\ &= \frac{\exp(s(\mathbf{x}_n, \mathbf{q}))}{\sum_{j=1}^N \exp(s(\mathbf{x}_j, \mathbf{q}))}\end{aligned}$$

根据 α 来计算输入信息的加权平均

$$\begin{aligned}\text{att}(\mathbf{X}, \mathbf{q}) &= \sum_{n=1}^N \alpha_n \mathbf{x}_n, \\ &= \mathbb{E}_{z \sim p(z | \mathbf{X}, \mathbf{q})}[\mathbf{x}_z]\end{aligned}$$



注意力打分函数 $s(x_n, q)$

加性模型

$$s(\mathbf{x}, \mathbf{q}) = \mathbf{v}^\top \tanh(\mathbf{W}\mathbf{x} + \mathbf{U}\mathbf{q}),$$

点积模型

$$s(\mathbf{x}, \mathbf{q}) = \mathbf{x}^\top \mathbf{q},$$

缩放点积模型

$$s(\mathbf{x}, \mathbf{q}) = \frac{\mathbf{x}^\top \mathbf{q}}{\sqrt{D}},$$

双线性模型

$$s(\mathbf{x}, \mathbf{q}) = \mathbf{x}^\top \mathbf{W} \mathbf{q},$$

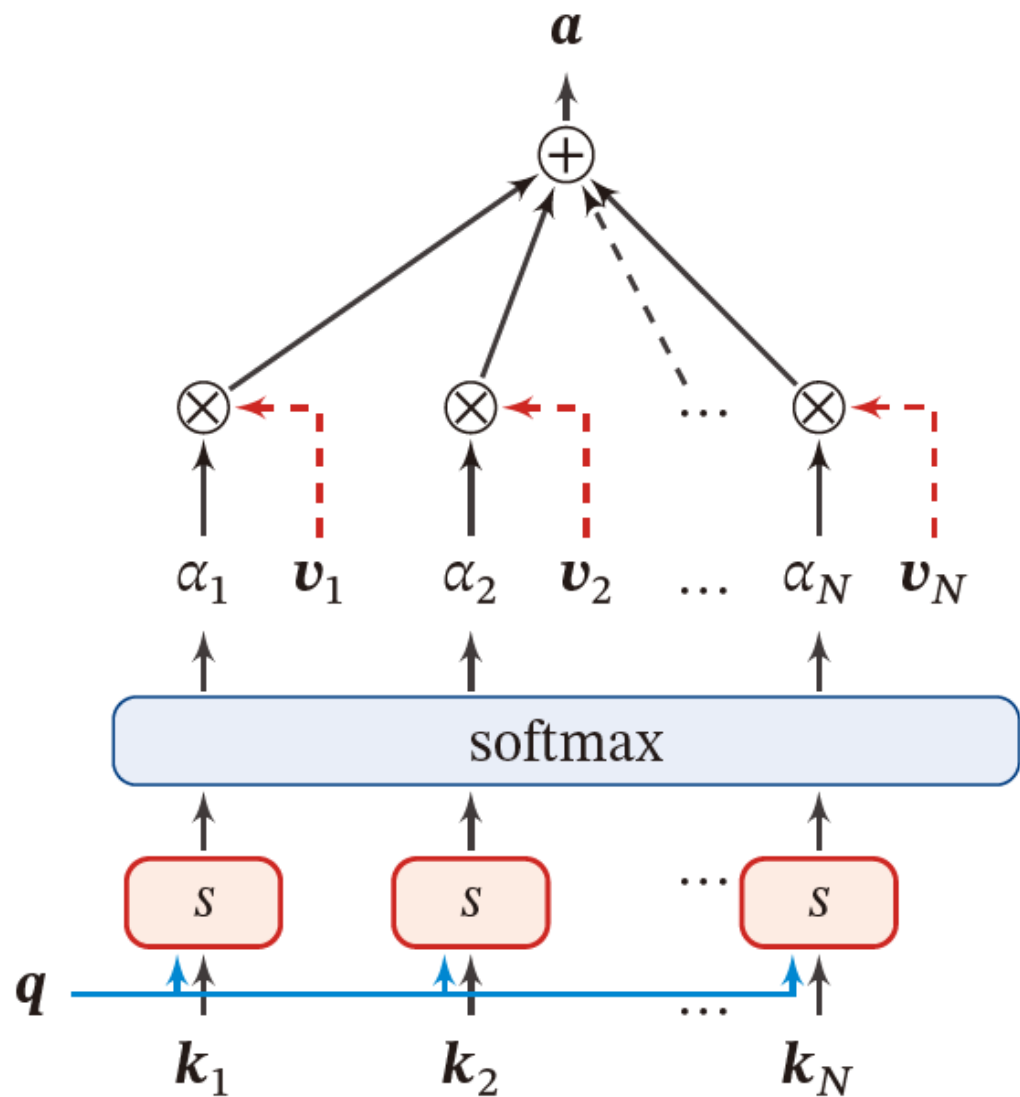
注意力机制的变体

硬性注意力 (hard attention)

键值对注意力 (key-value pair attention)

用 $(\mathbf{K}, \mathbf{V}) = [(\mathbf{k}_1, \mathbf{v}_1), \dots, (\mathbf{k}_N, \mathbf{v}_N)]$ 表示 N 组输入信息

$$\begin{aligned}\text{att}((\mathbf{K}, \mathbf{V}), \mathbf{q}) &= \sum_{n=1}^N \alpha_n \mathbf{v}_n, \\ &= \sum_{n=1}^N \frac{\exp(s(\mathbf{k}_n, \mathbf{q}))}{\sum_j \exp(s(\mathbf{k}_j, \mathbf{q}))} \mathbf{v}_n.\end{aligned}$$



注意力机制的变体

多头注意力 (multi-head attention)

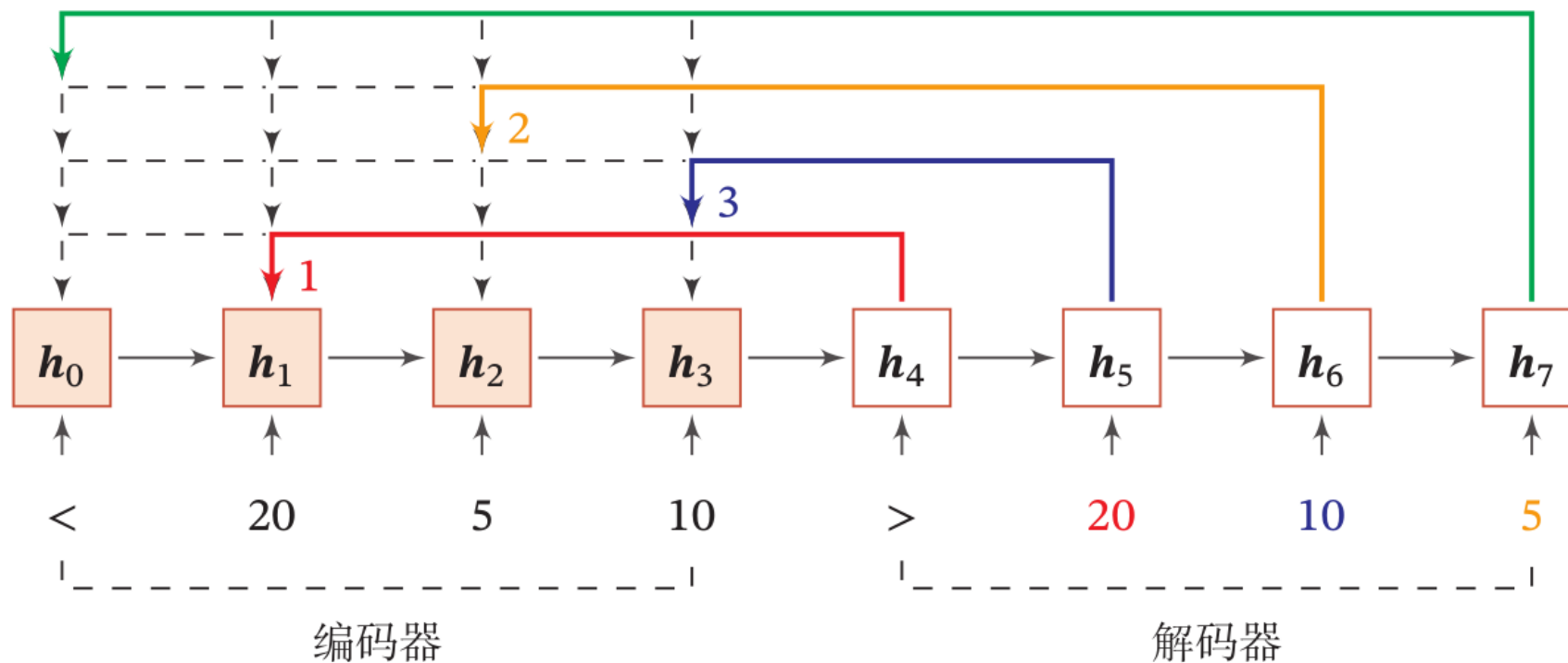
利用多个查询 $\mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_M]$ ，来并行地从输入信息中选取多组信息。每个注意力关注输入信息的不同部分

$$\text{att}((\mathbf{K}, \mathbf{V}), \mathbf{Q}) = \text{att}((\mathbf{K}, \mathbf{V}), \mathbf{q}_1) \oplus \dots \oplus \text{att}((\mathbf{K}, \mathbf{V}), \mathbf{q}_M)$$

结构化注意力 (structural attention)

指针网络（pointer network）

我们可以只利用注意力机制中的第一步，将注意力分布作为一个软性的指针（pointer）来指出相关信息的位置



注意力机制与外部记忆

注意力机制的应用

文本分类

The infantile cart is easy to use.

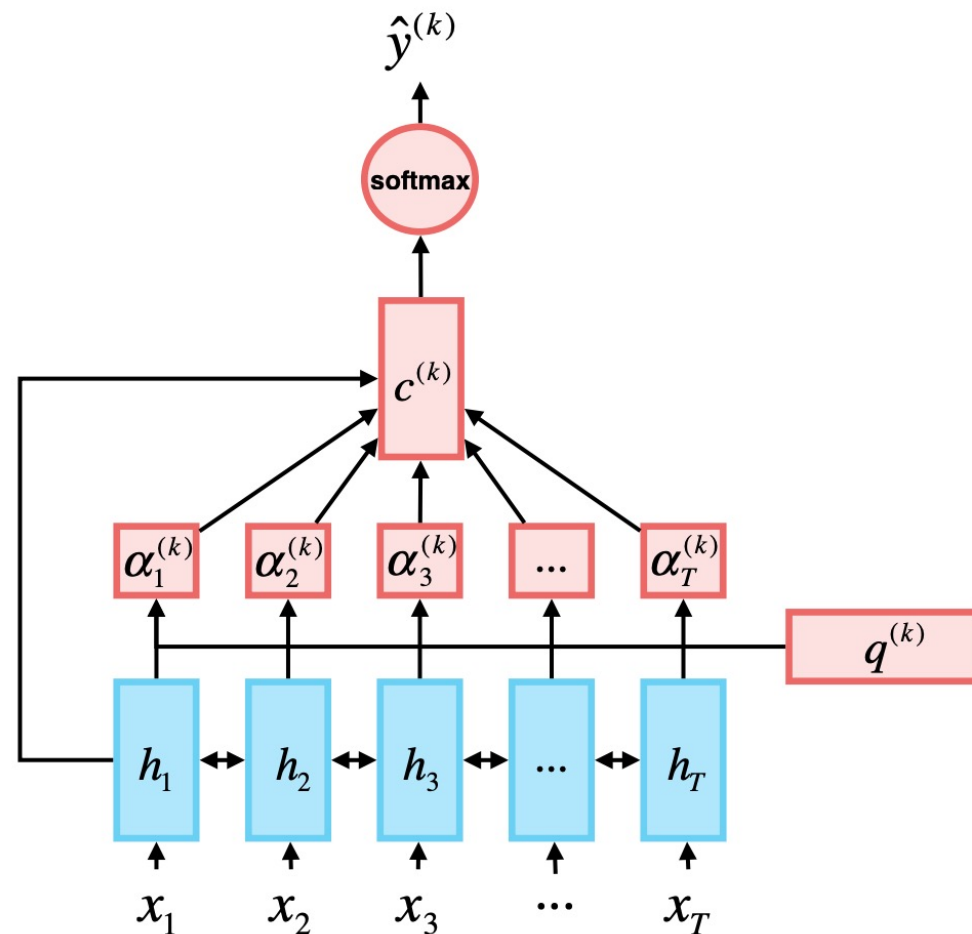


Figure 2: Static Task-Attentive Sentence Encoding

Same Representation, Different Attentions: Shareable Sentence Representation Learning from Multiple Tasks, IJCAI, 2018. [BibTeX][PDF]

Renjie Zheng, Junkun Chen, Xipeng Qiu.

文本分类

I have not read the original version of this work , but the translation lacks originality and art . A beautiful story , but the writing style lacks grace and creativity . This is the only time I have liked a movie better than the book . Do yourself a favor and skip the book . the movie is quite beautiful and moving

(a) Attention of task “Books” in SA-MTL, Output: Negative

I have not read the original version of this work , but the translation lacks originality and art . A beautiful story , but the writing style lacks grace and creativity . This is the only time I have liked a movie better than the book . Do yourself a favor and skip the book . the movie is quite beautiful and moving

(b) Attention of task “DVD” in SA-MTL, Output: Positive

文本分类

层次注意力模型

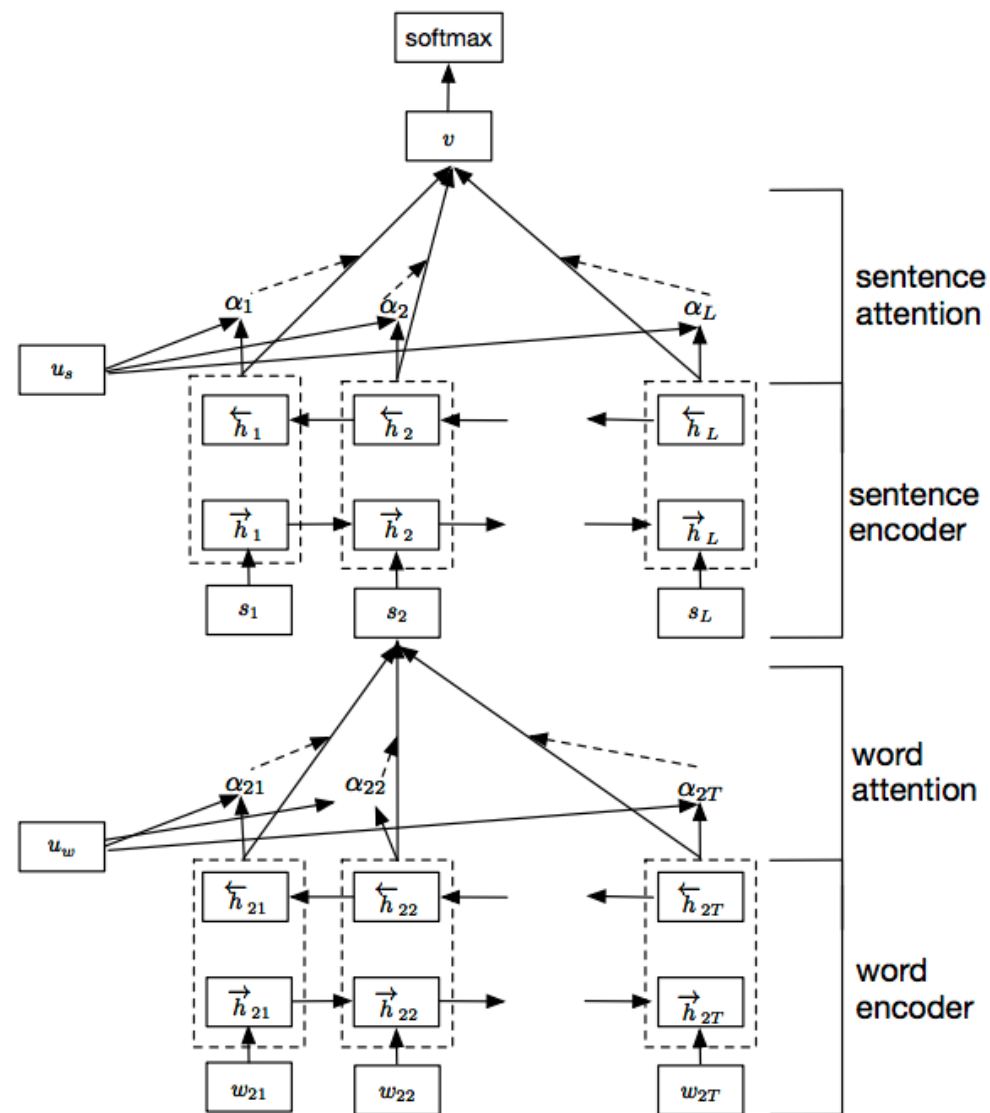
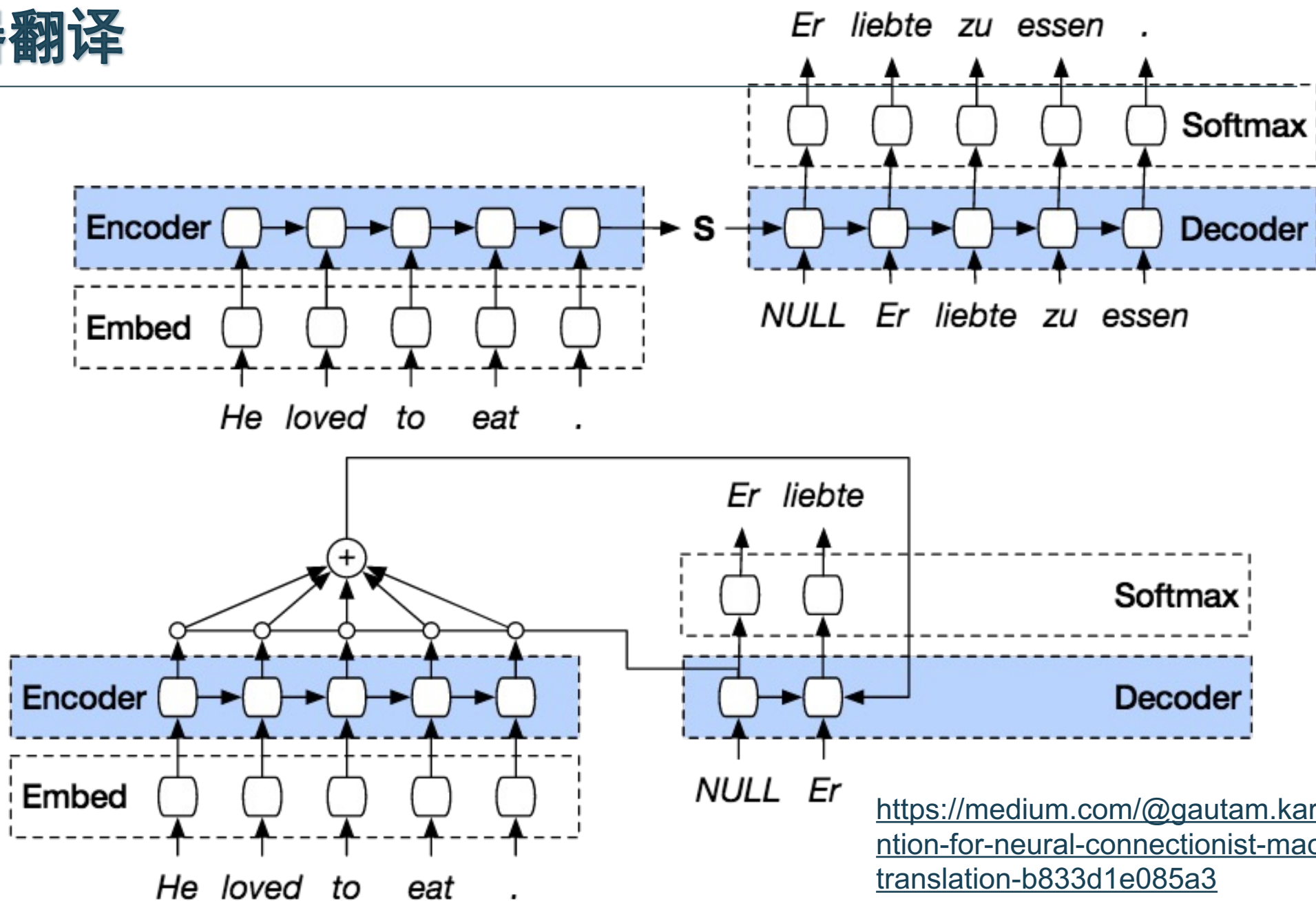


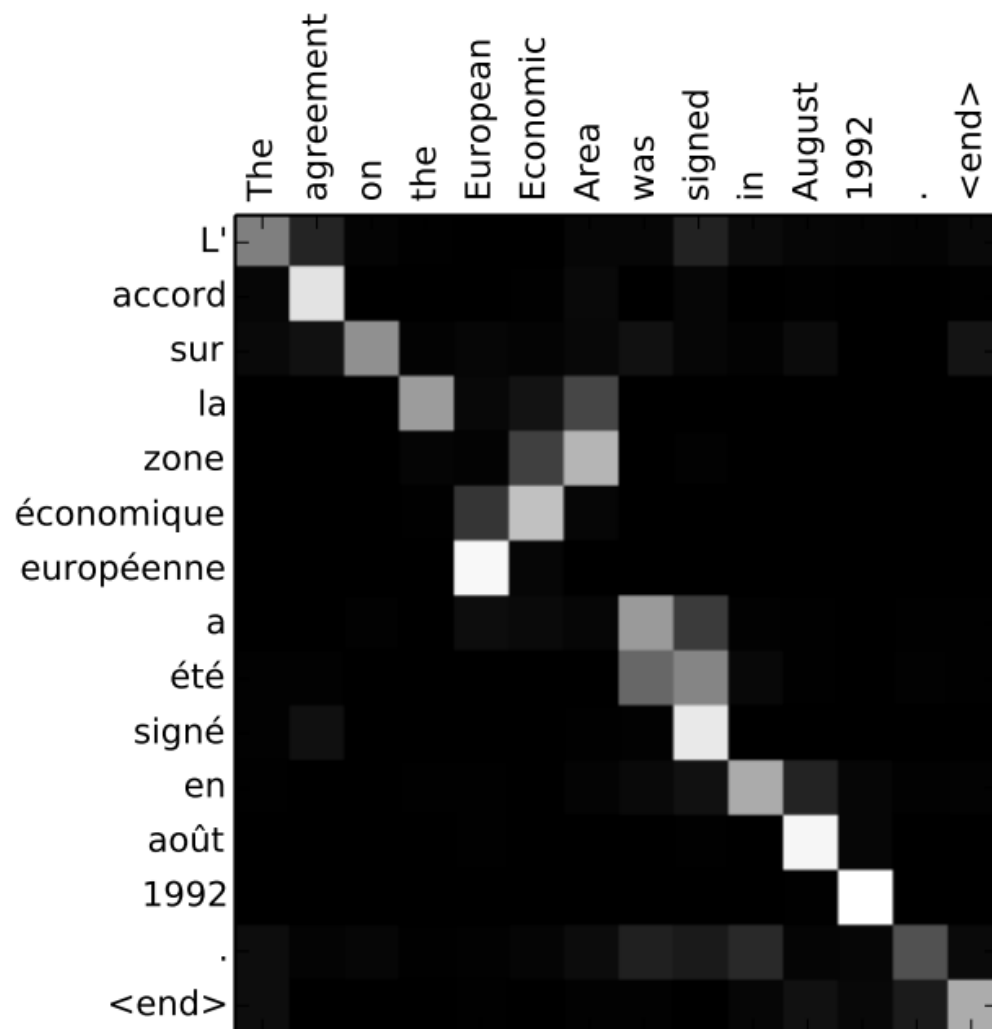
Figure 2: Hierarchical Attention Network.

机器翻译



<https://medium.com/@gautam.karmakar/attention-for-neural-connectionist-machine-translation-b833d1e085a3>

机器翻译



机器翻译

. la maison de Léa <end> .

Image Caption

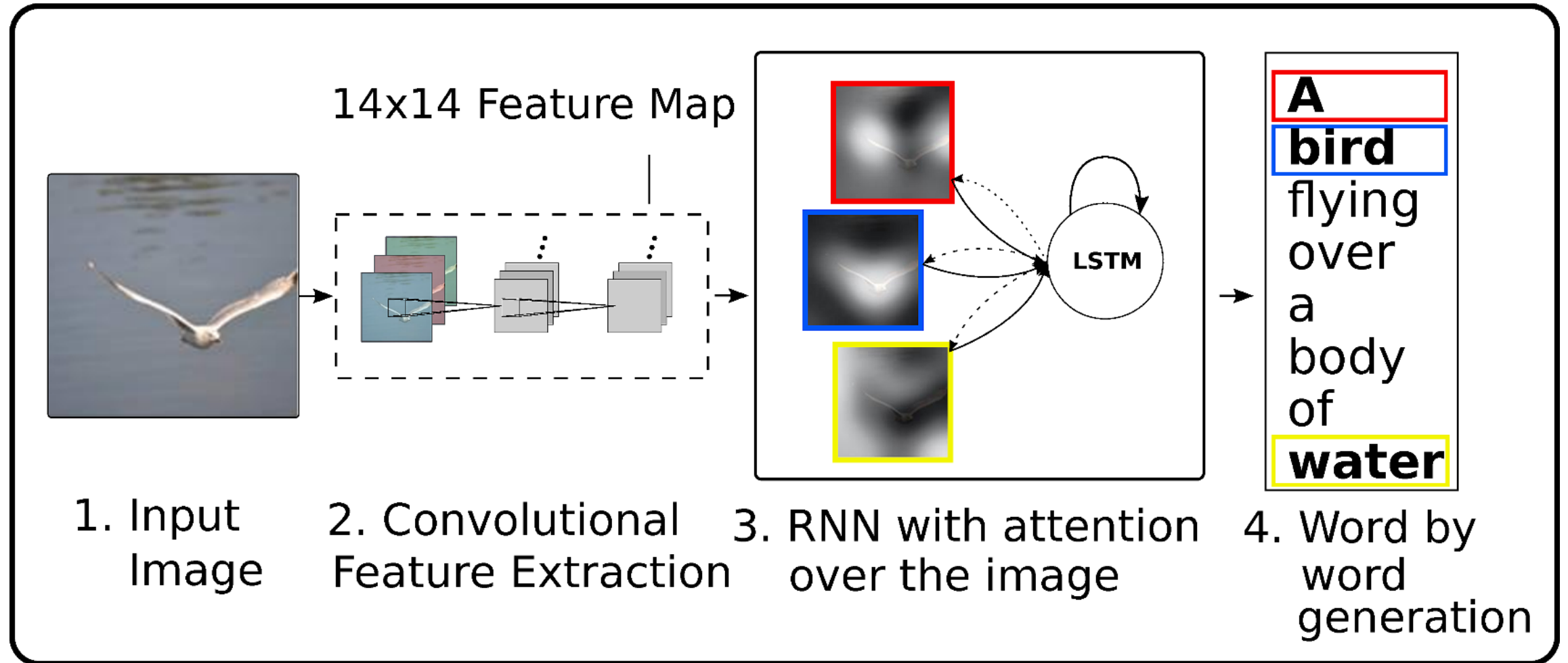
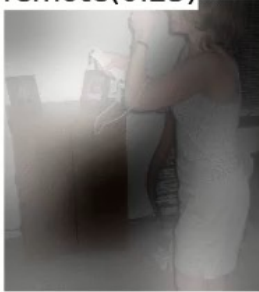


Image Caption

remote(0.23)



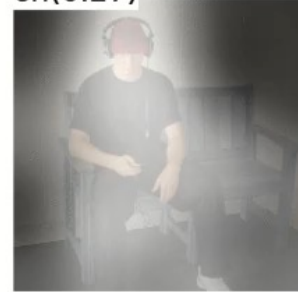
a(0.41)



sky(0.61)



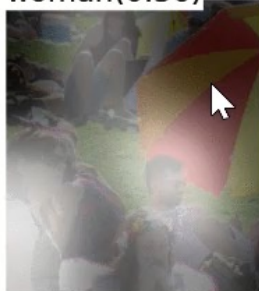
on(0.27)



red(0.38)



woman(0.36)



of(0.19)



A(0.99)



racket(0.16)



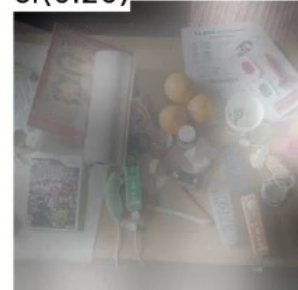
a(0.33)



sink(0.48)

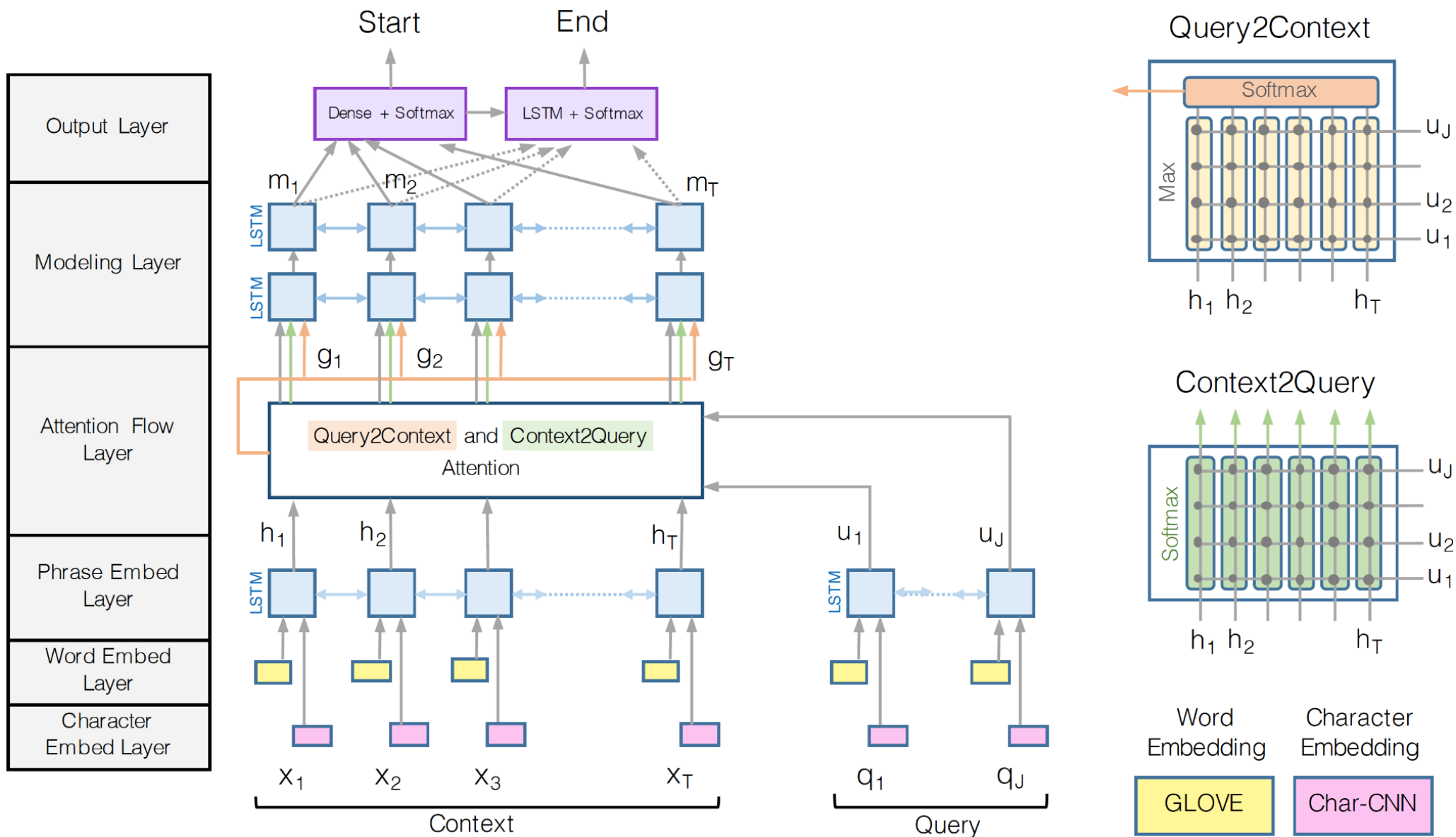


of(0.20)



阅读理解

<https://allenai.github.io/bi-att-flow/>

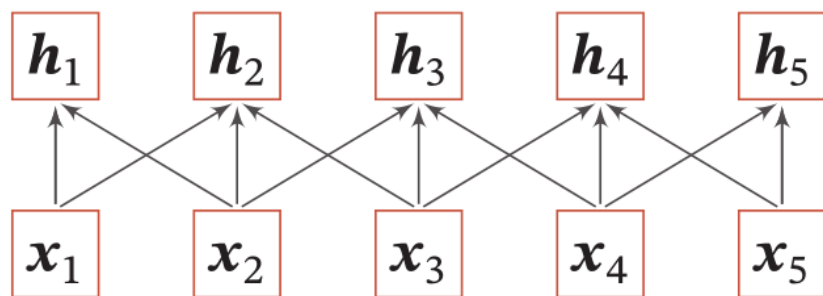


注意力机制与外部记忆

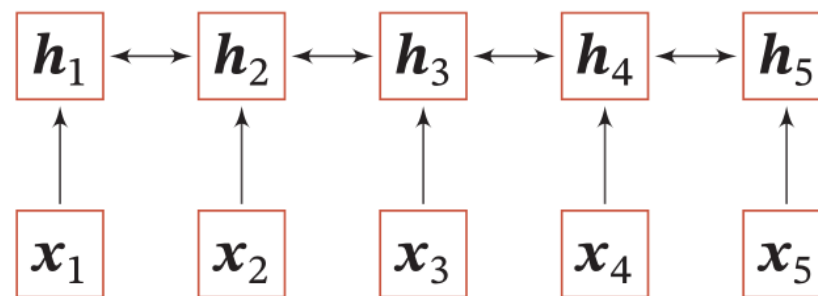
自注意力模型

变长序列的建模

当使用神经网络来处理一个变长的向量序列时，我们通常可以使用卷积网络或循环网络进行编码来得到一个相同长度的输出向量序列



(a) 卷积网络

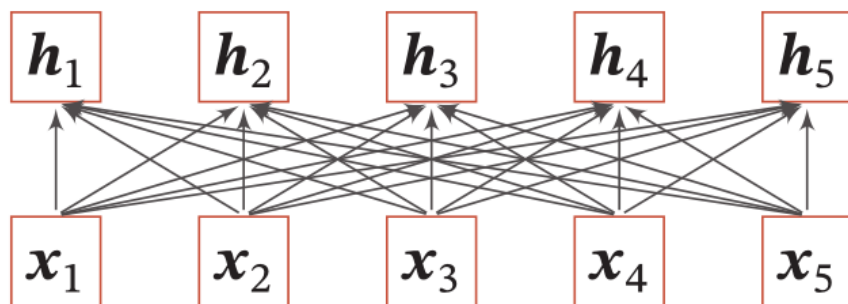


(b) 双向循环网络

只建模了输入信息的局部依赖关系

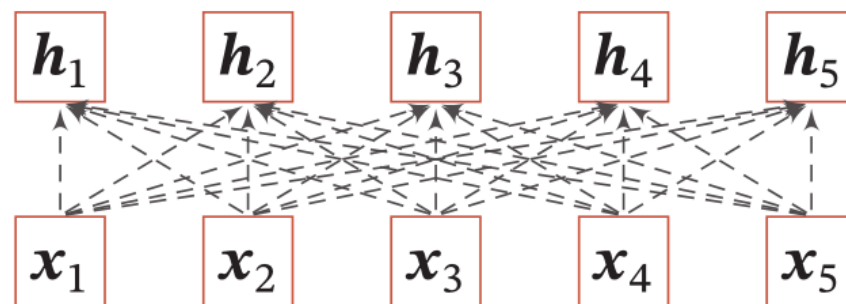
自注意力模型

如何建立非局部（Non-local）的依赖关系
全连接？



(a) 全连接模型

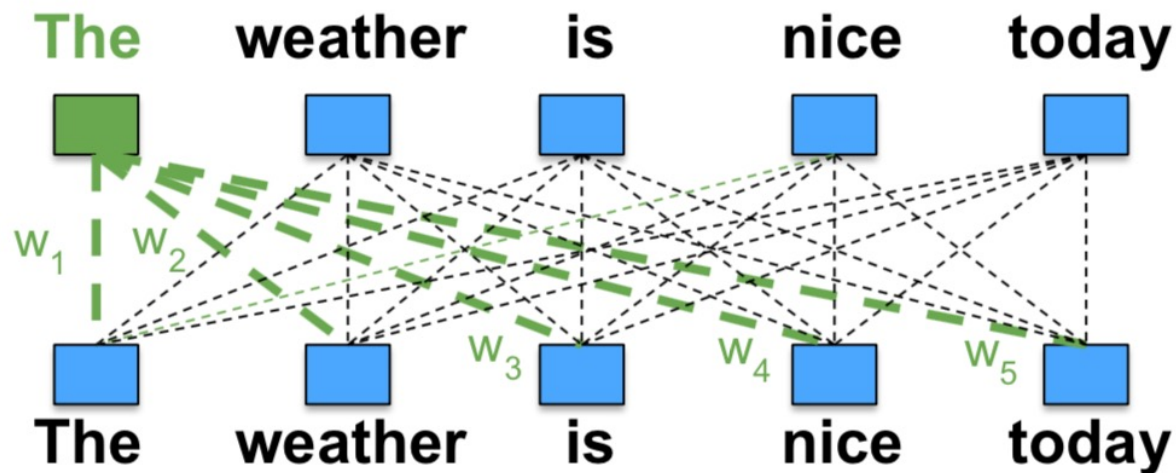
无法处理变长问题



(b) 自注意力模型

连接权重 α_{ij} 由注意力机制动态生成

自注意力示例



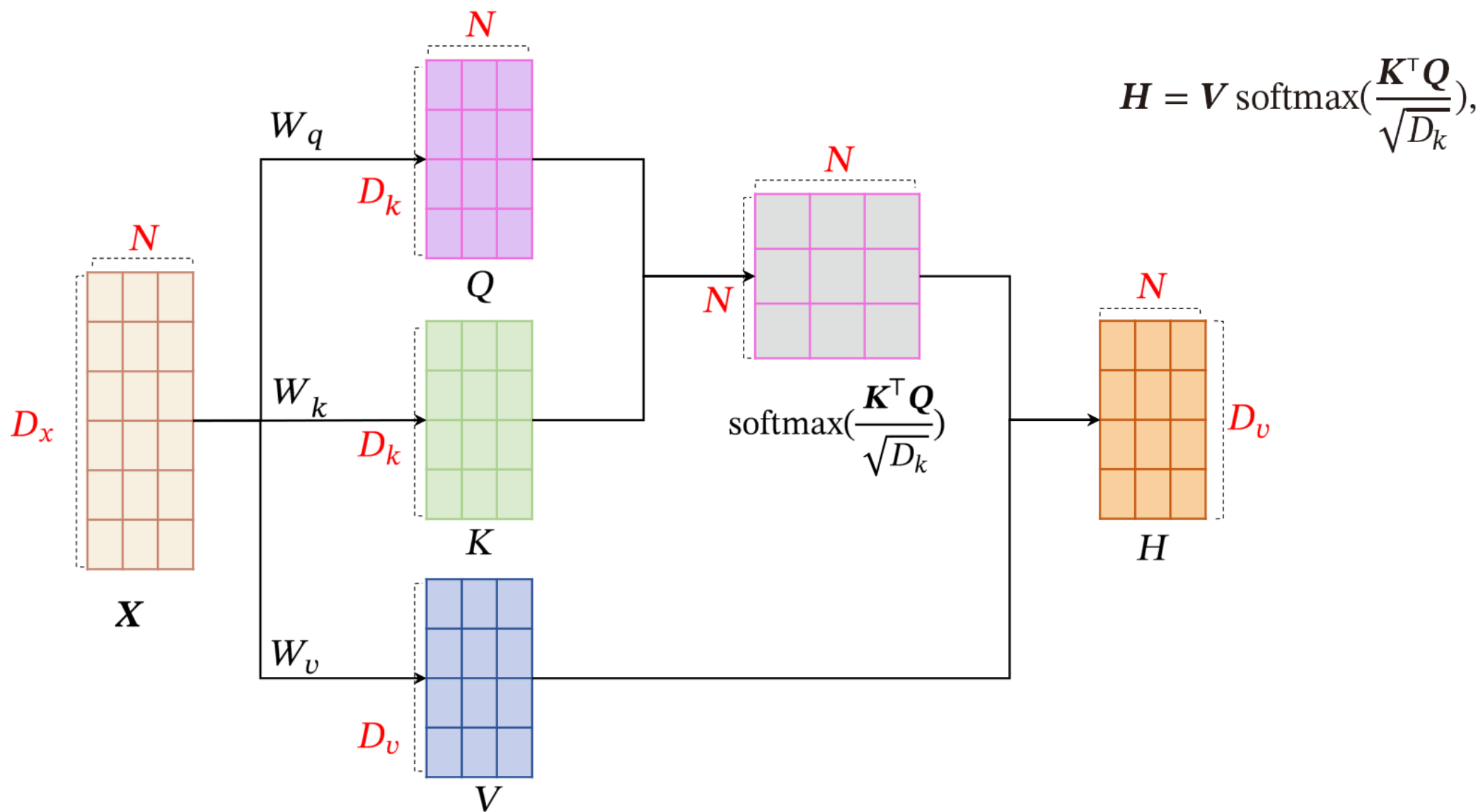
$$w_1, w_2, w_3, w_4, w_5 = \text{softmax} \left(\begin{bmatrix} 0.6 & 0.2 & 0.8 \end{bmatrix} \times \begin{bmatrix} 0.6 & 0.2 & 0.9 & 0.4 & 0.4 \\ 0.2 & 0.3 & 0.1 & 0.1 & 0.1 \\ 0.8 & 0.1 & 0.8 & 0.4 & 0.6 \end{bmatrix} \right)$$

The The weather is nice today

$$\begin{bmatrix} 1.8 \\ 2.3 \\ 0.4 \end{bmatrix} = w_1 \times \begin{bmatrix} 0.6 \\ 0.2 \\ 0.8 \end{bmatrix} + w_2 \times \begin{bmatrix} 0.2 \\ 0.3 \\ 0.1 \end{bmatrix} + w_3 \times \begin{bmatrix} 0.9 \\ 0.1 \\ 0.8 \end{bmatrix} + w_4 \times \begin{bmatrix} 0.4 \\ 0.1 \\ 0.4 \end{bmatrix} + w_5 \times \begin{bmatrix} 0.4 \\ 0.1 \\ 0.6 \end{bmatrix}$$

The The weather is nice today

自注意力模型的矩阵表示



自注意力模型的矩阵表示

输入序列为 $X = [x_1, \dots, x_N] \in \mathbb{R}^{D_x \times N}$ 首先生成三个向量序列

$$Q = W_q X \in \mathbb{R}^{D_k \times N},$$

$$K = W_k X \in \mathbb{R}^{D_k \times N},$$

$$V = W_v X \in \mathbb{R}^{D_v \times N},$$

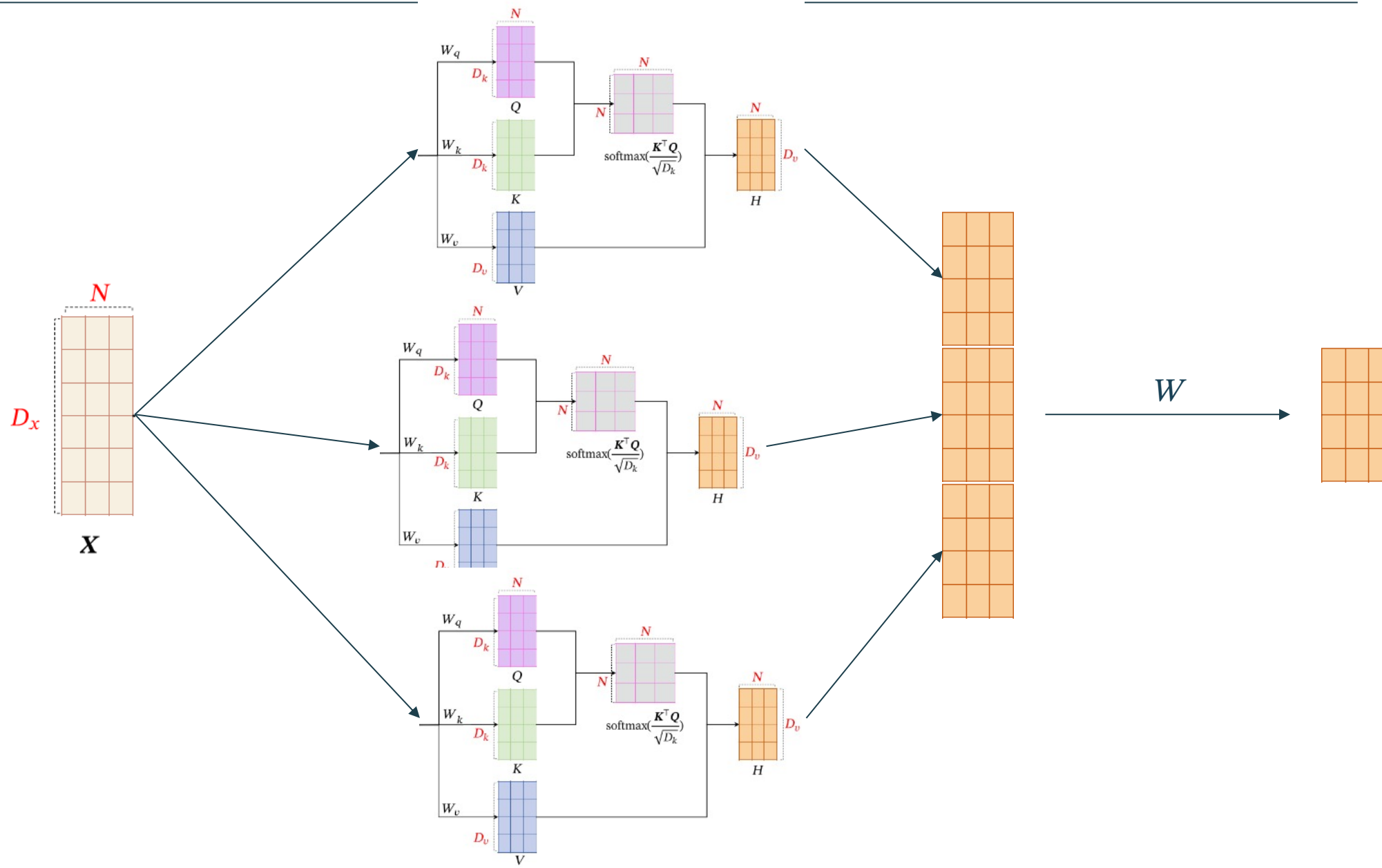
计算 h_n

$$h_n = \text{att}((K, V), q_n)$$

如果使用缩放点积来作为注意力打分函数，输出向量序列可以简写为

$$H = V \text{softmax}\left(\frac{K^\top Q}{\sqrt{D_k}}\right),$$

多头 (multi-head) 自注意力模型



注意力机制与外部记忆

Transformer

Transformer Encoder

仅仅自注意力还不够

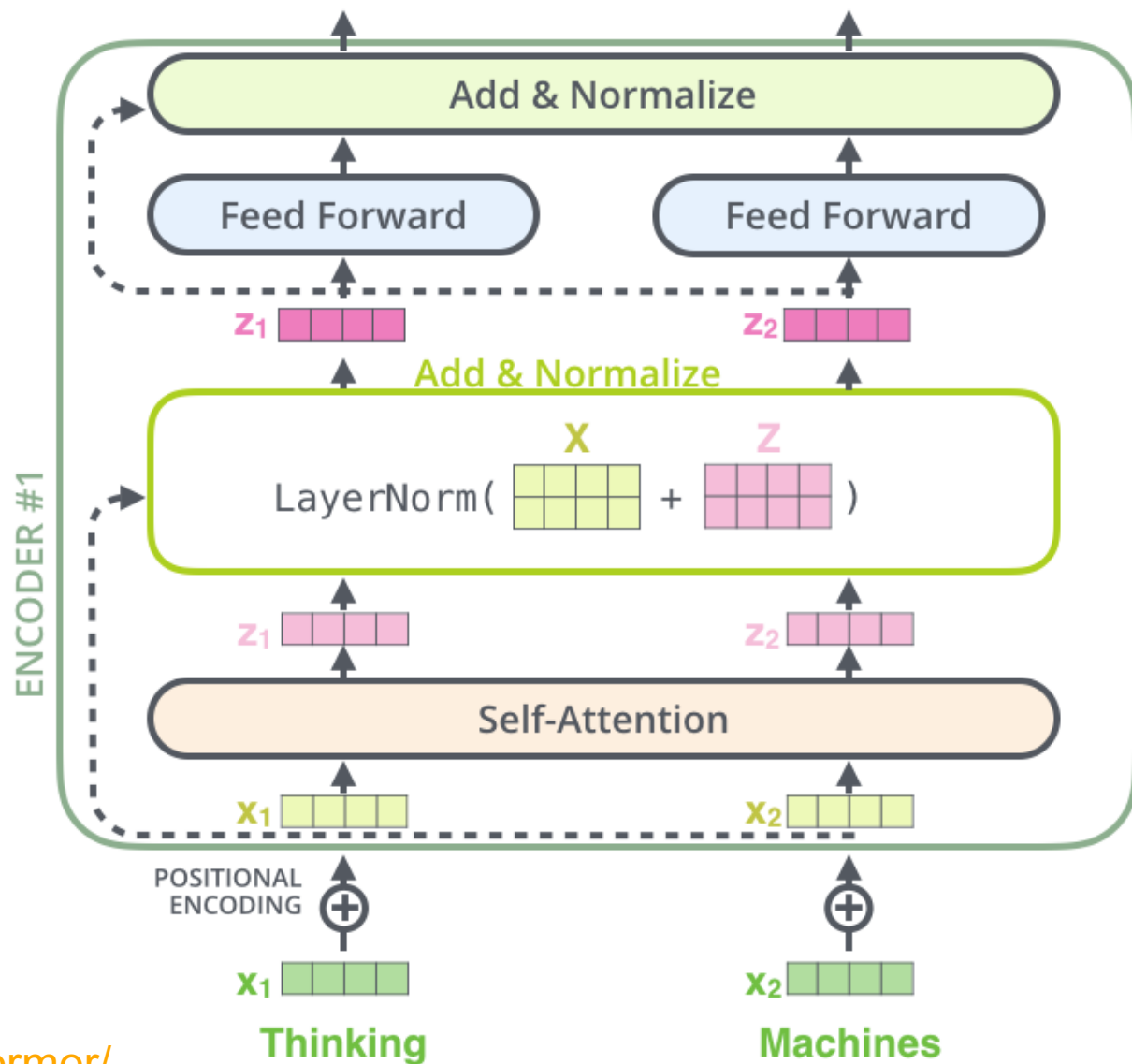
其它操作

位置编码

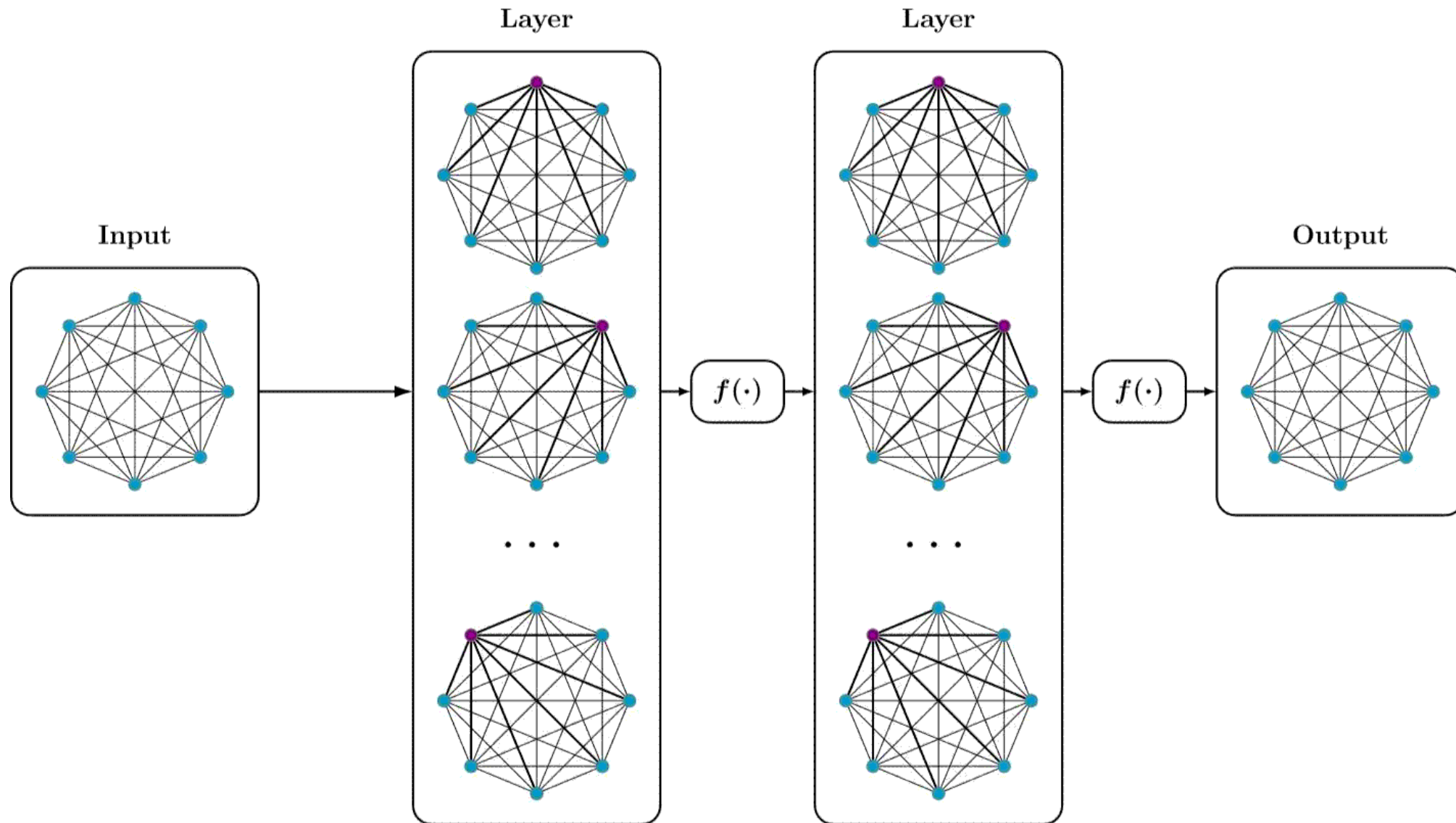
层归一化

直连边

逐位的FNN



Transformer



复杂度分析

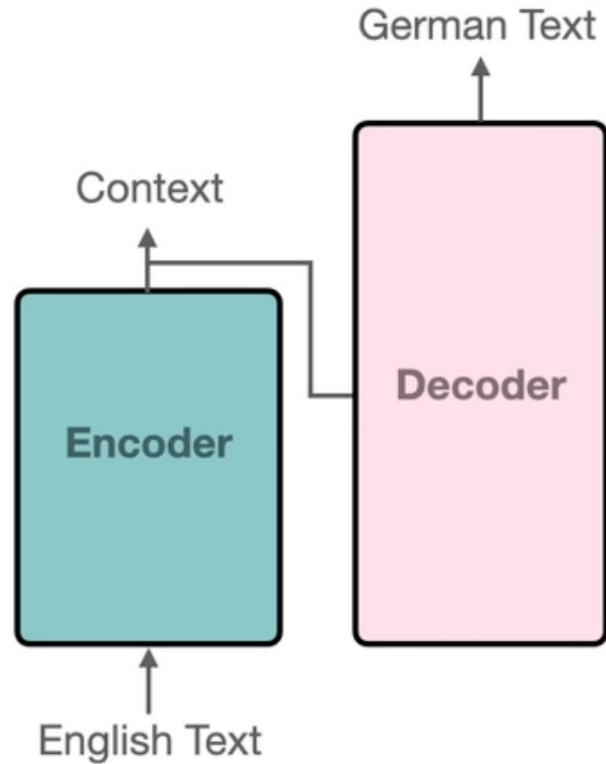
模型	每层复杂度	序列操作数	最大路径长度
CNN	$O(kLd^2)$	$O(1)$	$O(\log_k(L))$
RNN	$O(Ld^2)$	$O(L)$	$O(L)$
Transformer	$O(L^2d)$	$O(1)$	$O(1)$

k 卷积核大小 L 序列长度 d 维度

Transformer

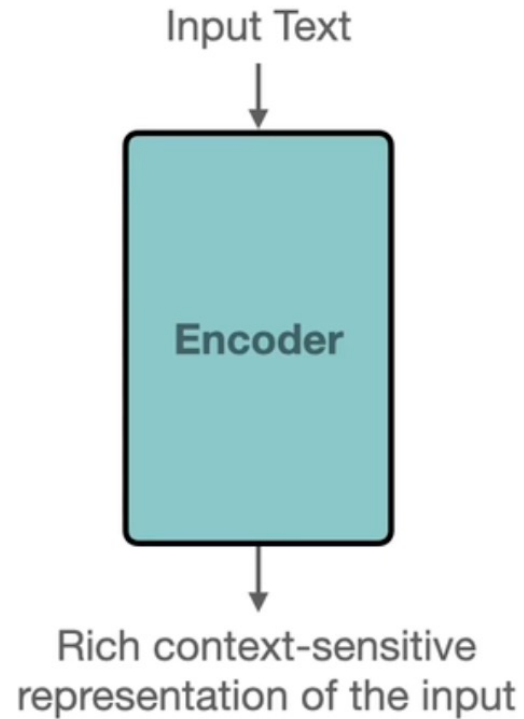
Transformer

Original Transformer



Translating English into German

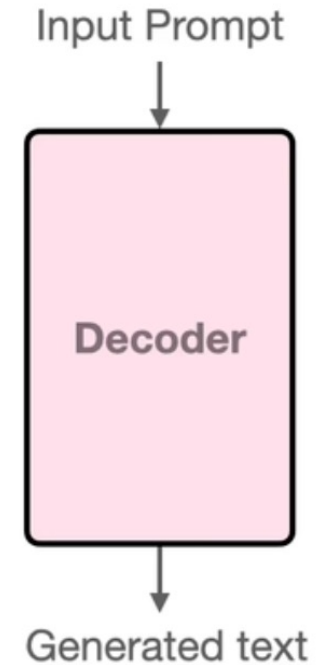
Encoder Model



Basis:

- BERT model
- Most of the embedding models

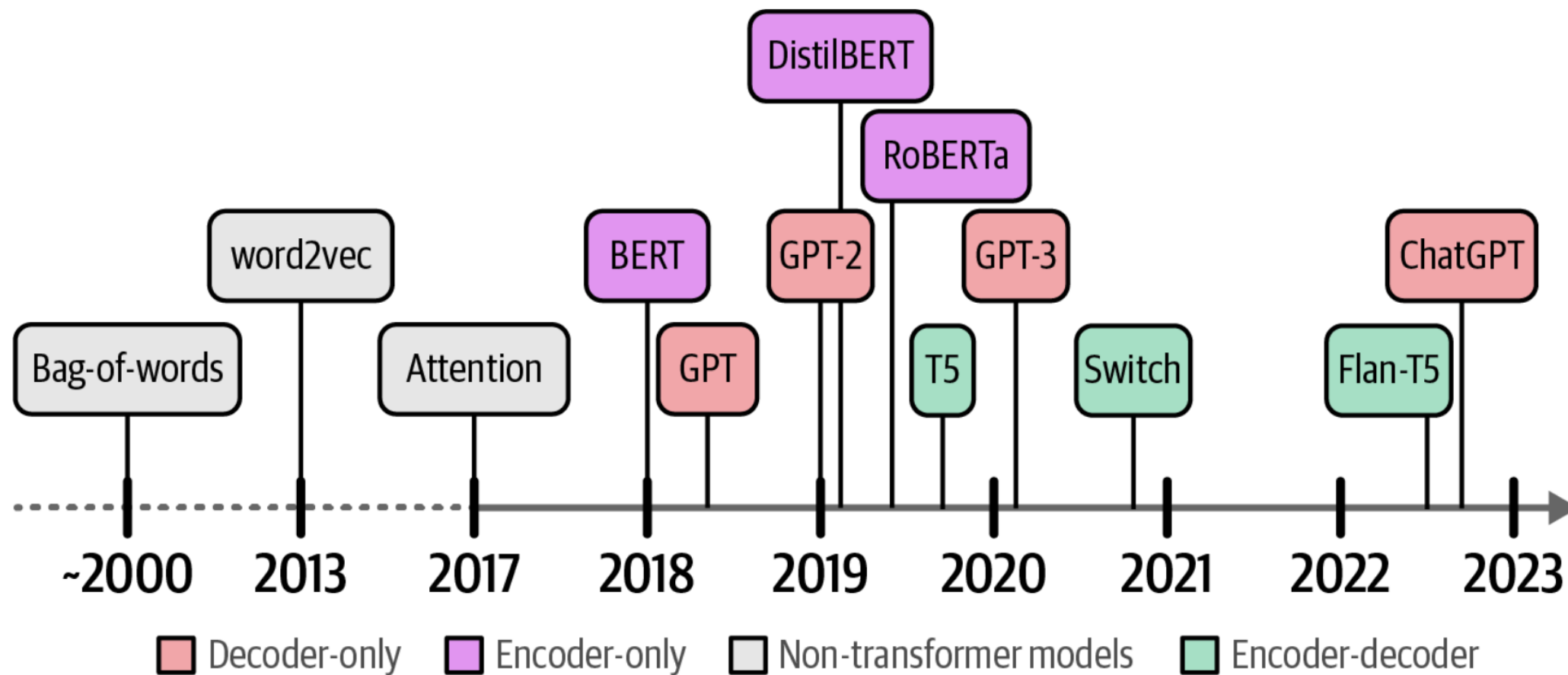
Decoder Model



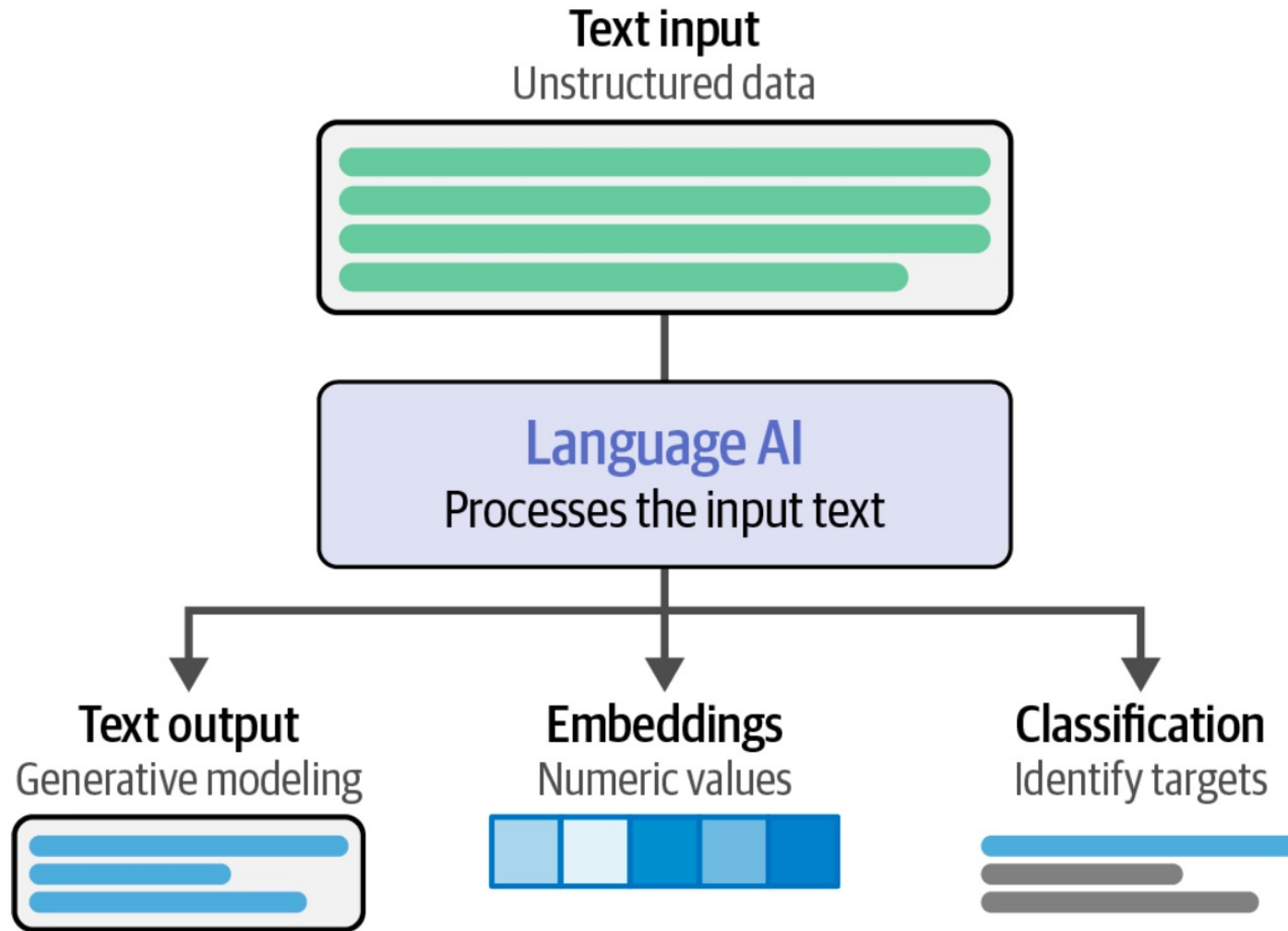
Basis:

Popular LLMs from OpenAI, Anthropic, Cohere and Meta

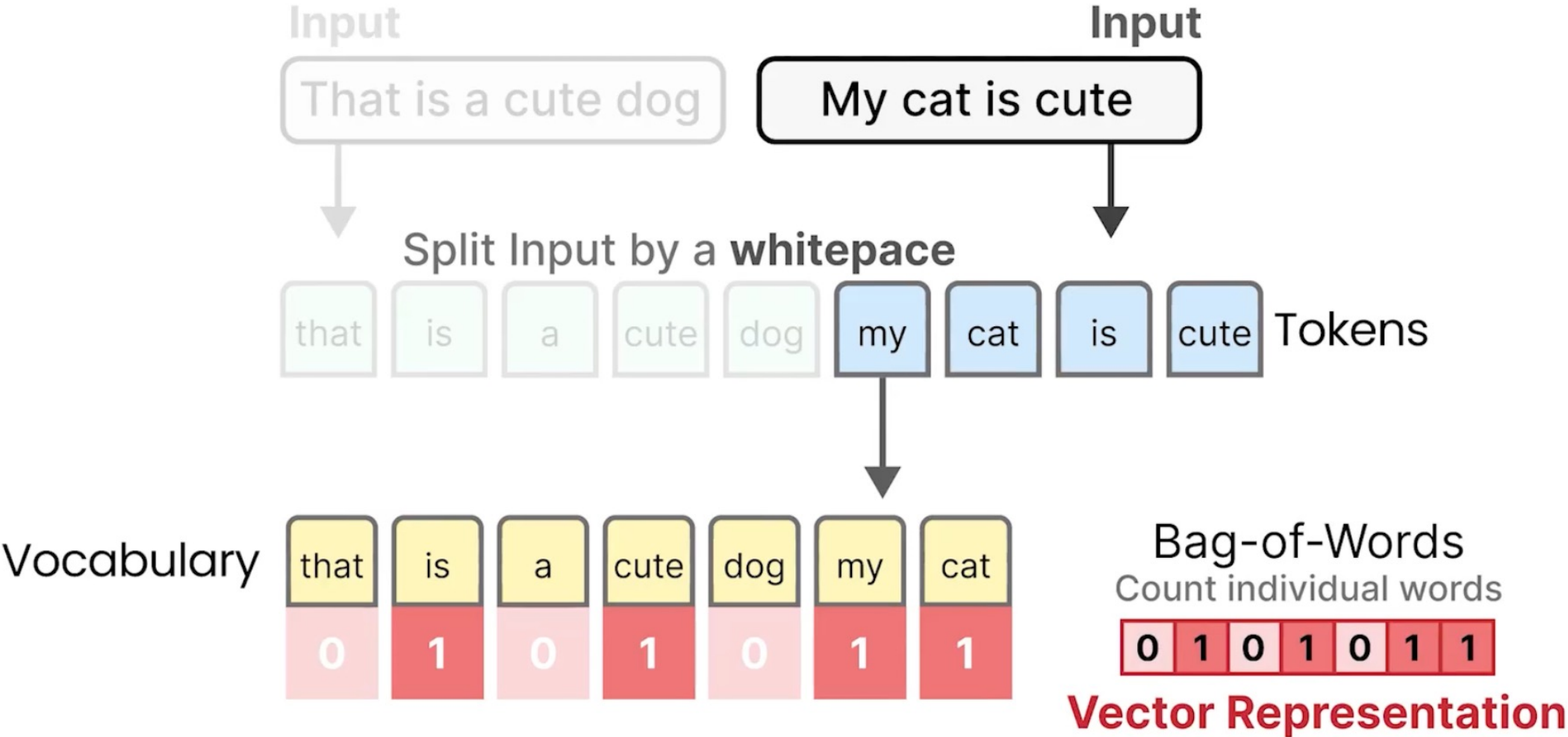
Language AI



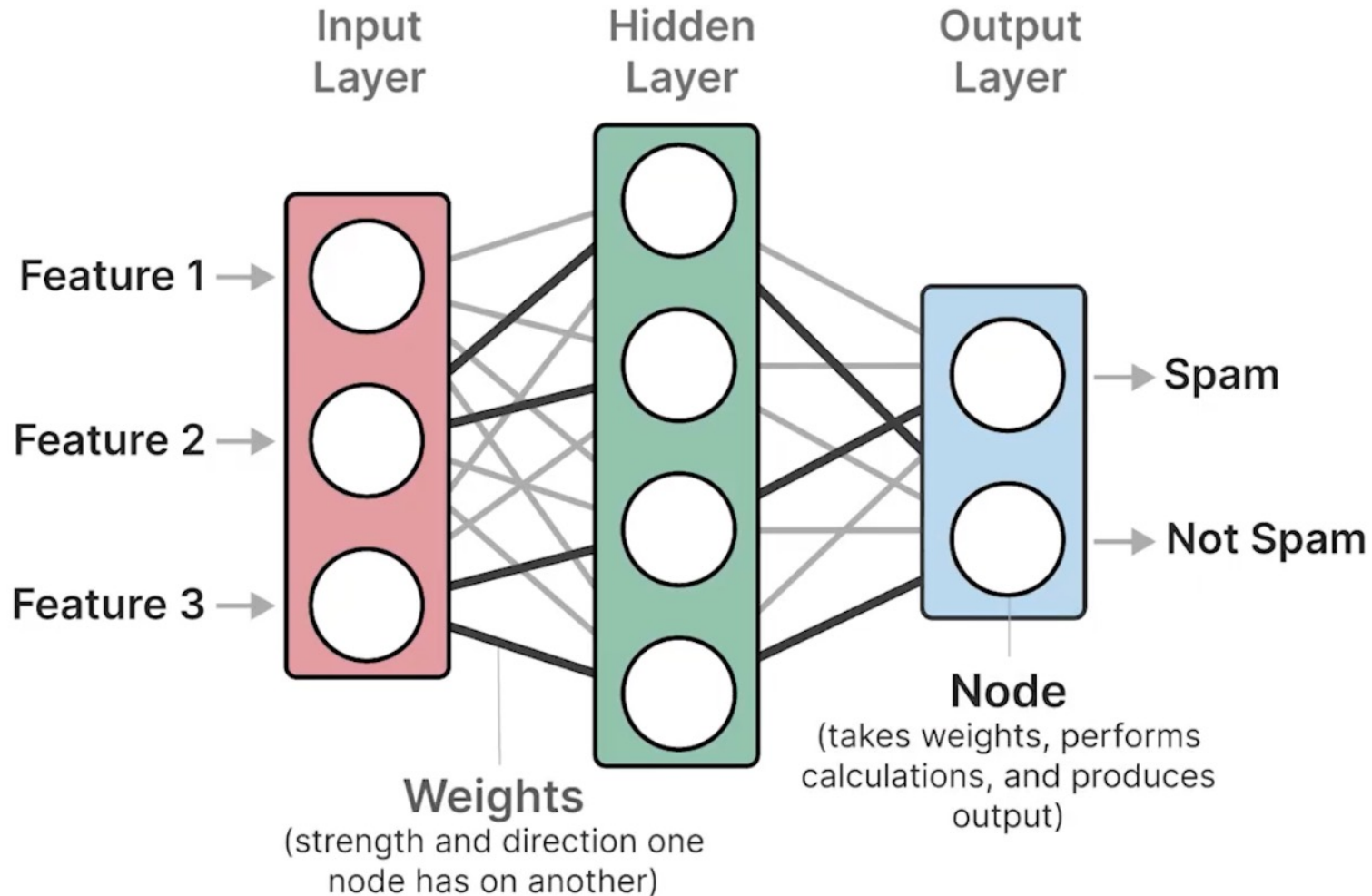
Language AI



Bag of words

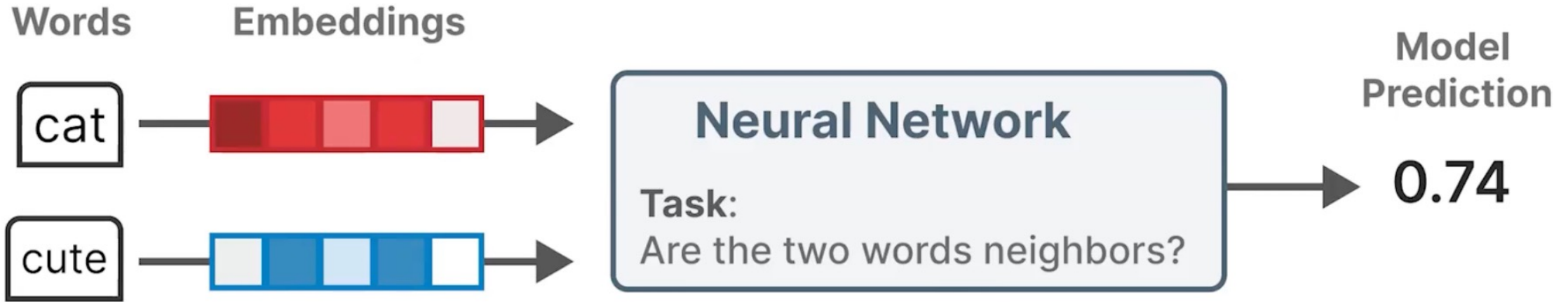


Vector Embeddings

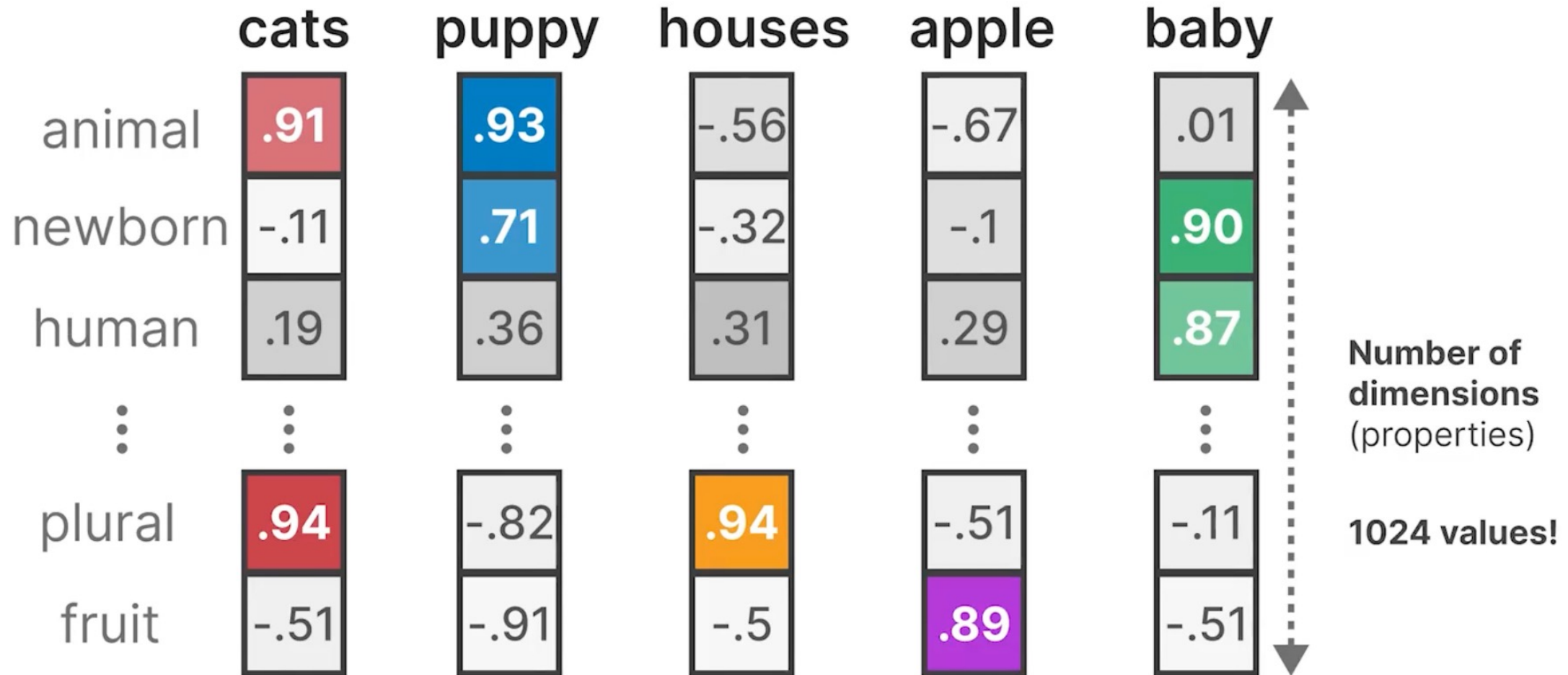


- **Bag-of-Words** does not consider the semantic nature of text
- **Word2Vec** could capture meaning of words in vector embeddings through neural networks.

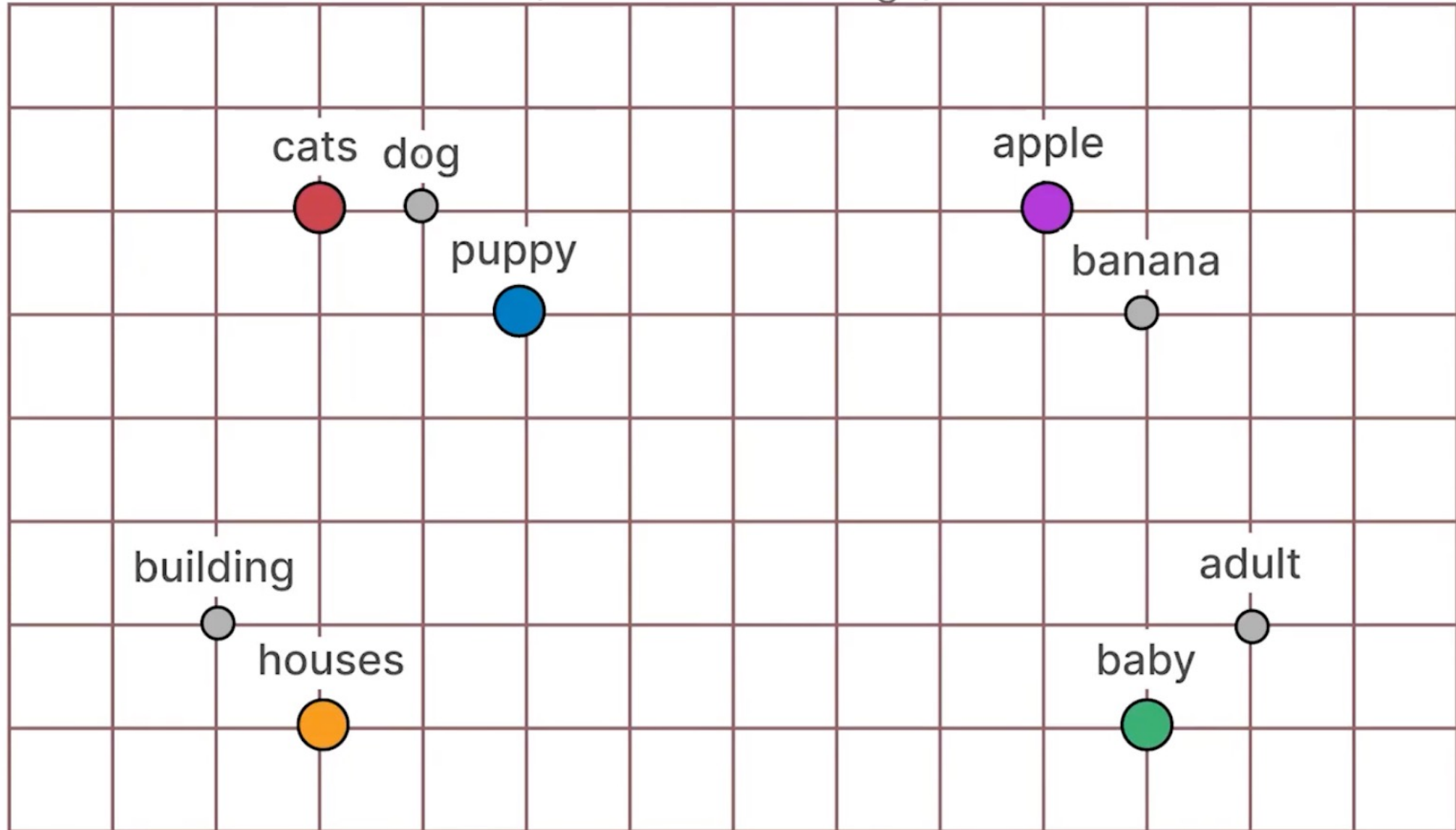
Vector Embeddings



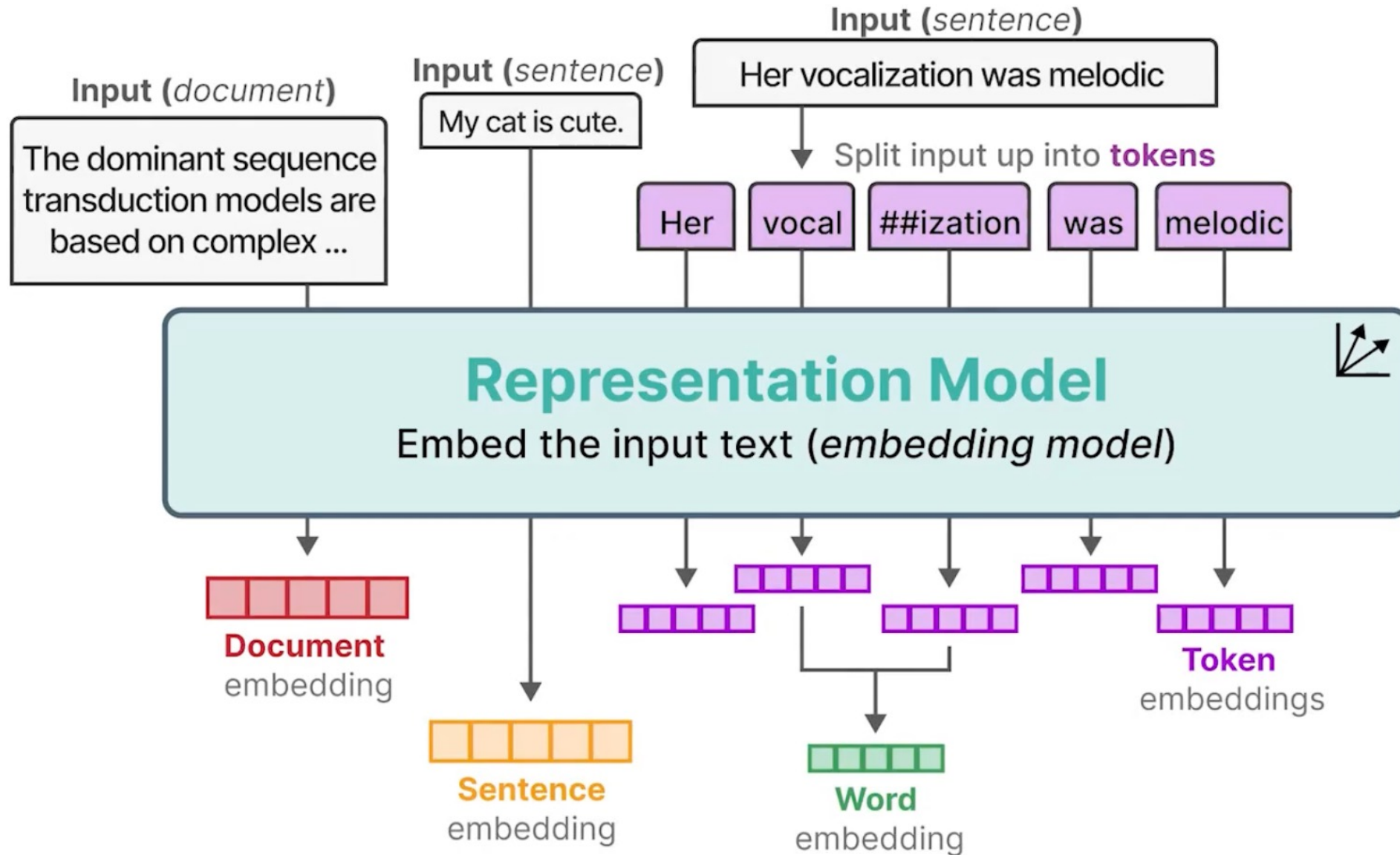
Vector Embeddings



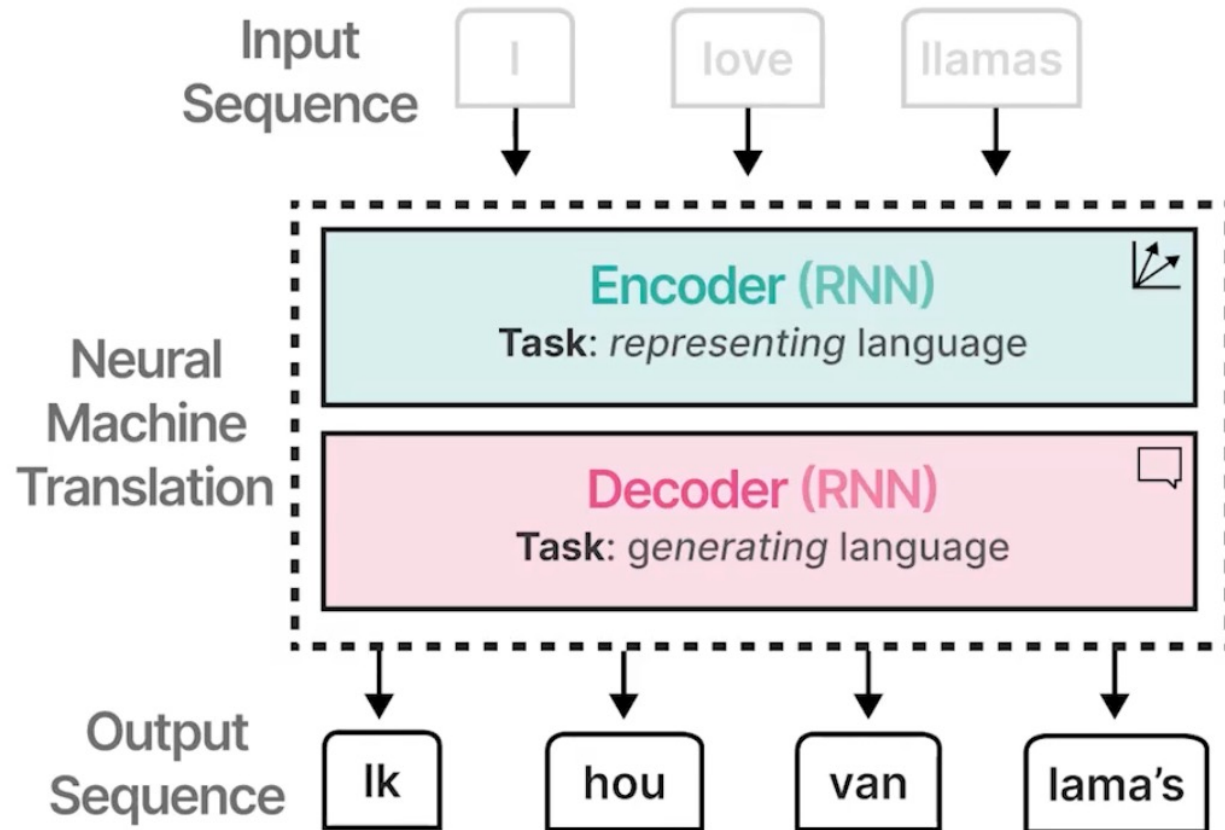
Vector Embeddings



Vector Embeddings

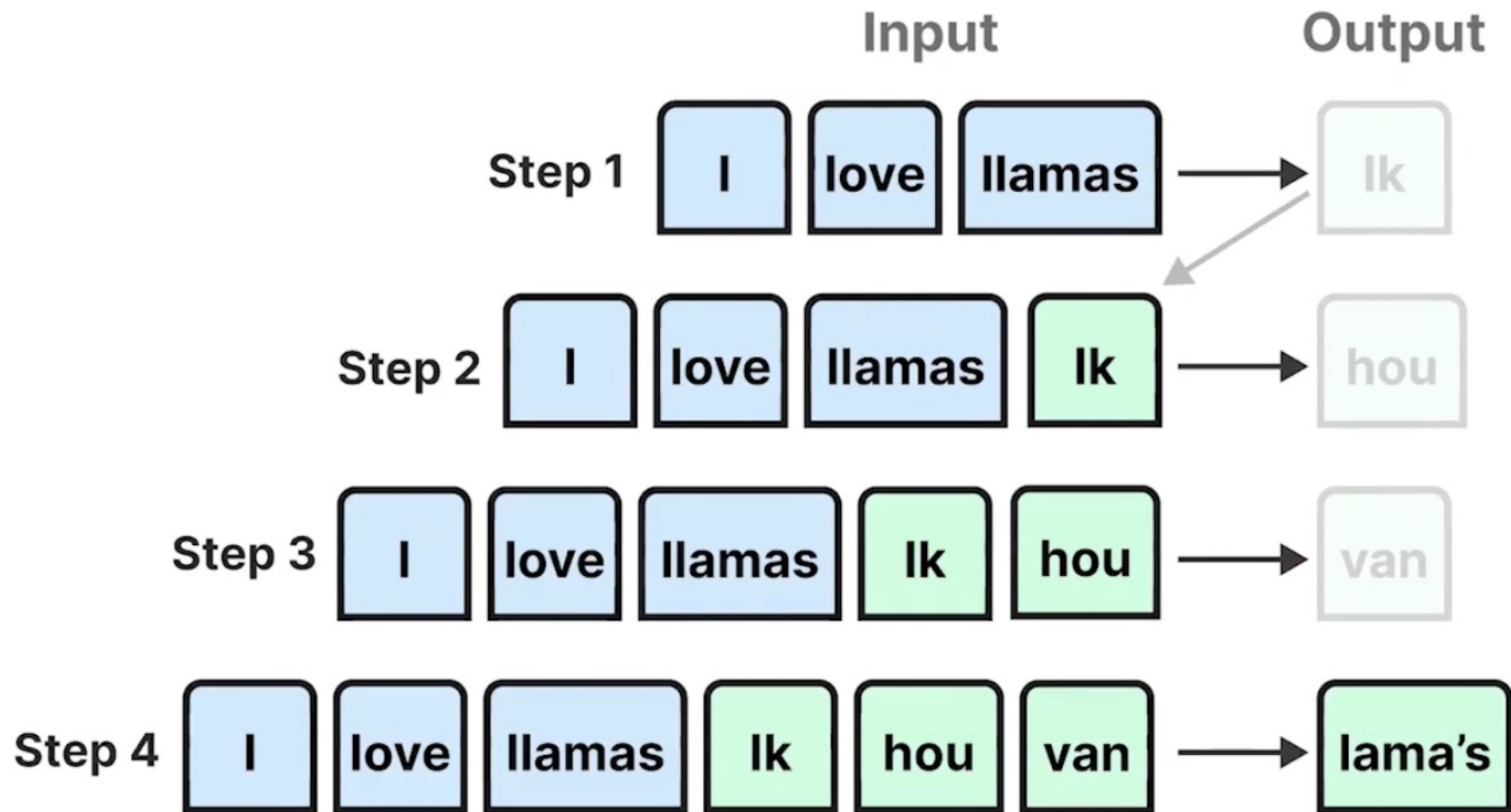


Attention

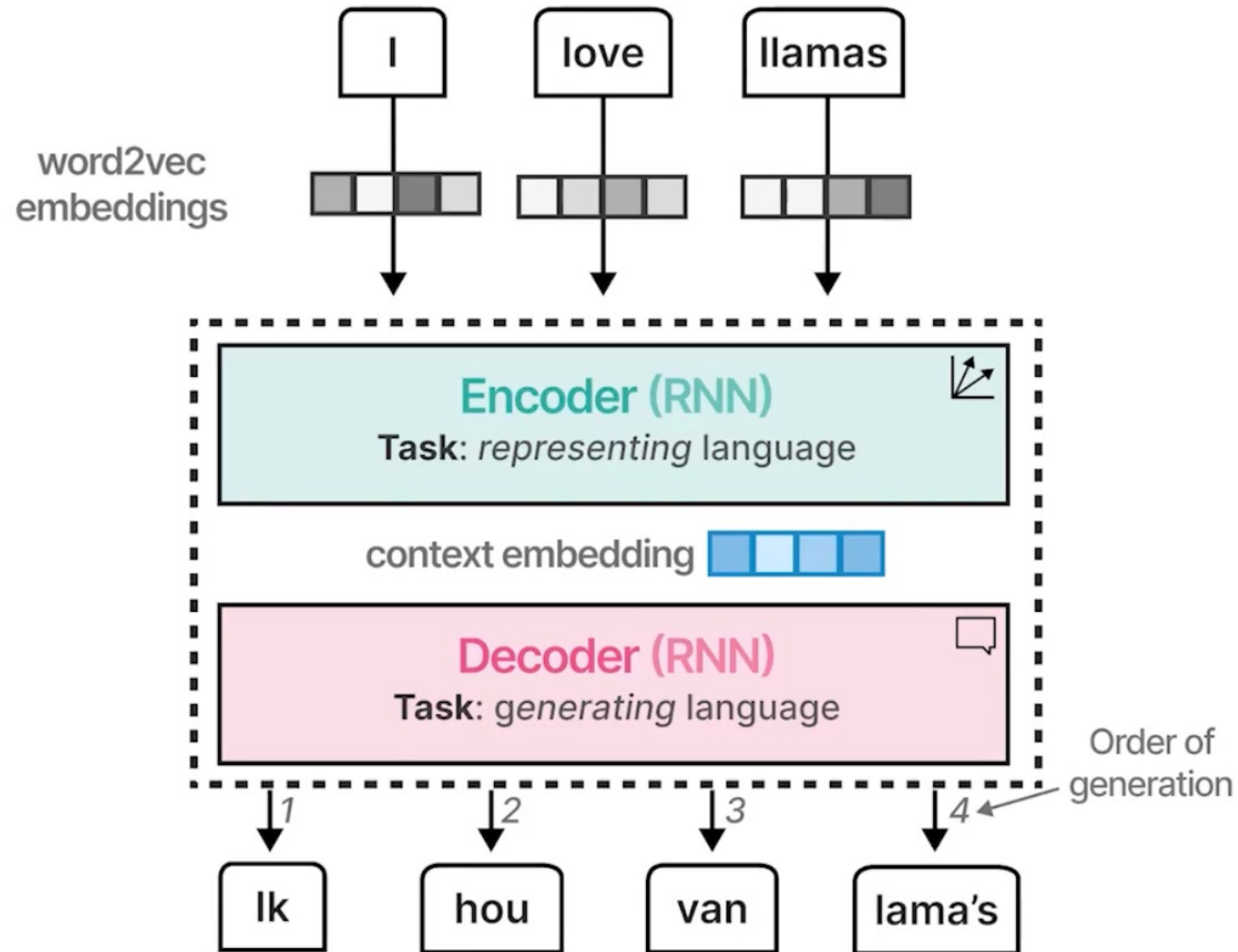


- **Word2Vec** creates *static* embeddings.
 - *The same embedding is generated for the word "**bank**" regardless of the context.*
- **Recurrent Neural Networks (RNNs)** can be used to model entire sequences

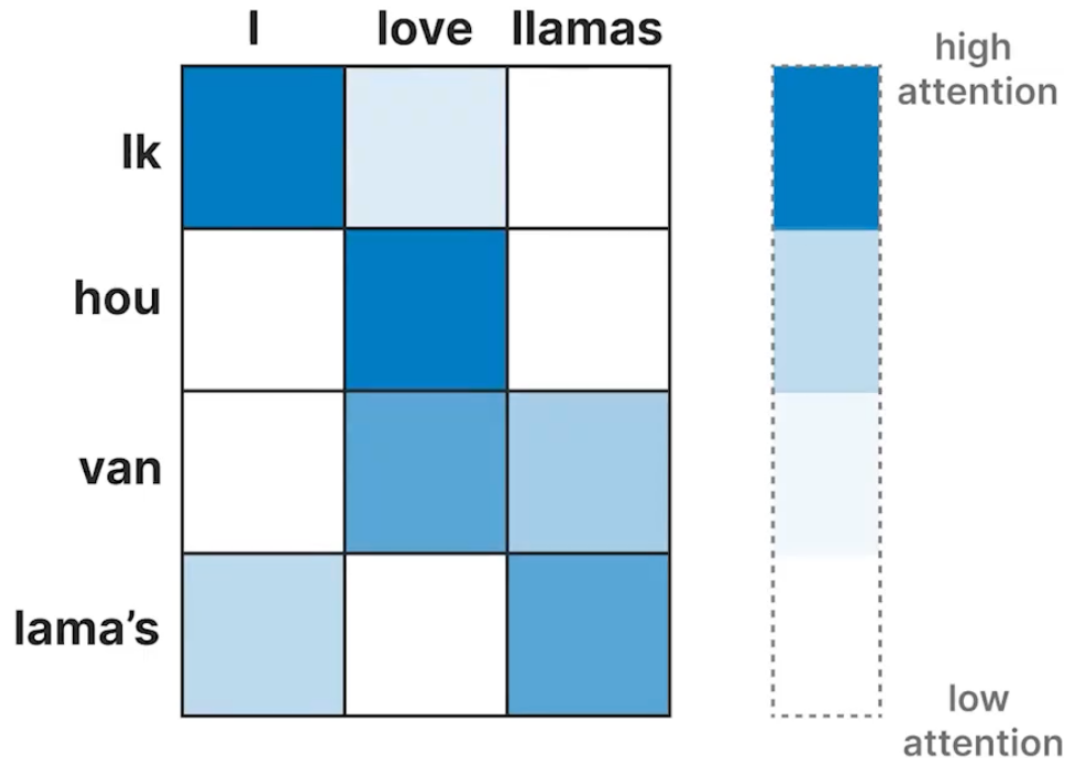
Attention



Attention

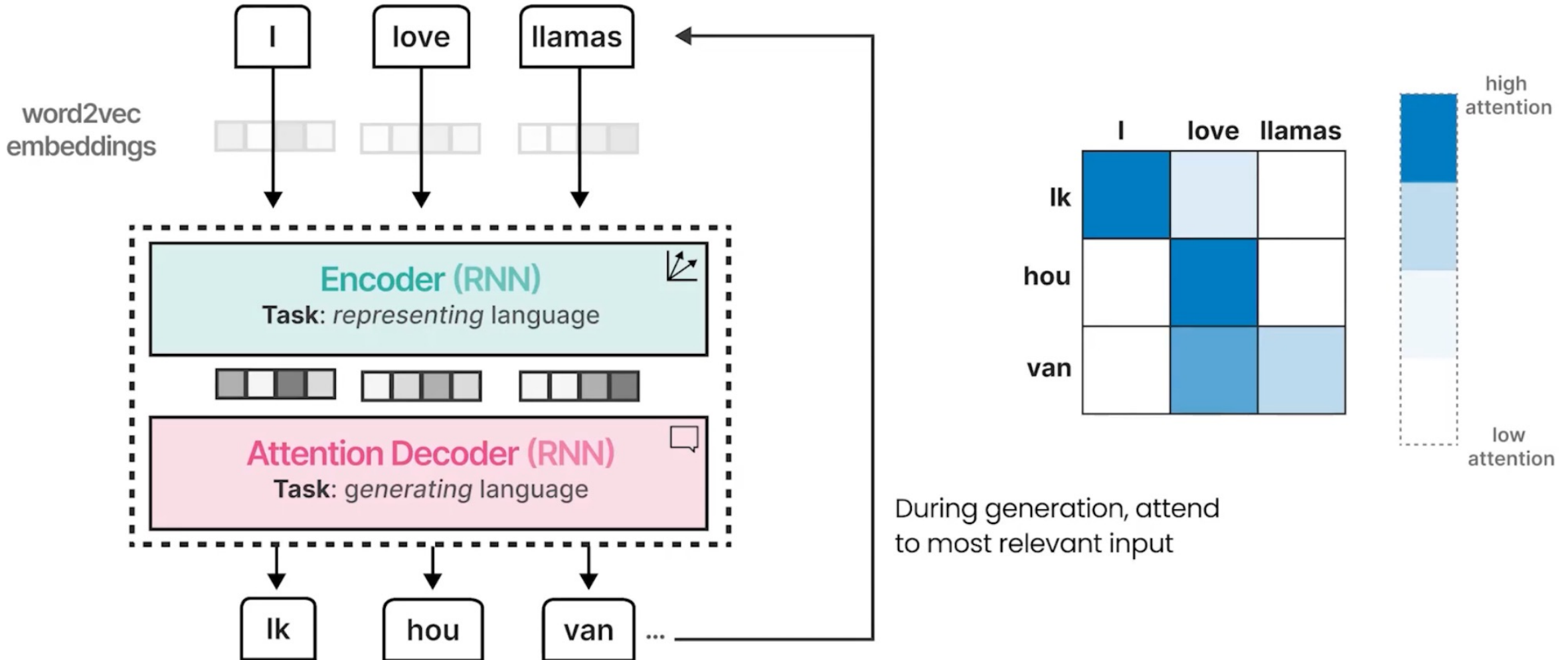


Attention



- **Attention** allows a model to focus on parts of the input that are relevant to one another
- "Attend" to each other and amplify their signal

Attention



Attention

Attention Is All You Need

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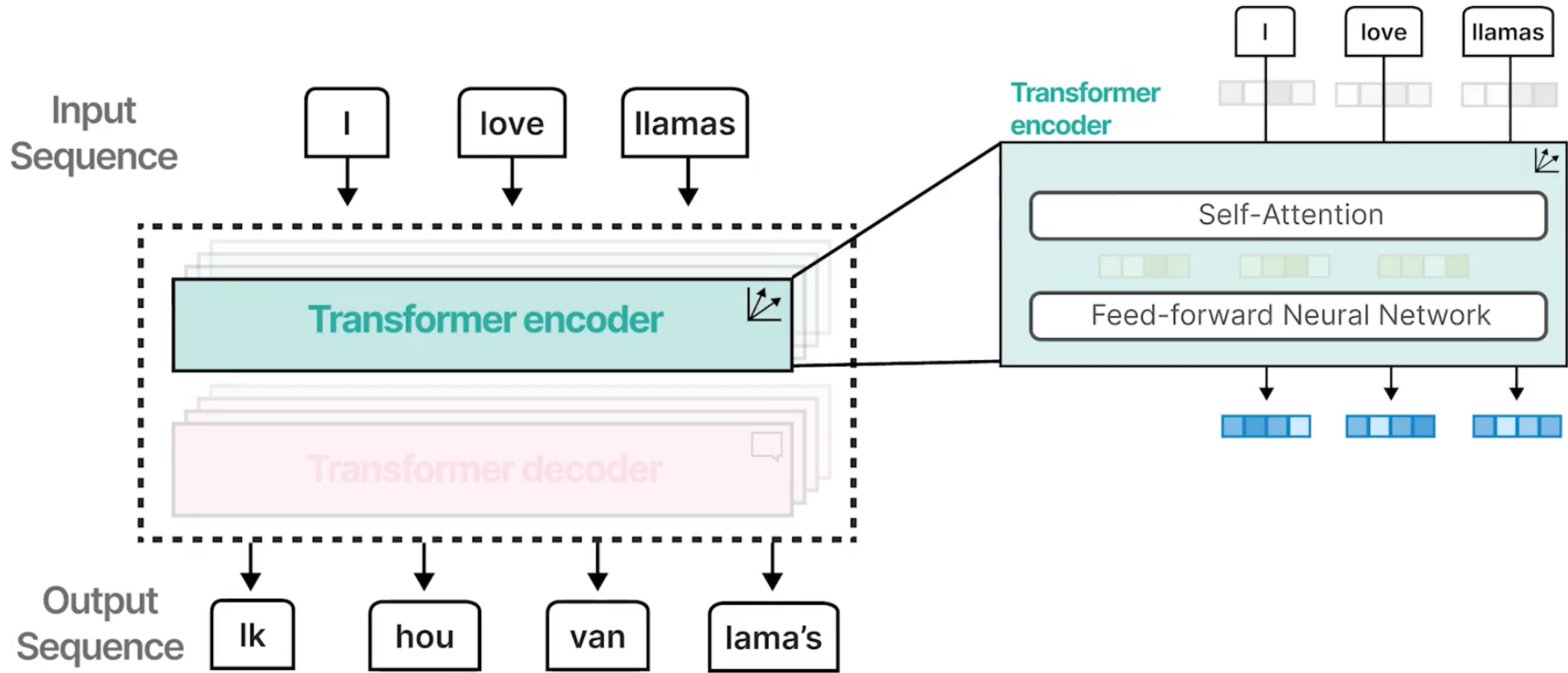
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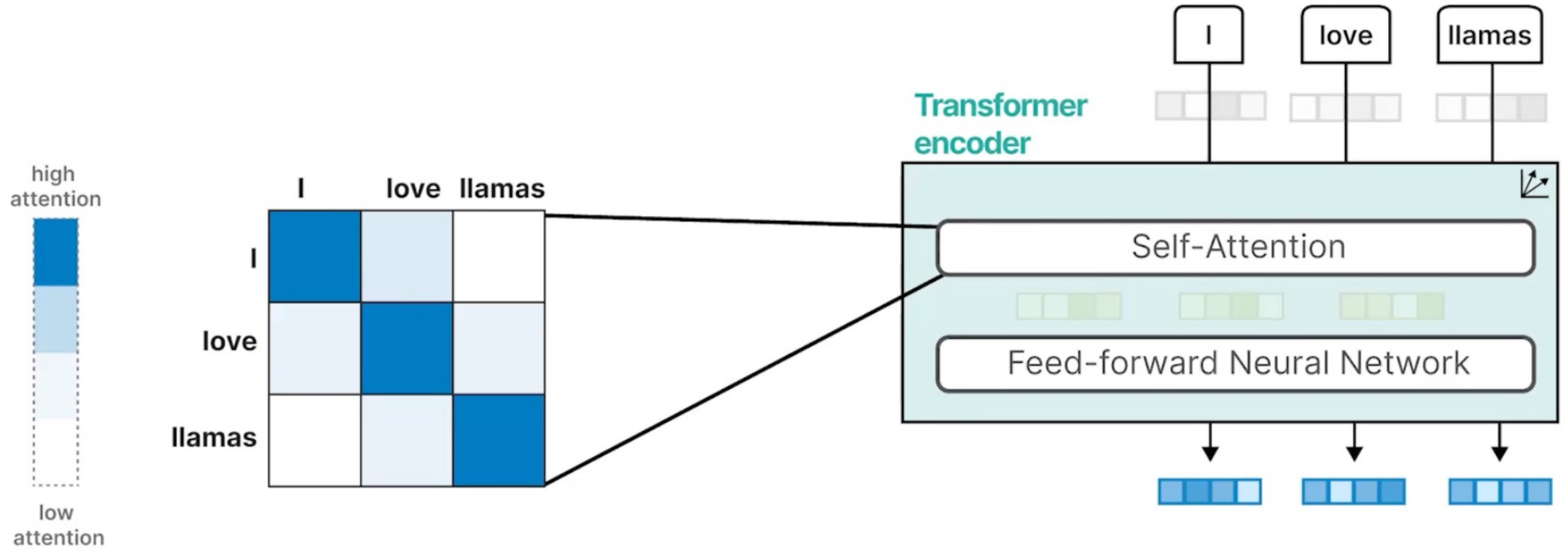
Illia Polosukhin* ‡
illia.polosukhin@gmail.com

- **Transformer** – A new architecture solely based on attention and without the RNN.
- Could be trained in parallel which speeds up calculation significantly

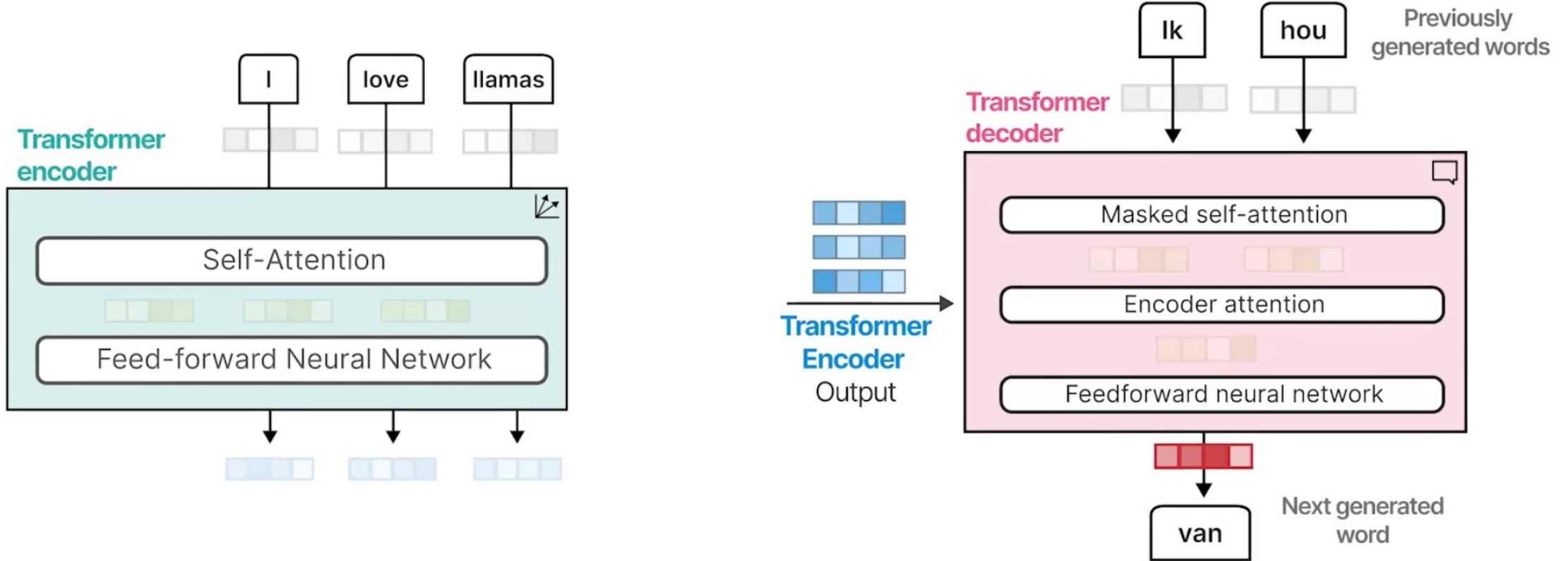
Attention



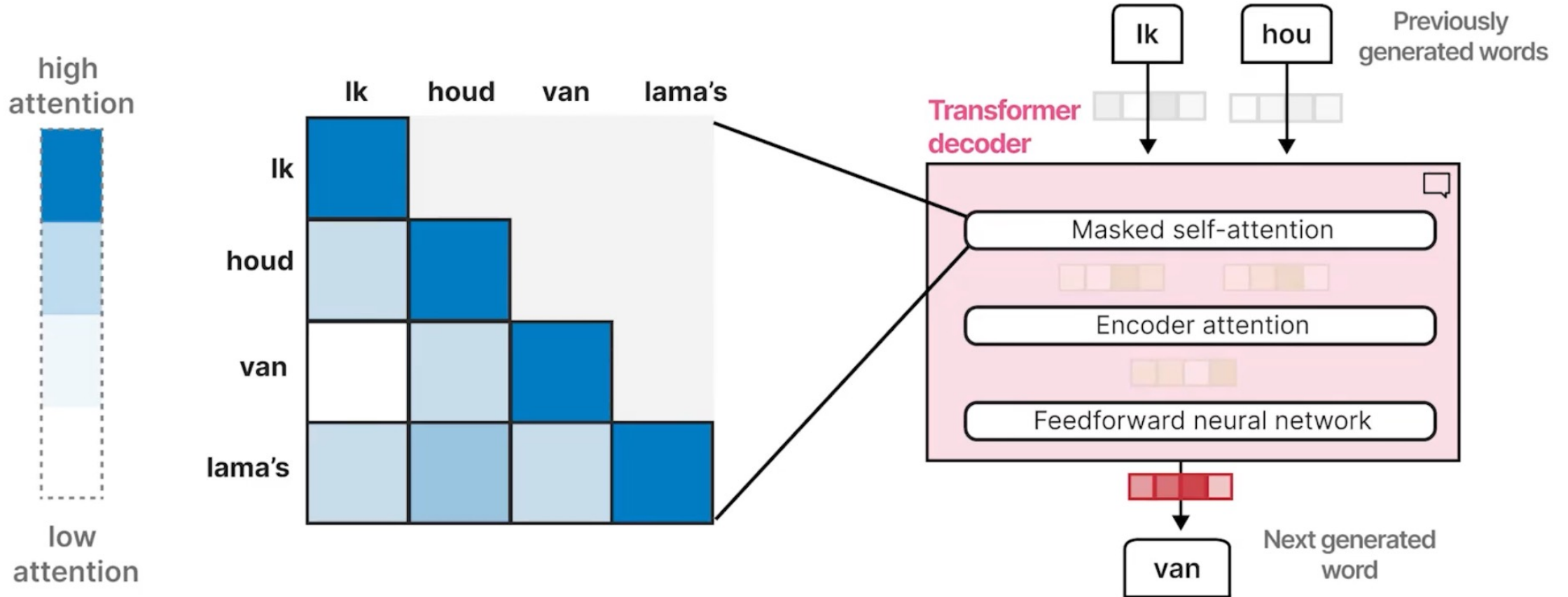
Attention



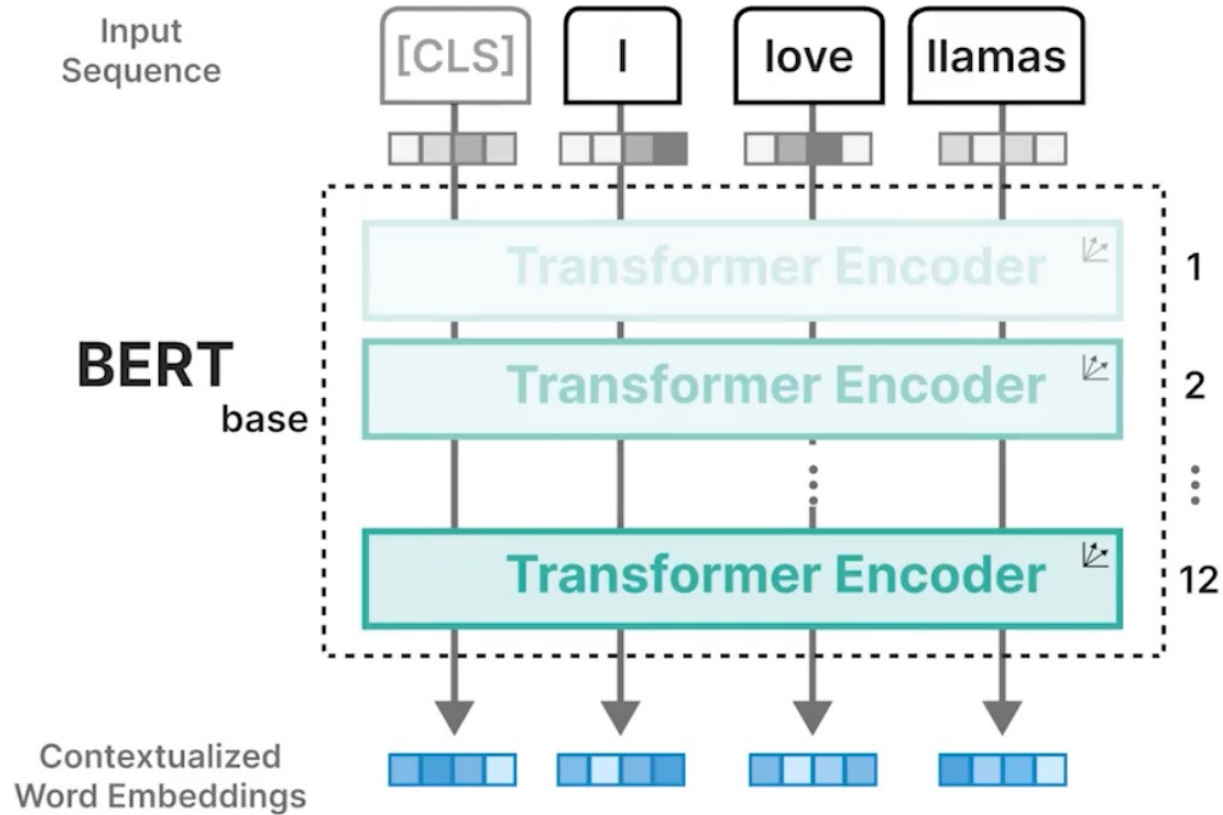
Attention



Attention

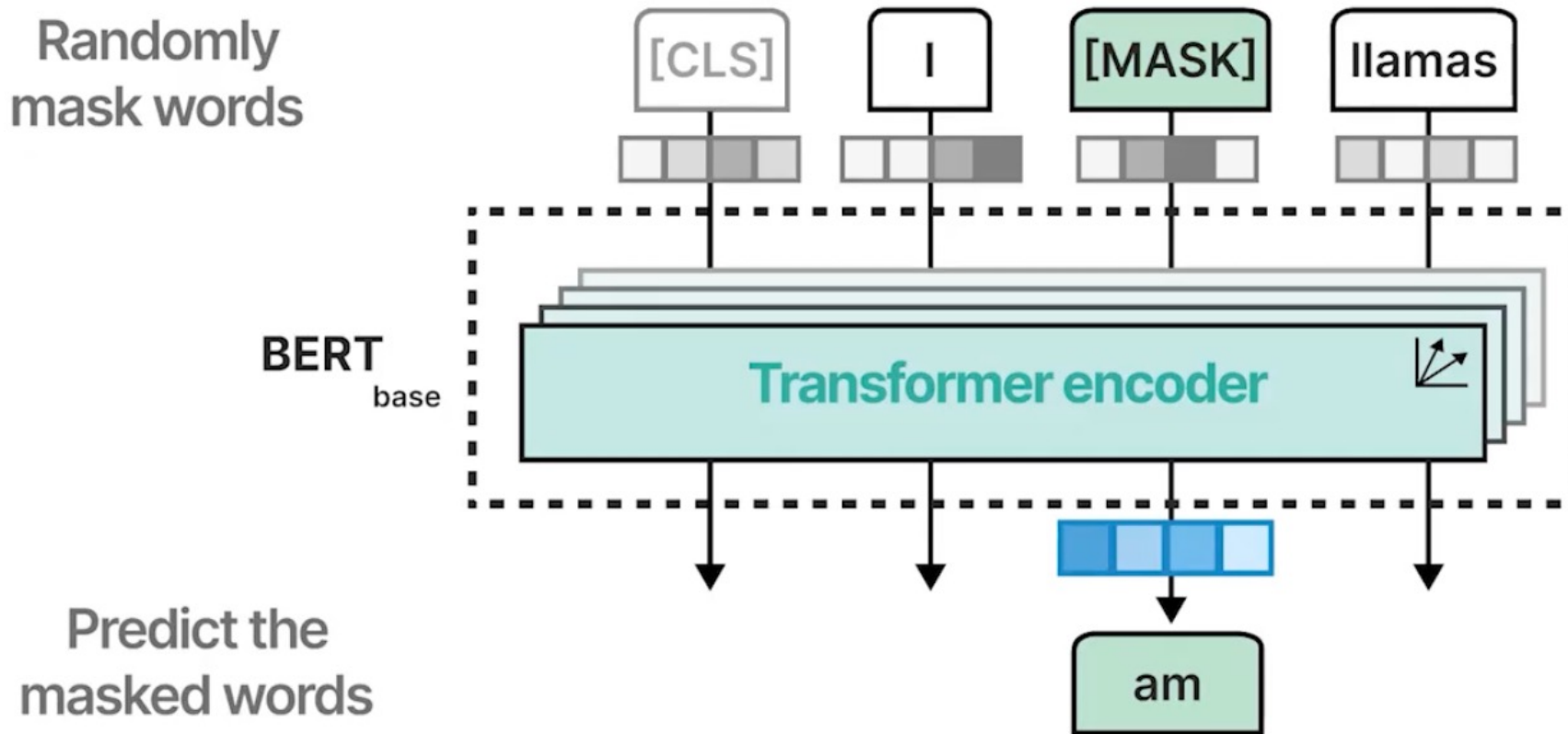


Representation Models



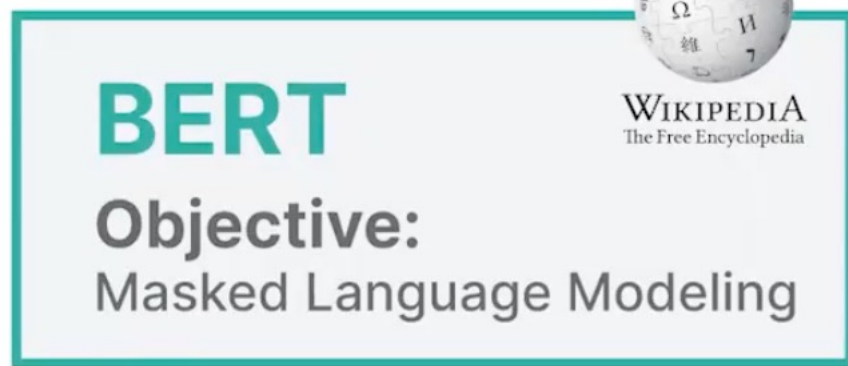
Bidirectional Encoder Representations from Transformers (BERT)

Representation Models



Representation Models

1 Pre-train on large dataset



2 Fine-tune for downstream task

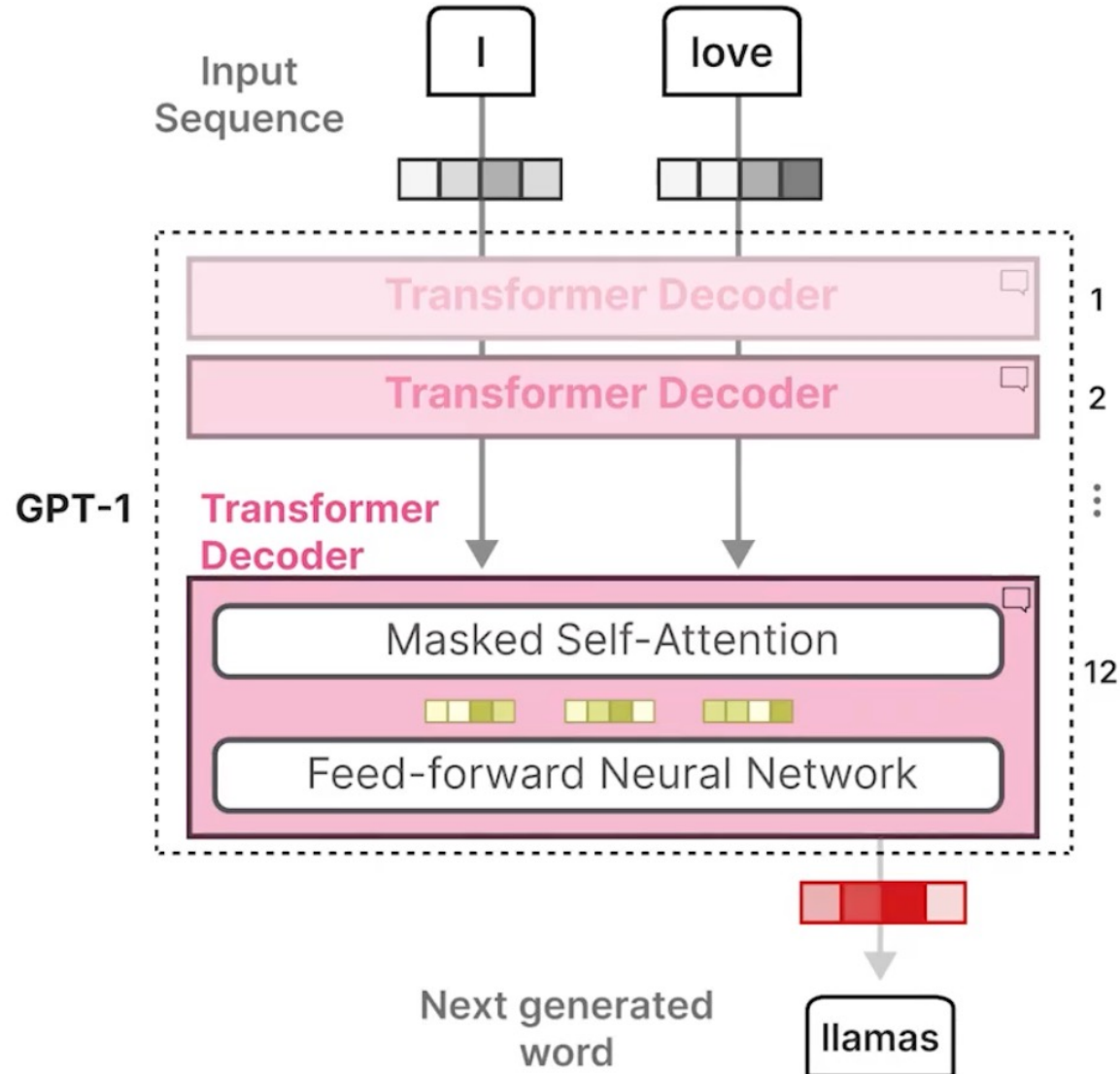


Classification

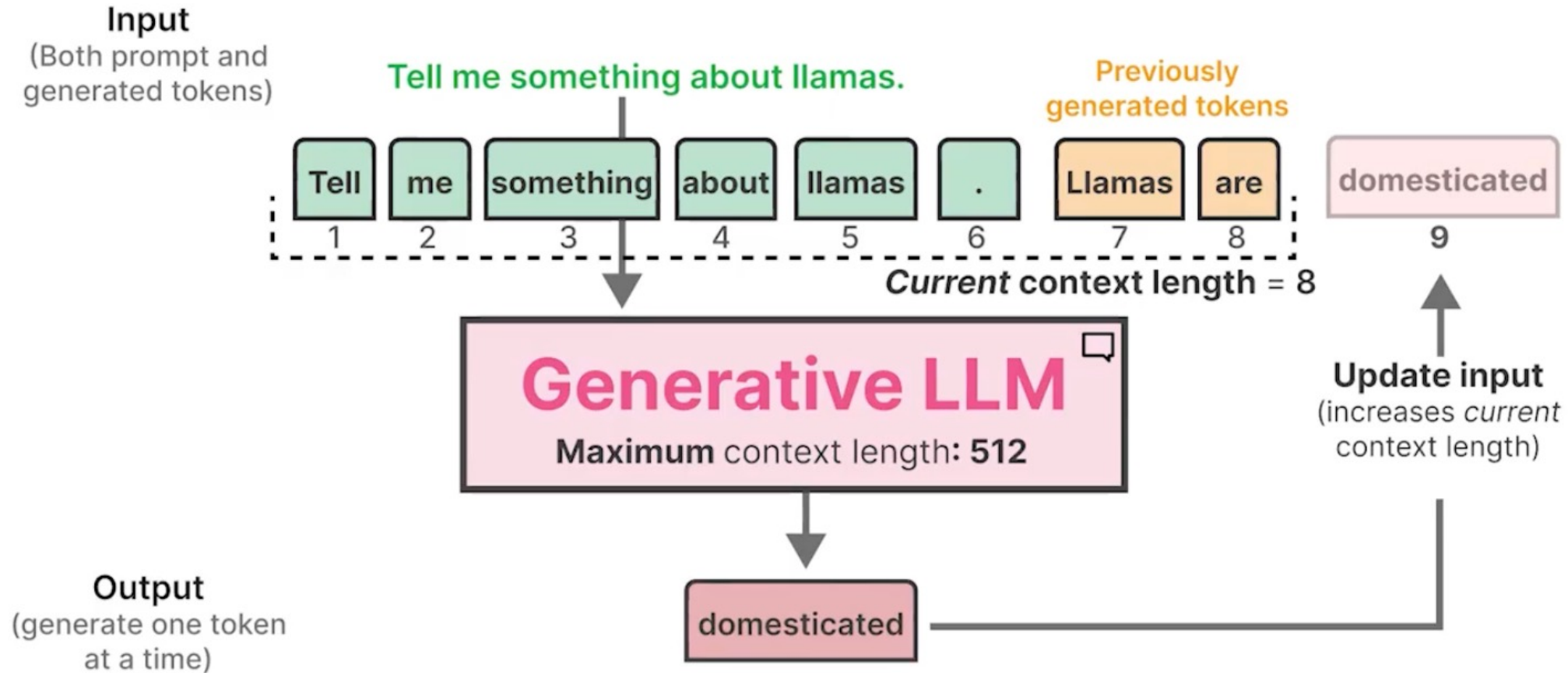
Named Entity
Recognition

Paraphrase
Identification

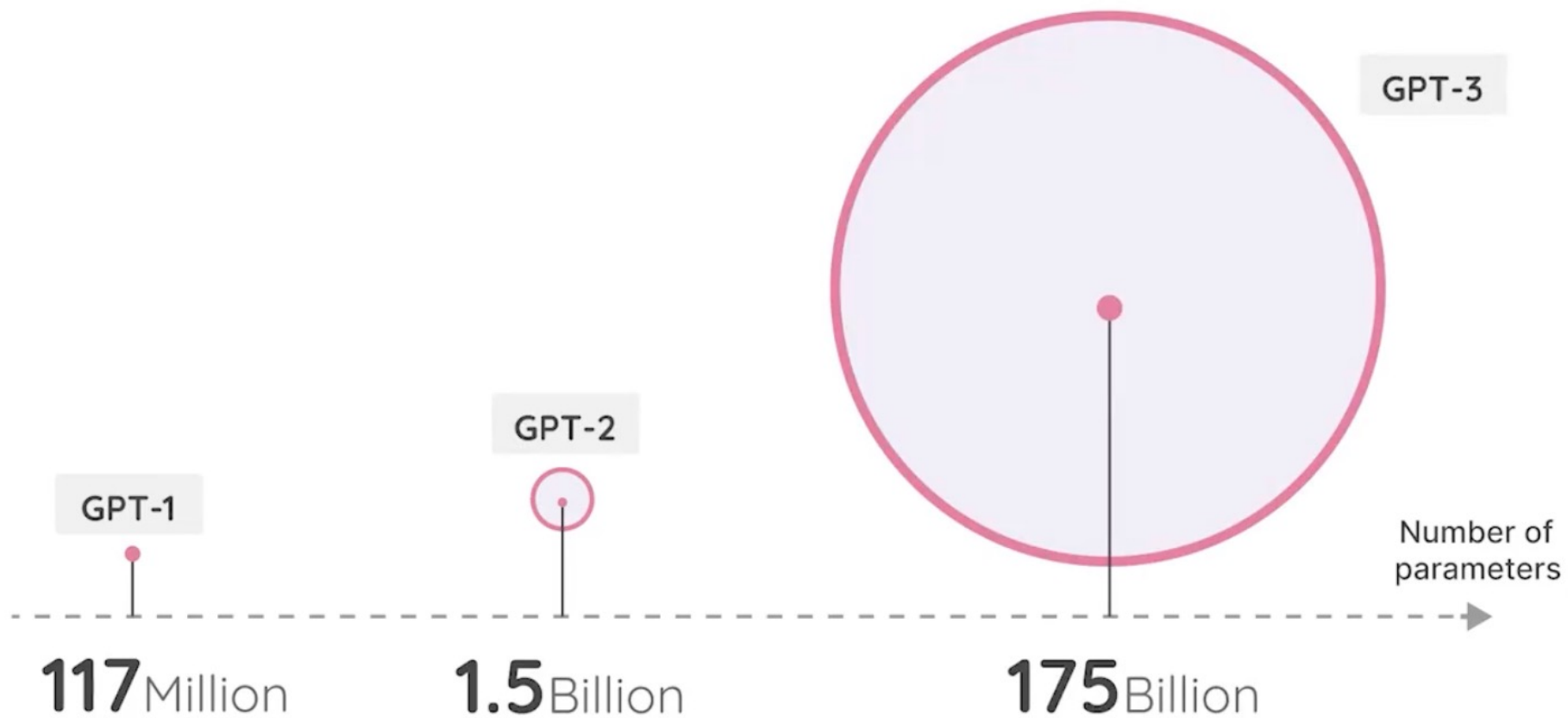
Generative Models



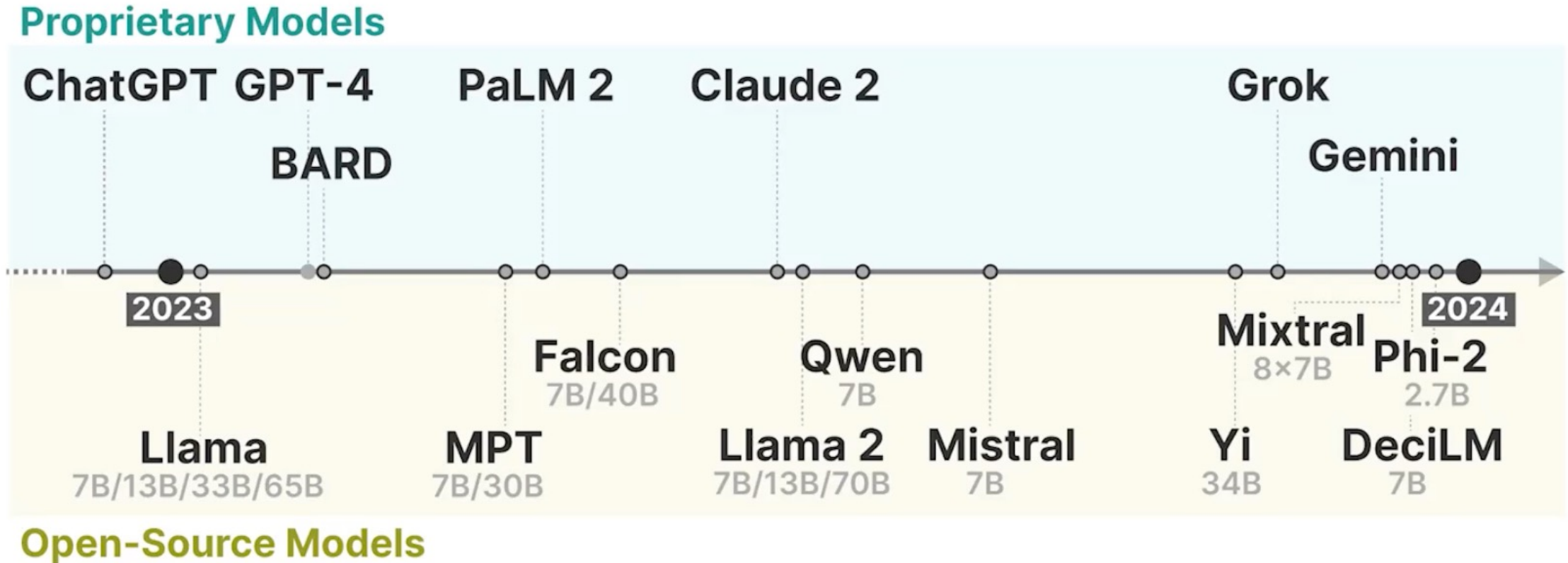
Context Length



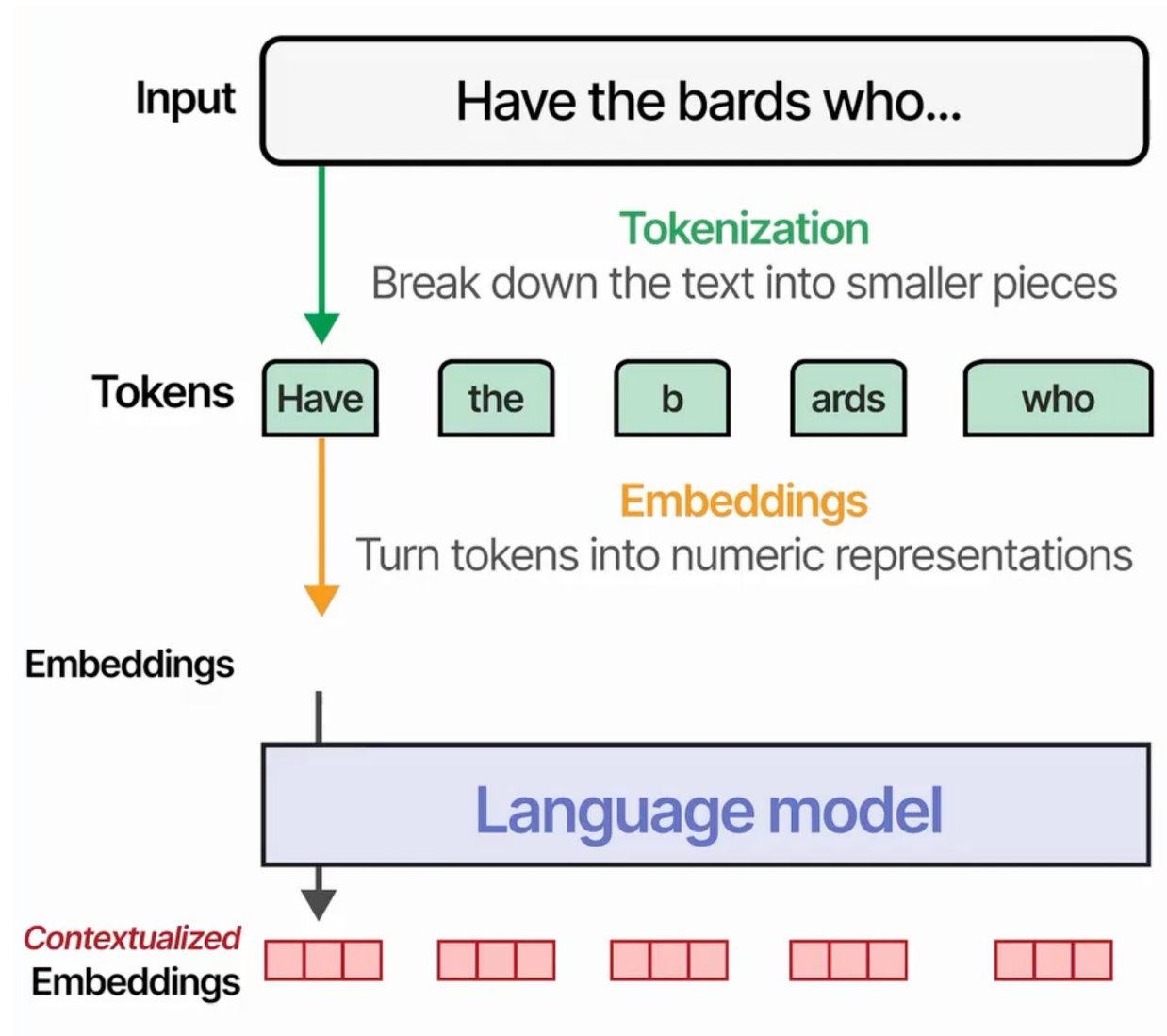
Parameters



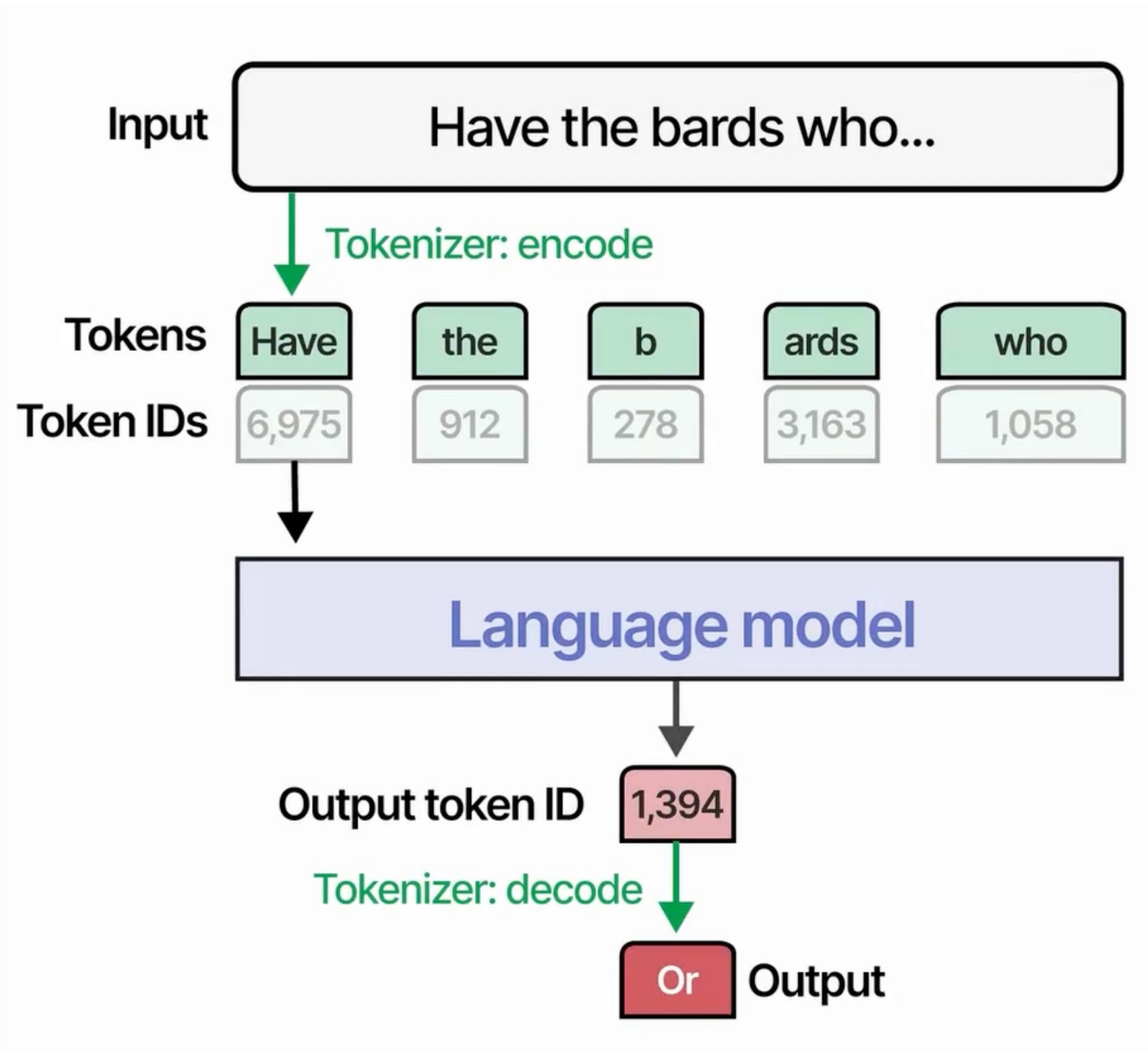
Year of Generative AI



Tokenization



Tokenization



Tokenization

Text Have the 🎵 bards who preceded...

Word
tokens

Have	the	🎵	bards	who	preceded	...
------	-----	---	-------	-----	----------	-----

Subword
tokens

Have	the	🎵	b	ards	who	preced	ed	...
------	-----	---	---	------	-----	--------	----	-----

Character
tokens

H	a	v	e		t	h	e		🎵		b	a	r	d	s	..
---	---	---	---	--	---	---	---	--	---	--	---	---	---	---	---	----

Byte
tokens

H a v e <space> t h e <space> 🎵 <space>																
0	0	0	0	0	0	0	0	0	1	1	1	1	0	...		
1	1	1	1	0	1	1	1	0	1	0	0	0	0			
0	1	1	1	1	1	1	1	1	1	0	0	1	1			
0	0	1	0	0	1	0	0	0	1	1	0	1	0			
1	0	0	0	0	0	1	0	0	0	1	1	0	0			
0	0	1	1	0	1	0	1	0	0	1	1	1	0			
0	0	1	0	0	0	0	0	0	0	1	1	0	0			
0	1	0	1	0	0	0	1	0	0	1	0	1	0			

注意力机制与外部记忆

Transformer 的细节

Architectural Overview

Prompt

Write an email apologizing to Sarah
for the tragic gardening mishap.
Explain how it happened.



Transformer LLM

Generation

Dear

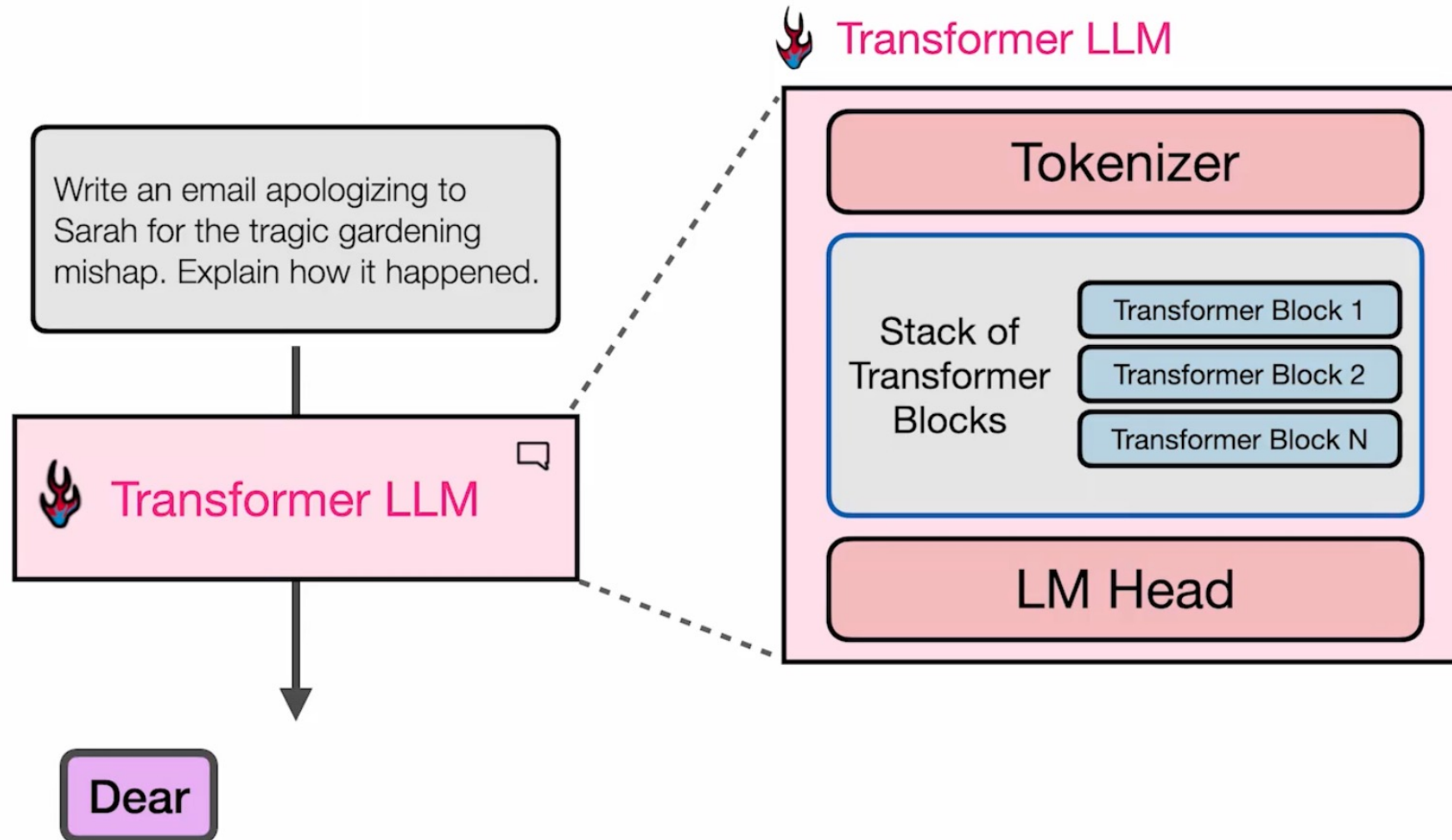
Sarah

.

\n

Architectural Overview

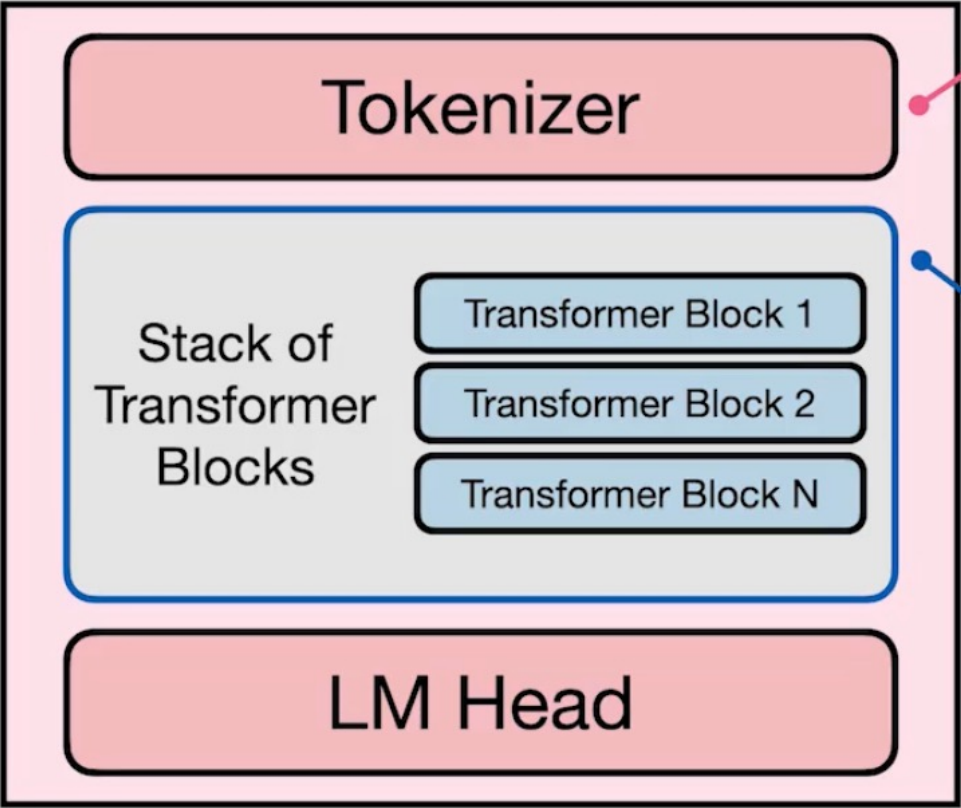
Three major components: Tokenizer, Transformer Blocks, and LM Head



Architectural Overview

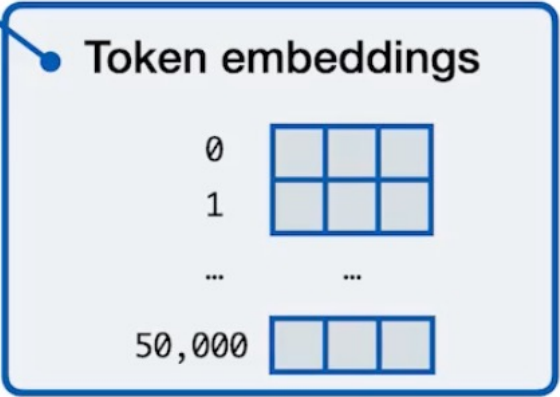


Transformer LLM



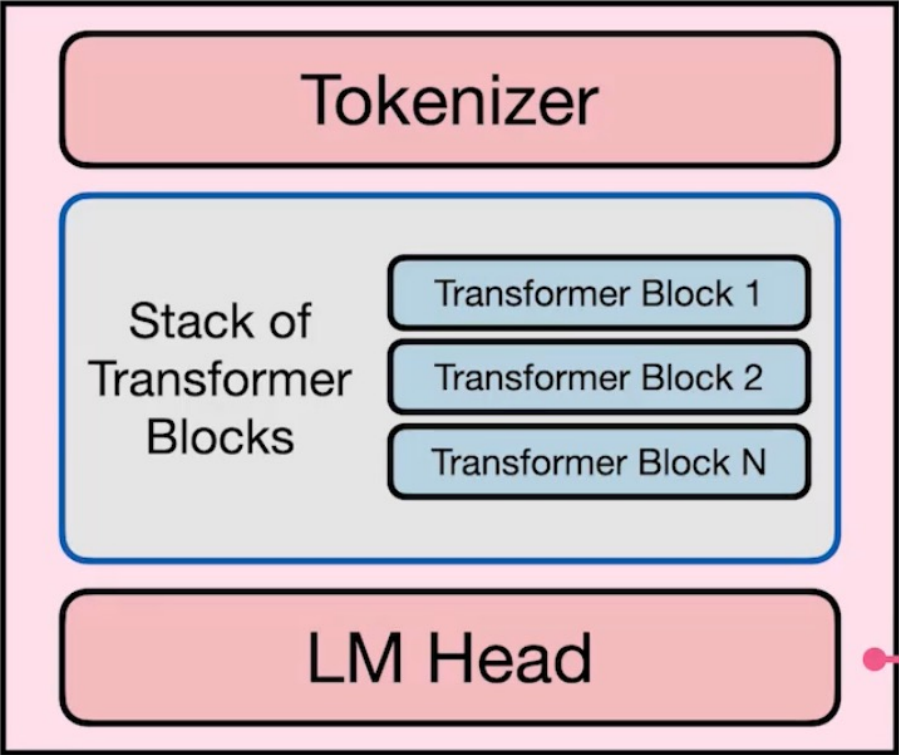
Token vocabulary

Token ID	Token
0	!
1	"
...	...
50,000	Zyzzyyva



Architectural Overview

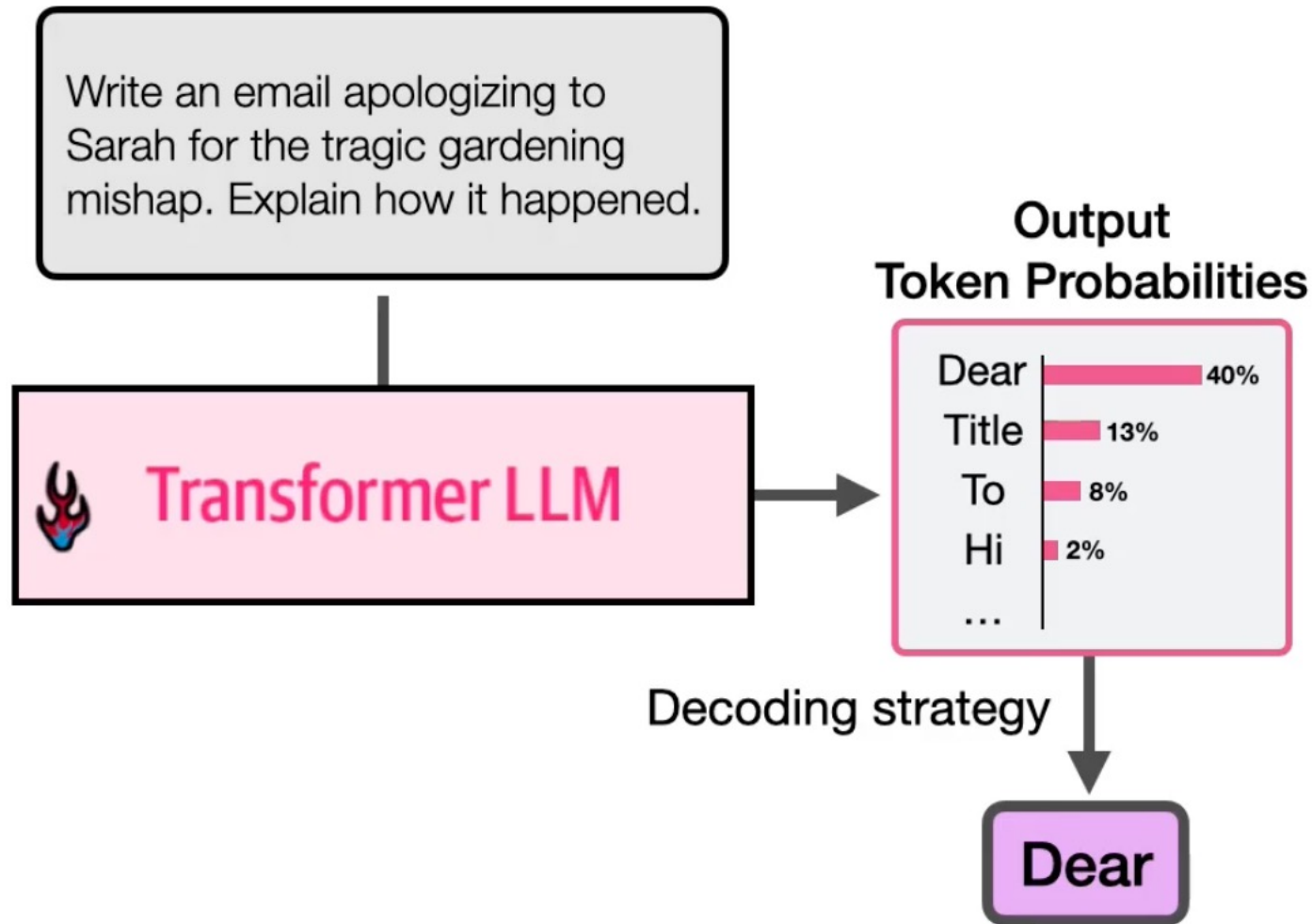
Transformer LLM



Output



Architectural Overview

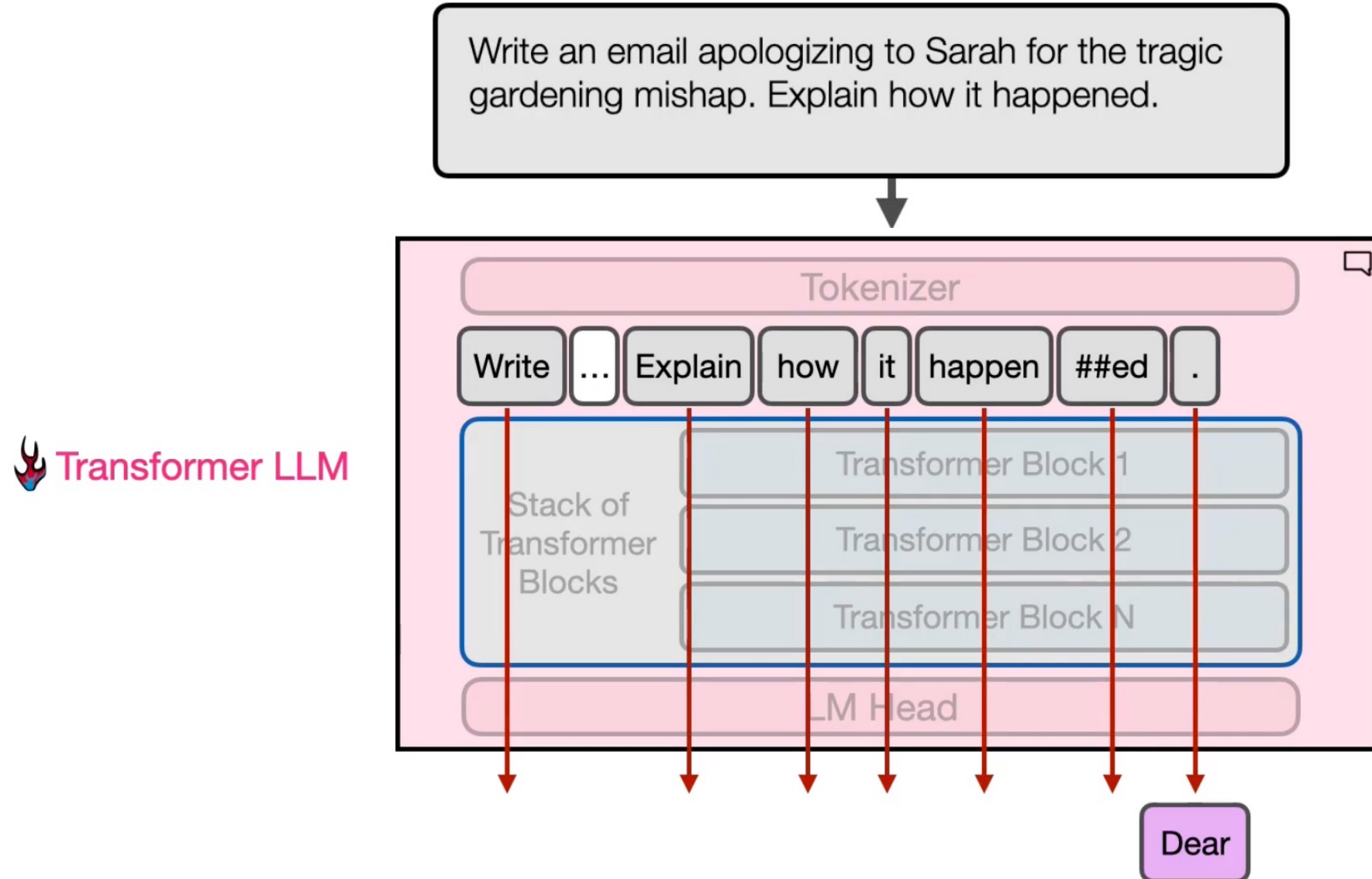


Choose the highest:

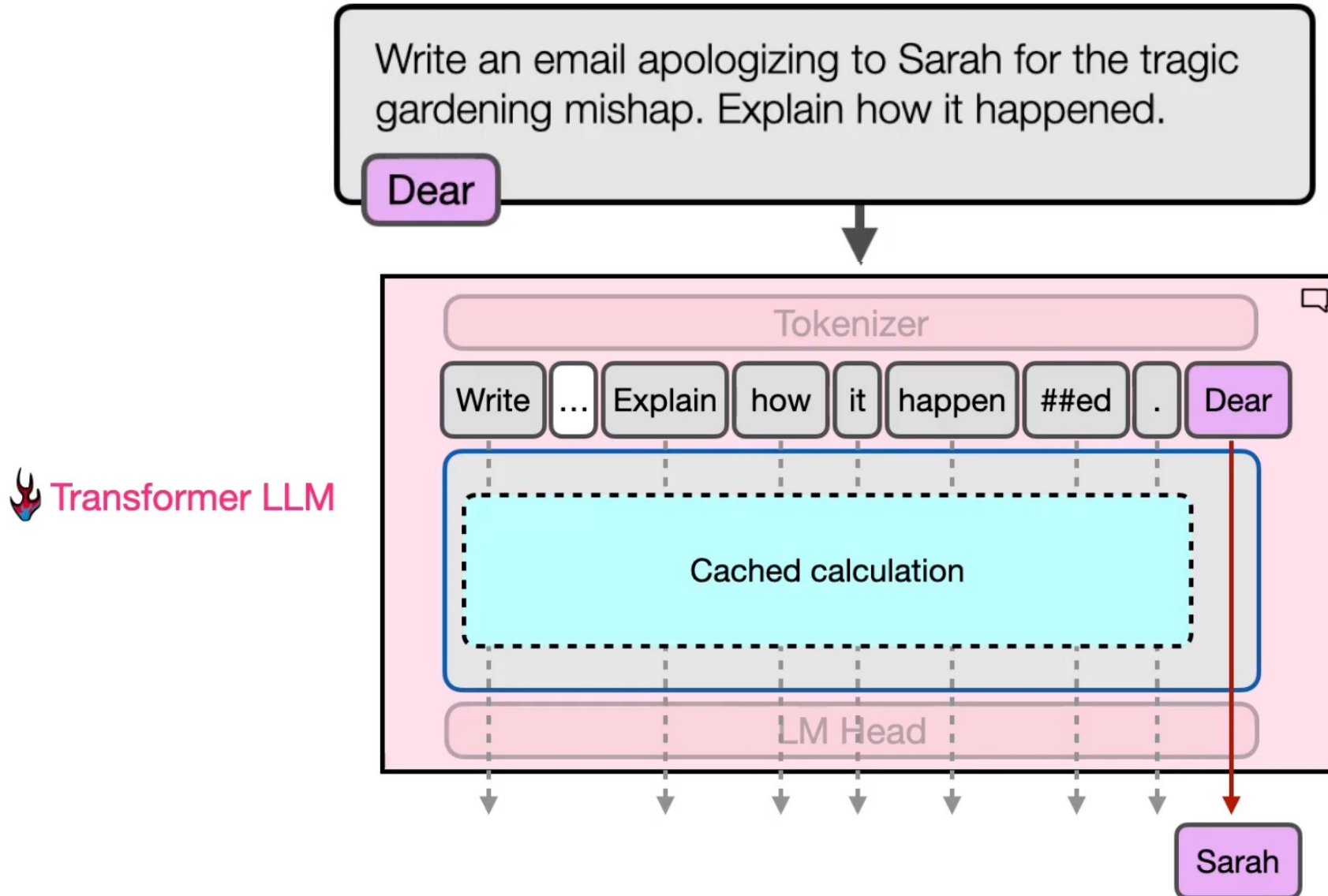
- `temperature = 0`
- Greedy decoding

- Choose `top_p`
- Add randomness
(`temperature > 0`)

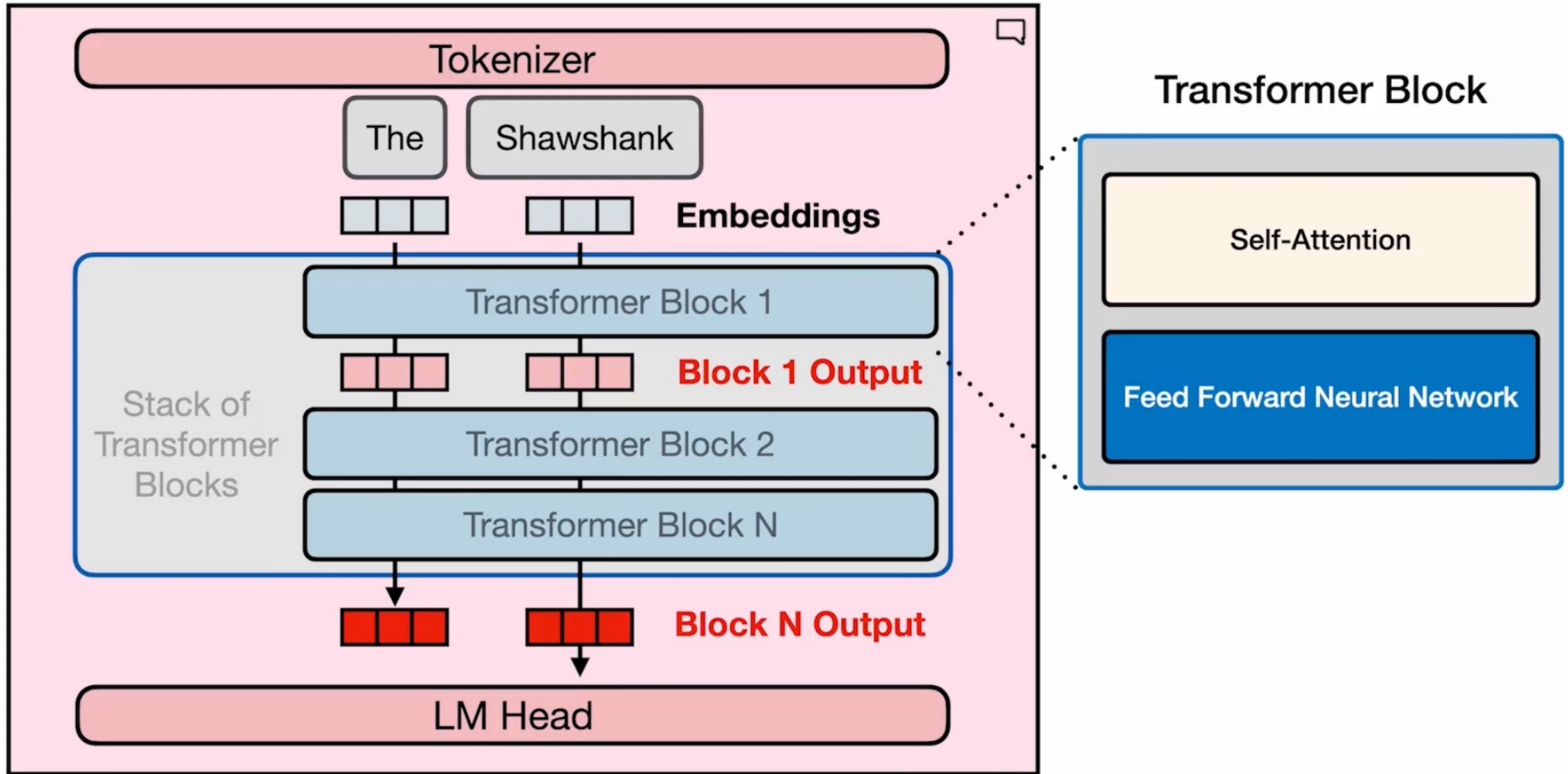
Architectural Overview



Architectural Overview

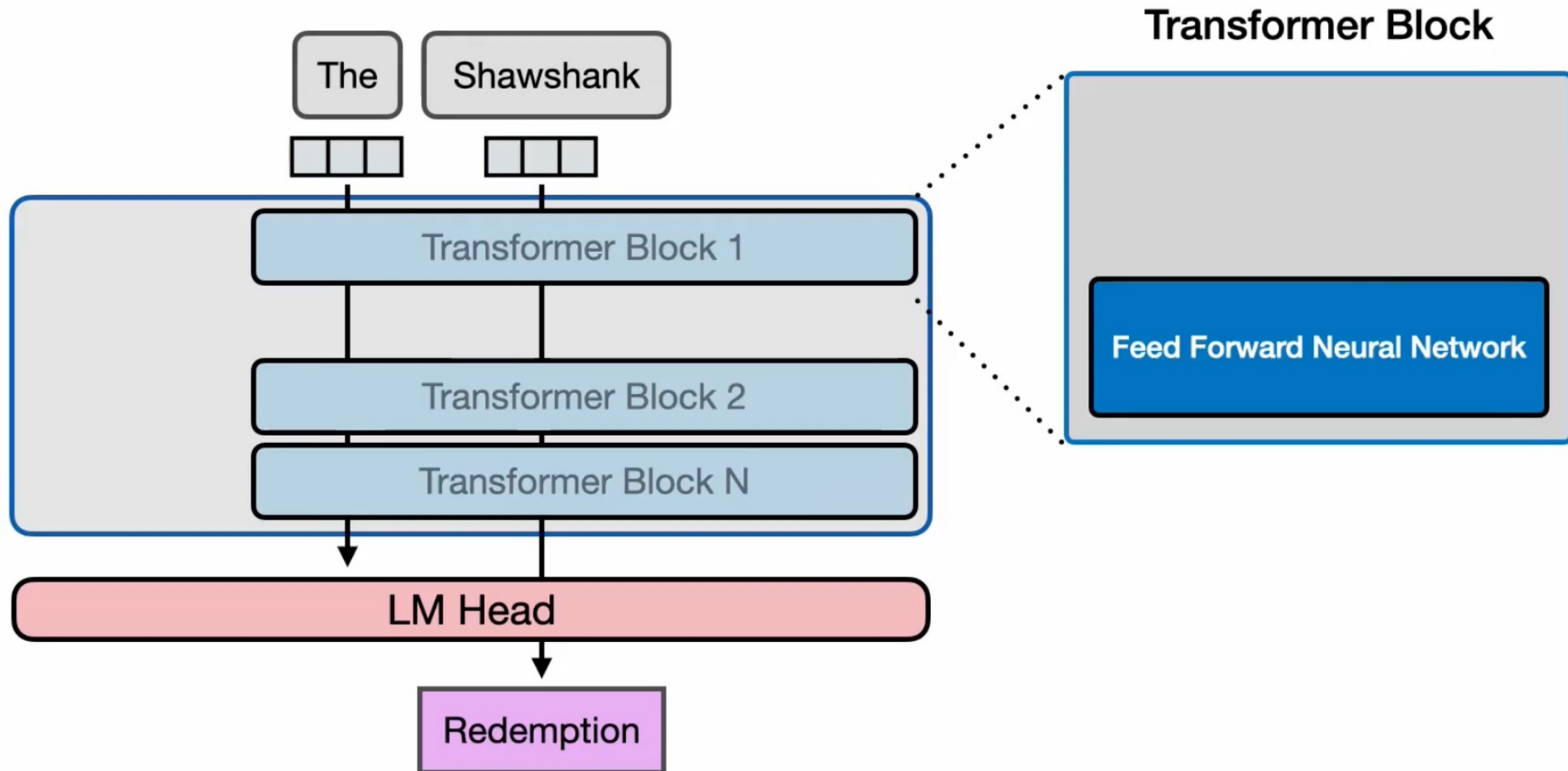


Transformer Block



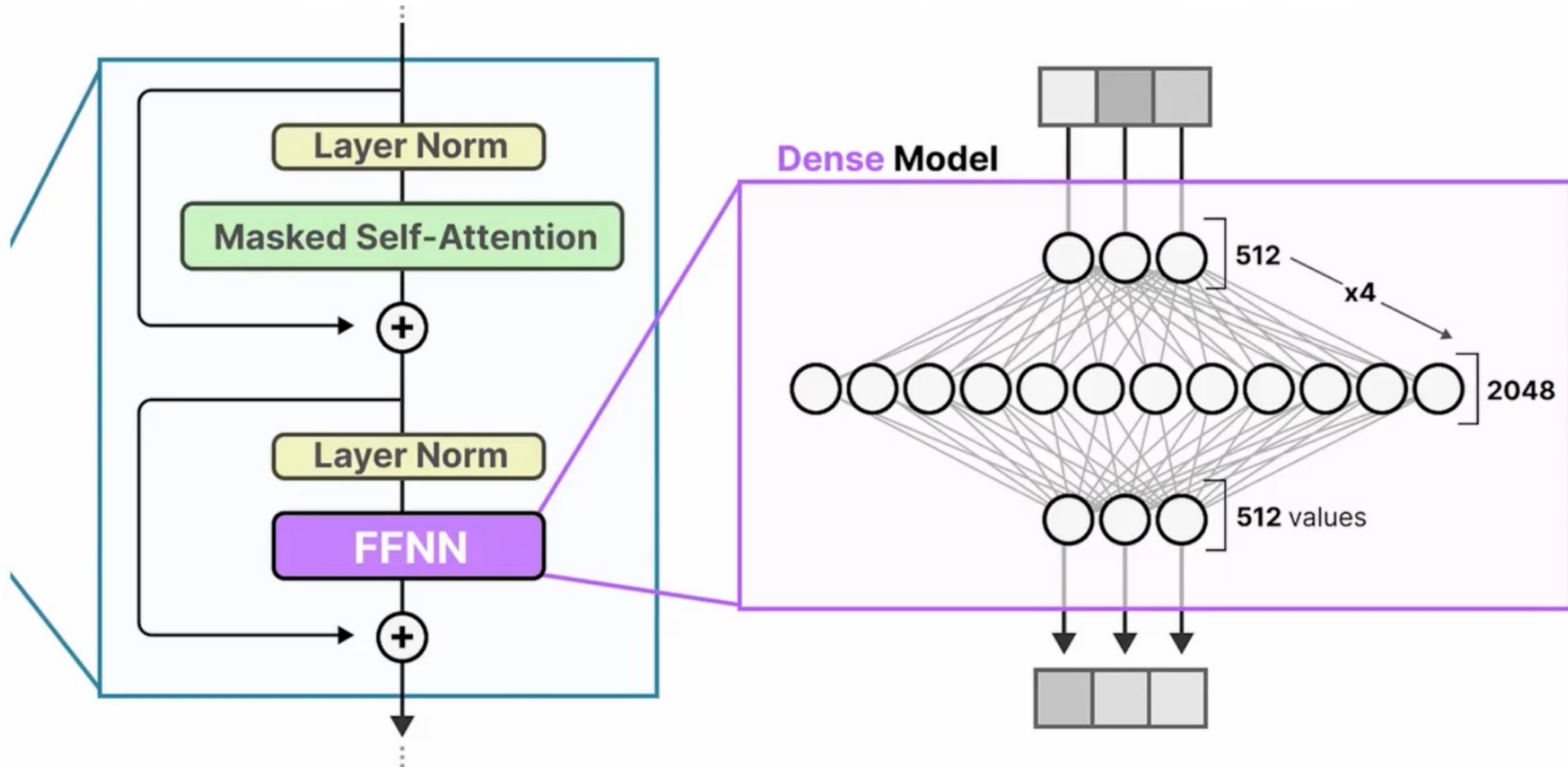
Transformer Block

Transformer Block: Feedforward Neural Network Intuition



Transformer Block

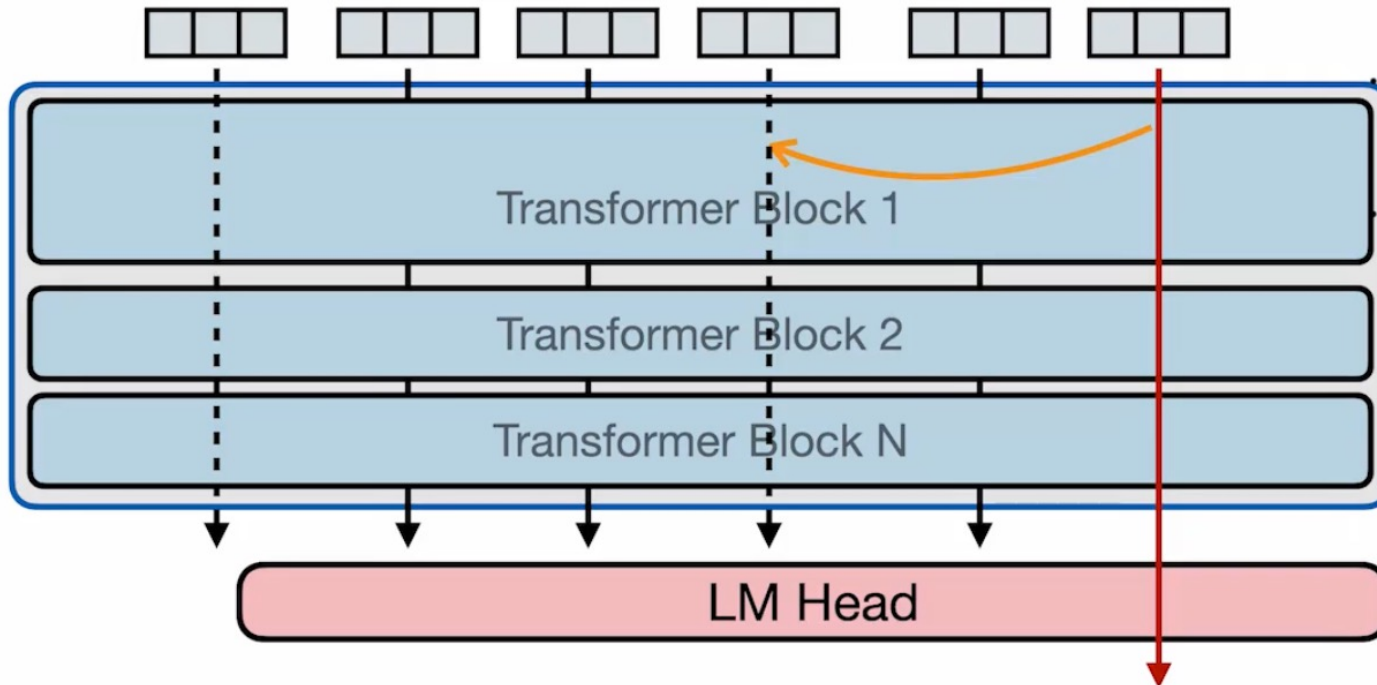
Transformer Block: Feedforward Neural Network Intuition



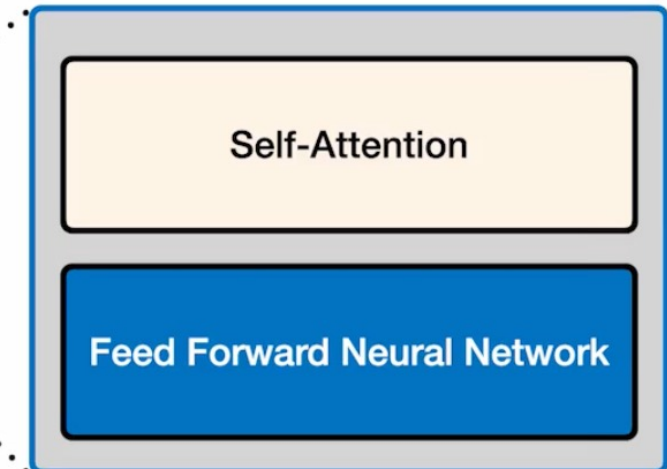
Transformer Block

Transformer Block: Attention allows us to incorporate relevant information from the context

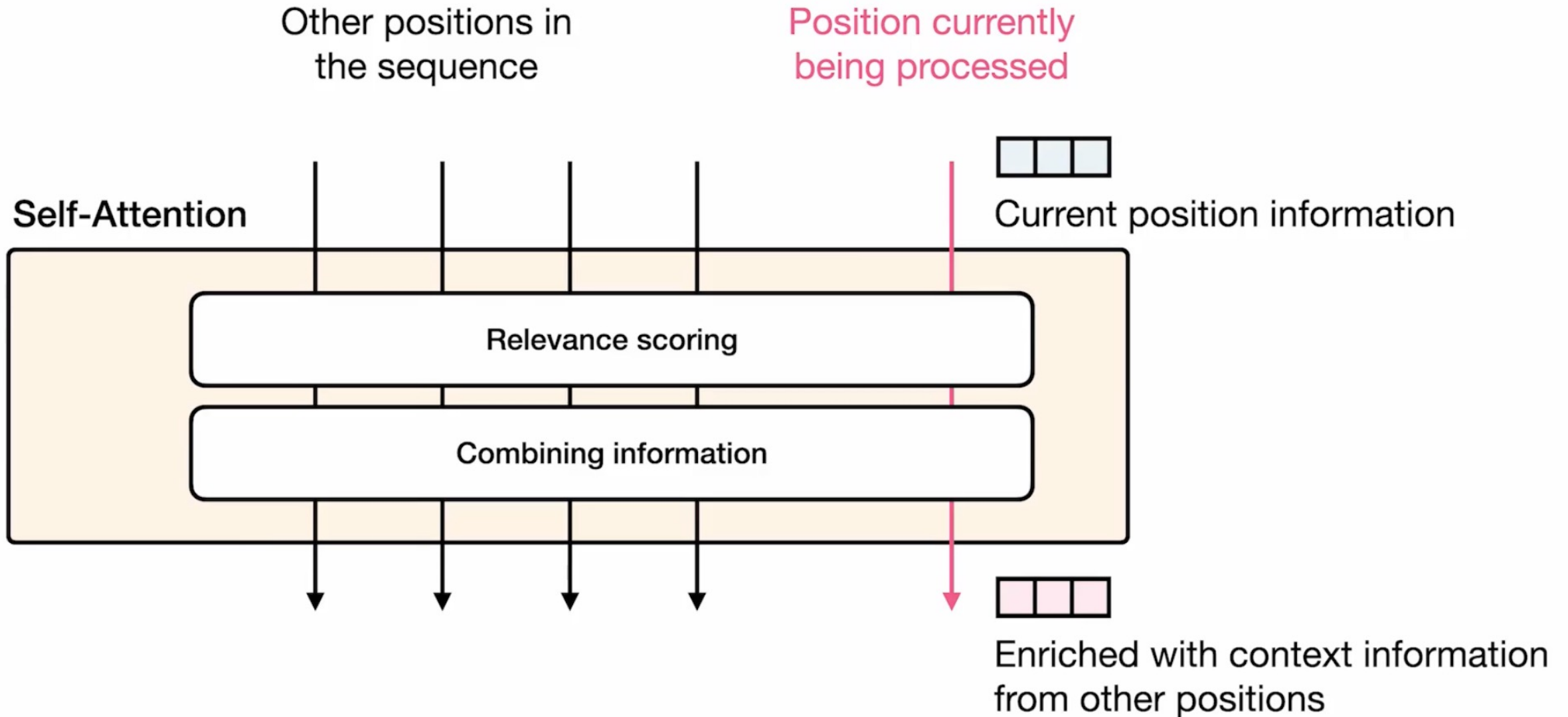
The dog chased the llama because it



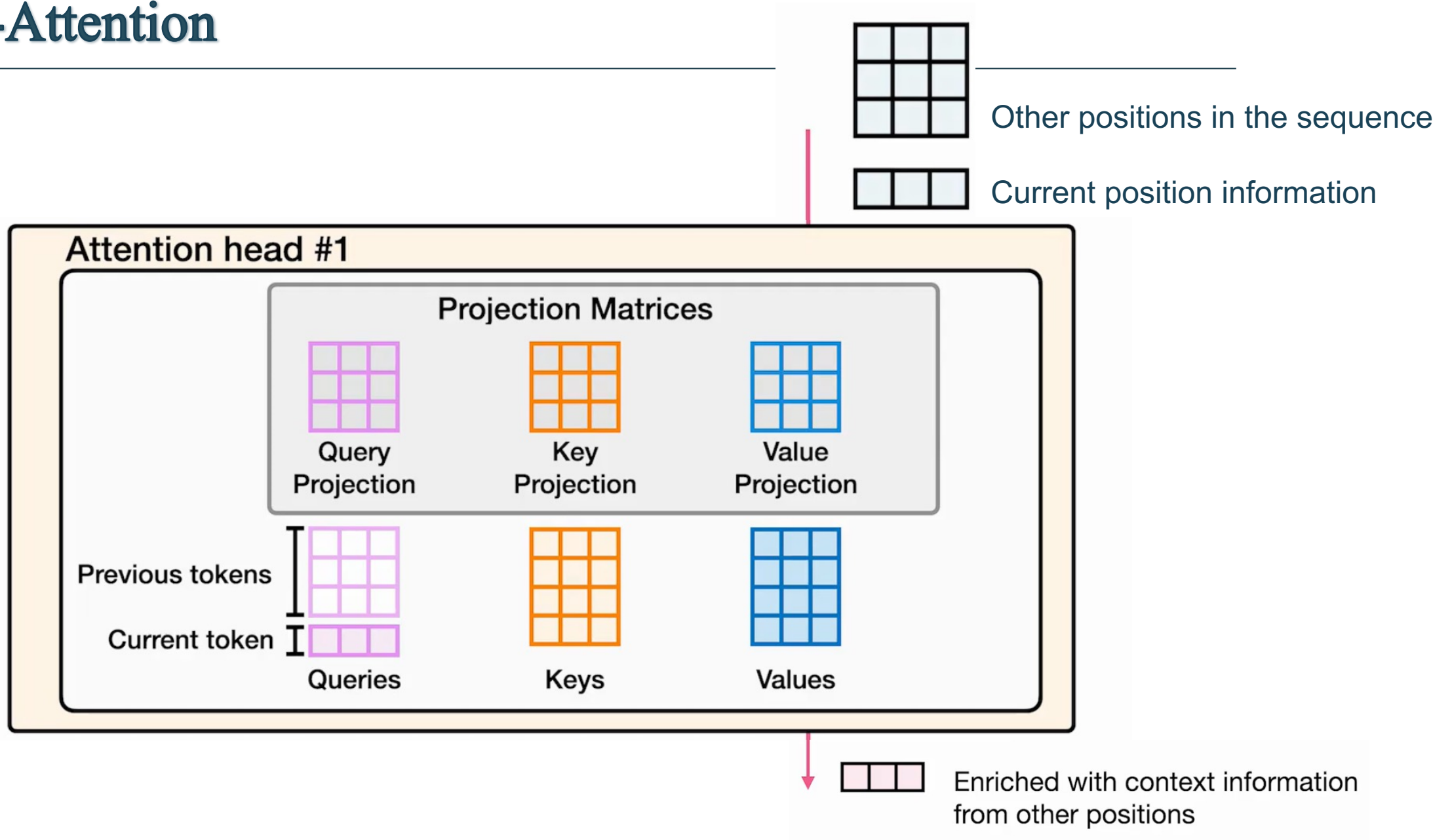
Transformer Block



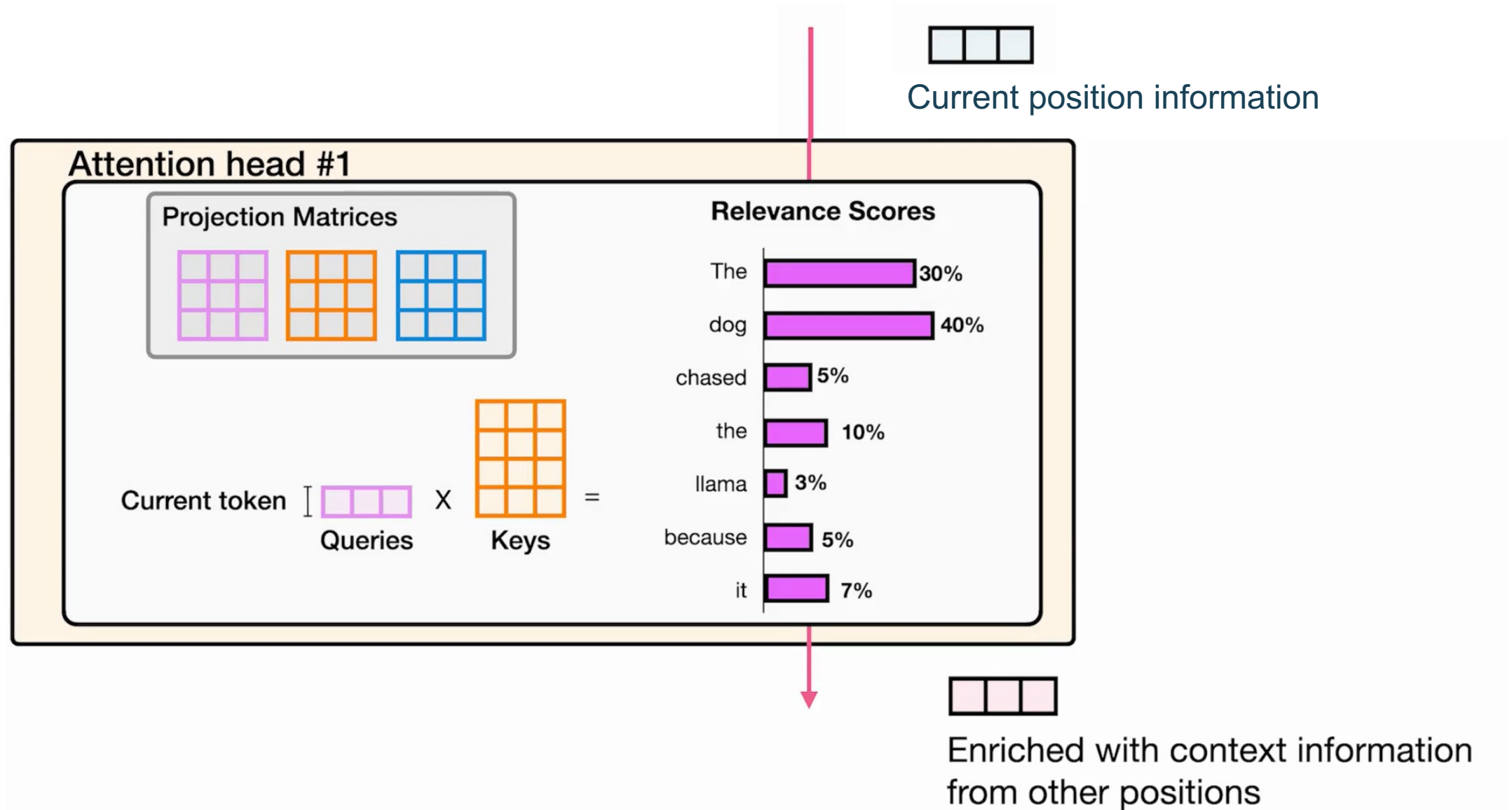
Transformer Block



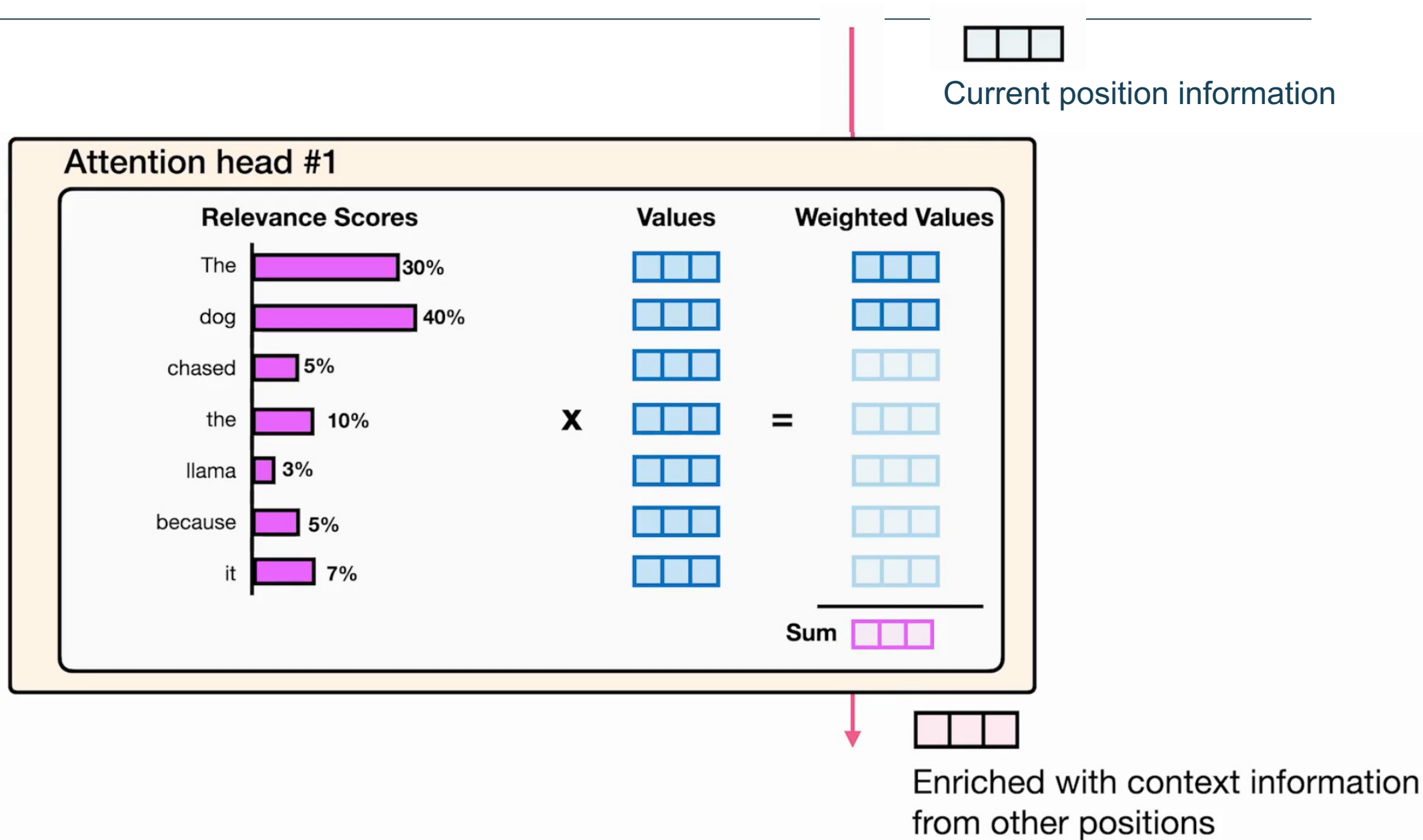
Self-Attention



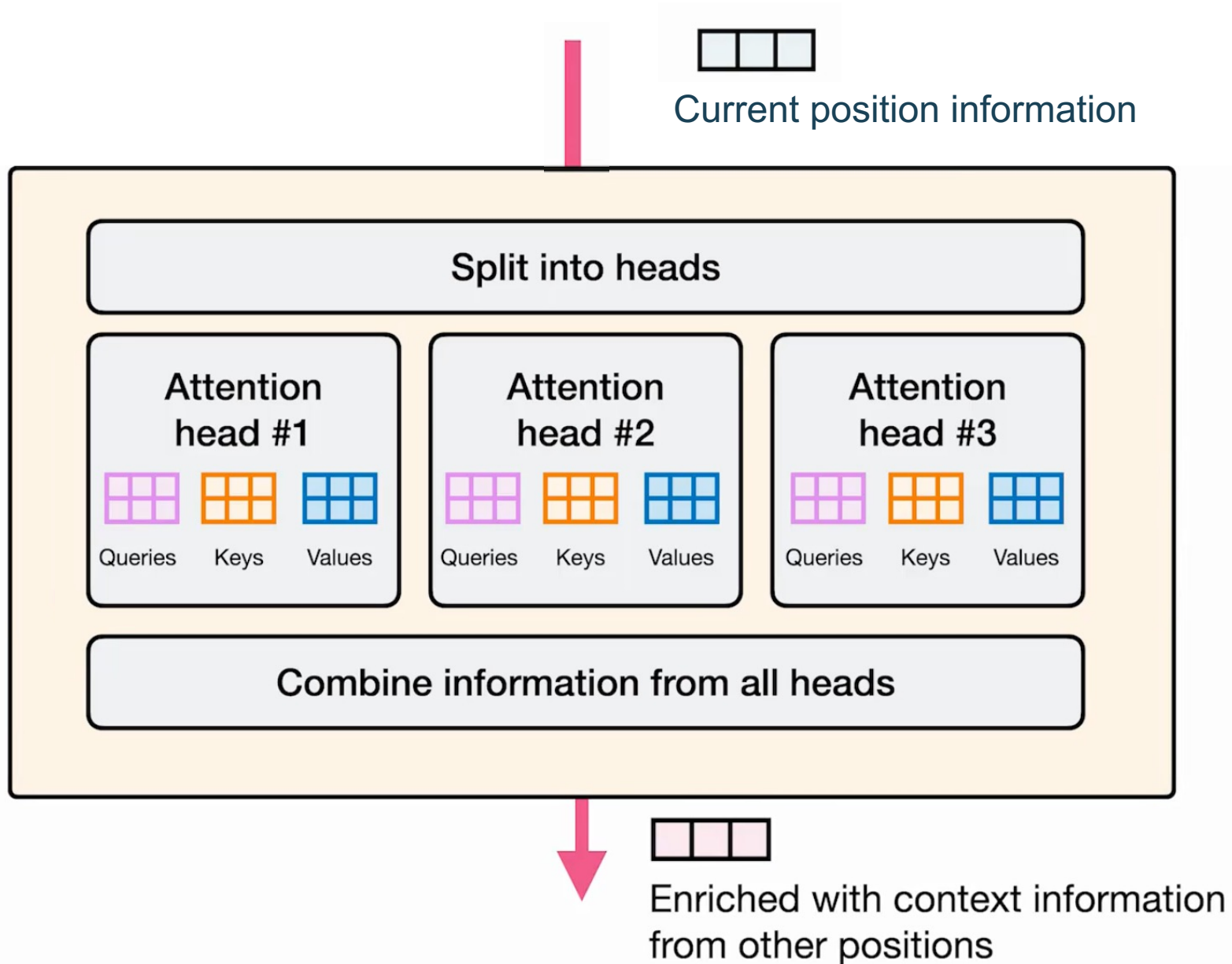
Self-Attention



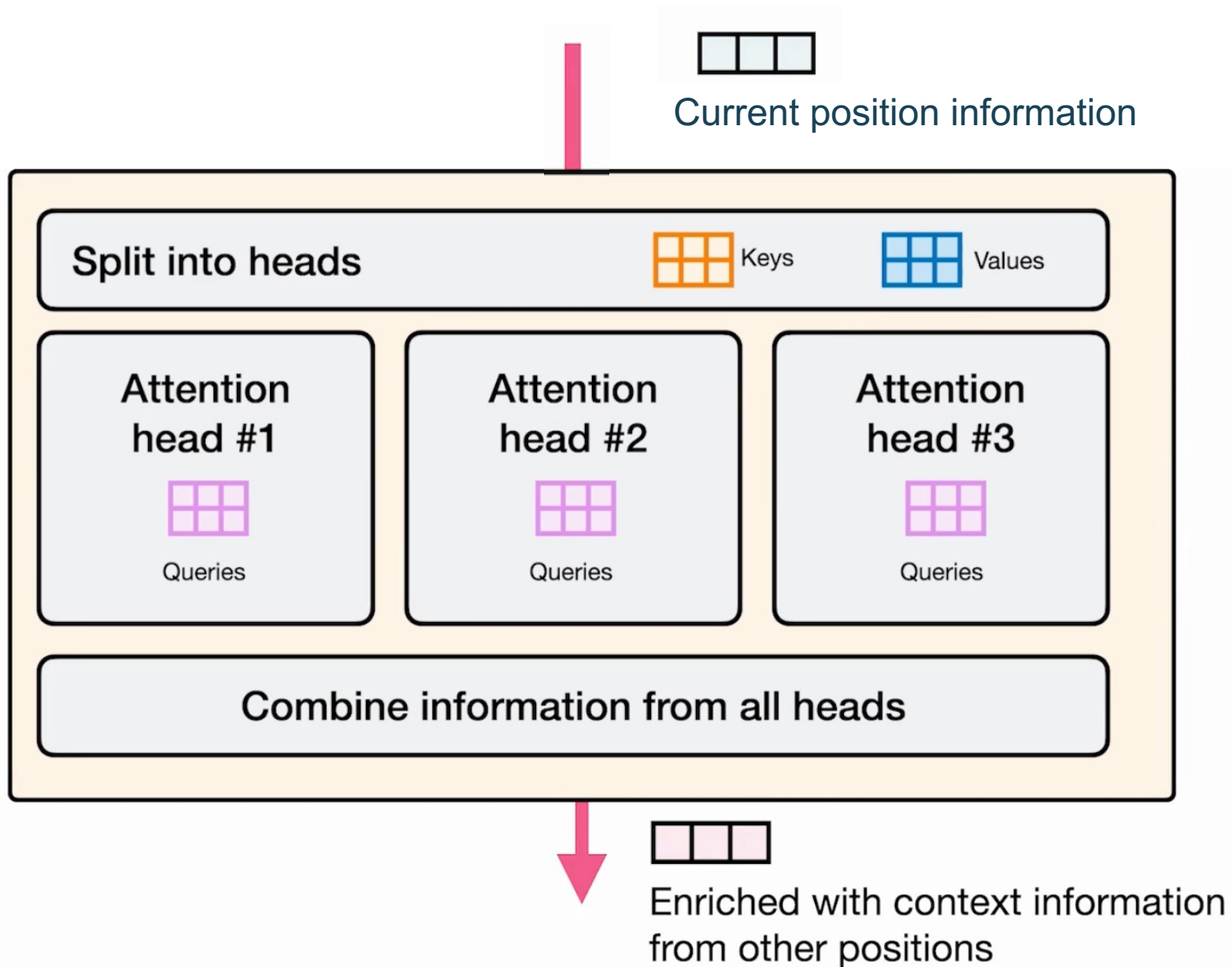
Self-Attention



Self-Attention

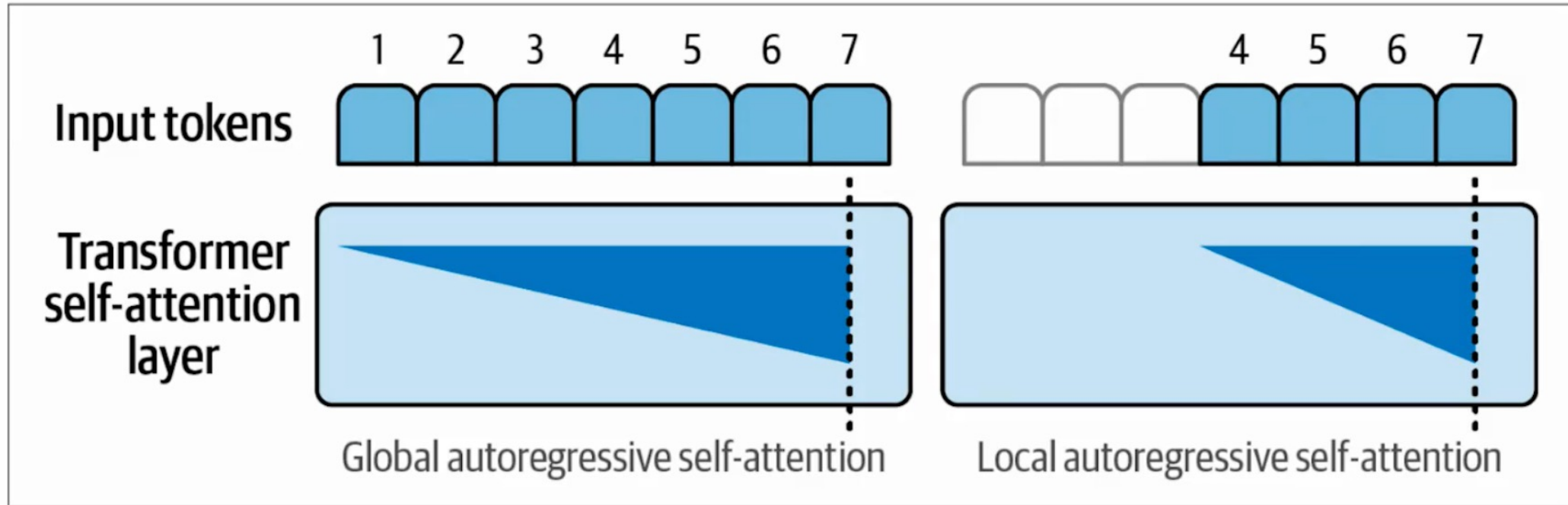


Self-Attention



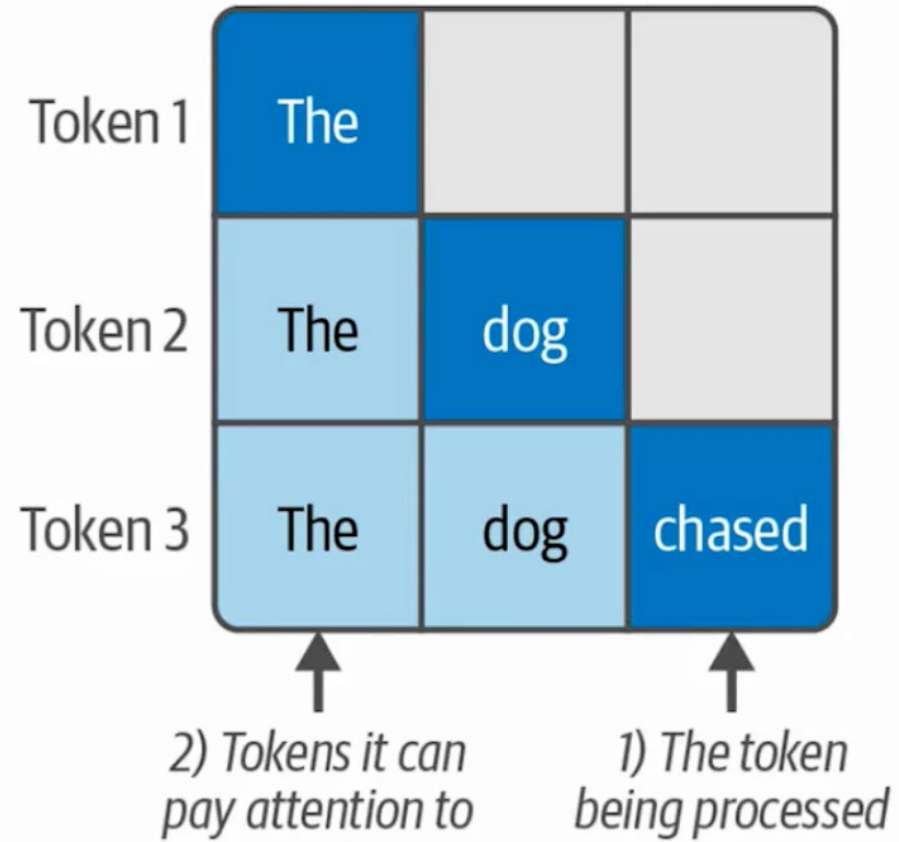
Self-Attention

More attention efficiency: sparse attention

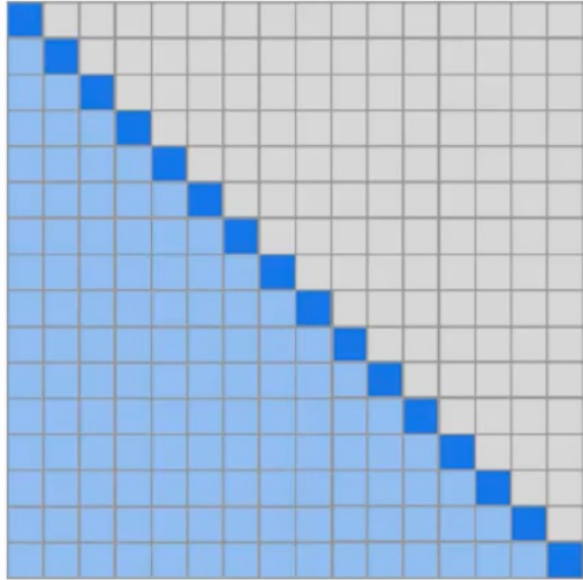


Local attention boosts performance by only paying attention to a small number of previous positions.

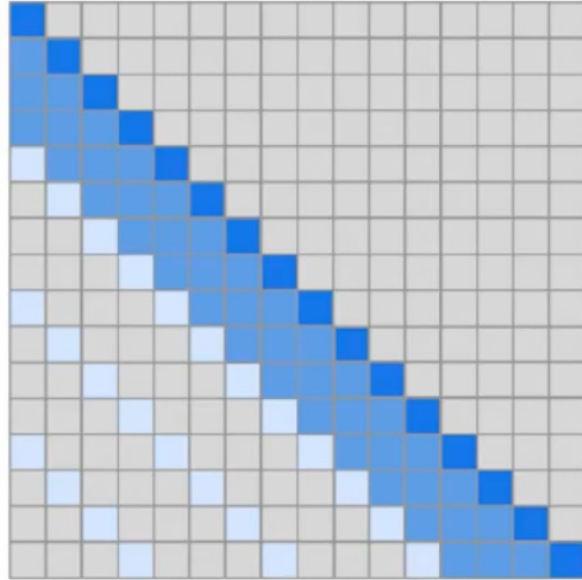
Self-Attention



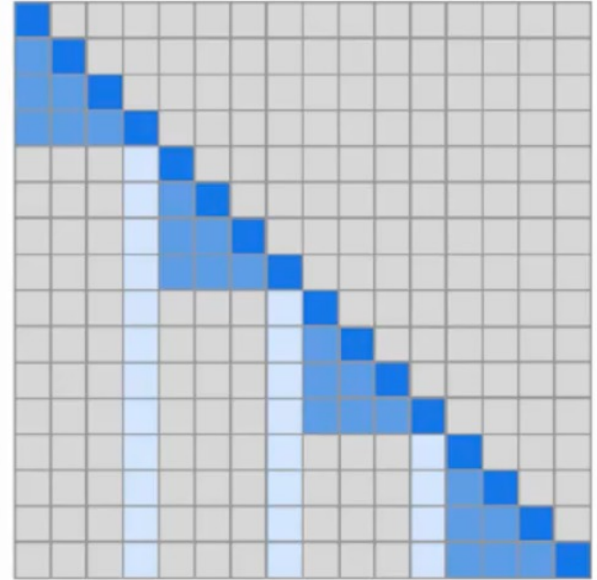
Self-Attention



(a) Transformer



(b) Sparse Transformer (strided)



(c) Sparse Transformer (fixed)

Full attention versus sparse attention.

(Source: “Generating long sequences with sparse transformers”(<https://arxiv.org/pdf/1904.1Fdfd0509>).)

Self-Attention

Paper: The Llama 3 Herd of Models

3.2 Model Architecture

Llama 3 uses a standard, dense Transformer architecture (Vaswani et al., 2017). It does not deviate significantly from Llama and Llama 2 (Touvron et al., 2023a,b) in terms of model architecture; our performance gains are primarily driven by improvements in data quality and diversity as well as by increased training scale.

We make a few small modifications compared to Llama 2:

- We use grouped query attention (GQA; Ainslie et al. (2023)) with 8 key-value heads to improve inference speed and to reduce the size of key-value caches during decoding.
- We use an attention mask that prevents self-attention between different documents within the same sequence. We find that this change had limited impact during in standard pre-training, but find it to be important in continued pre-training on very long sequences.

	8B	70B	405B
Layers	32	80	126
Model Dimension	4,096	8192	16,384
FFN Dimension	14,336	28,672	53,248
Attention Heads	32	64	128
Key/Value Heads	8	8	8
Peak Learning Rate	3×10^{-4}	1.5×10^{-4}	8×10^{-5}
Activation Function	SwiGLU		
Vocabulary Size	128,000		
Positional Embeddings	RoPE ($\theta = 500,000$)		

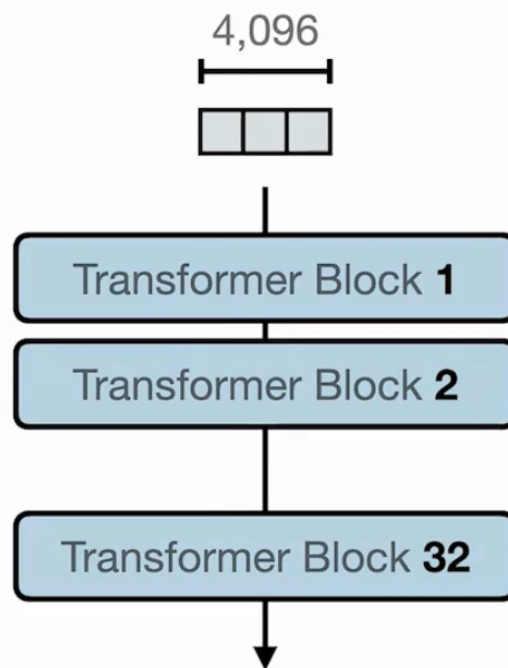
Table 3 Overview of the key hyperparameters of Llama 3. We display settings for 8B, 70B, and 405B language models.

Self-Attention

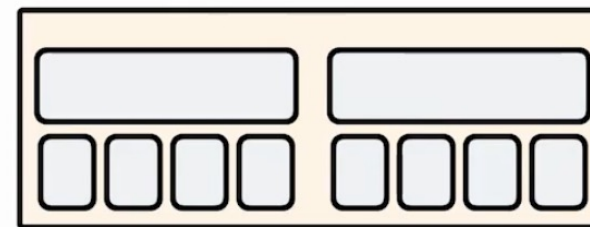
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Vocabulary Size		128,000	
Positional Embeddings		RoPE ($\theta = 500,000$)	

Token vocabulary

Token ID	Token
0	!
1	"
...	...
128,000	Zyzyva

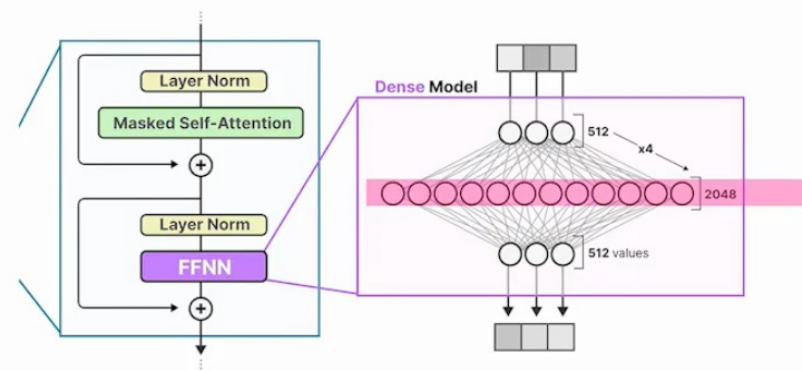


Self-Attention



n_groups = 8

n_attention_heads = 32



A Survey of Transformers

Tianyang Lin, Yuxin Wang, Xiangyang Liu, [Xipeng Qiu](#)

Transformers have achieved great success in many artificial intelligence fields, such as natural language processing, computer vision, and audio processing. Therefore, it is natural to attract lots of interest from academic and industry researchers. Up to the present, a great variety of Transformer variants (a.k.a. X-formers) have been proposed, however, a systematic and comprehensive literature review on these Transformer variants is still missing. In this survey, we provide a comprehensive review of various X-formers. We first briefly introduce the vanilla Transformer and then propose a new taxonomy of X-formers. Next, we introduce the various X-formers from three perspectives: architectural modification, pre-training, and applications. Finally, we outline some potential directions for future research.

Pre-trained Models for Natural Language Processing: A Survey

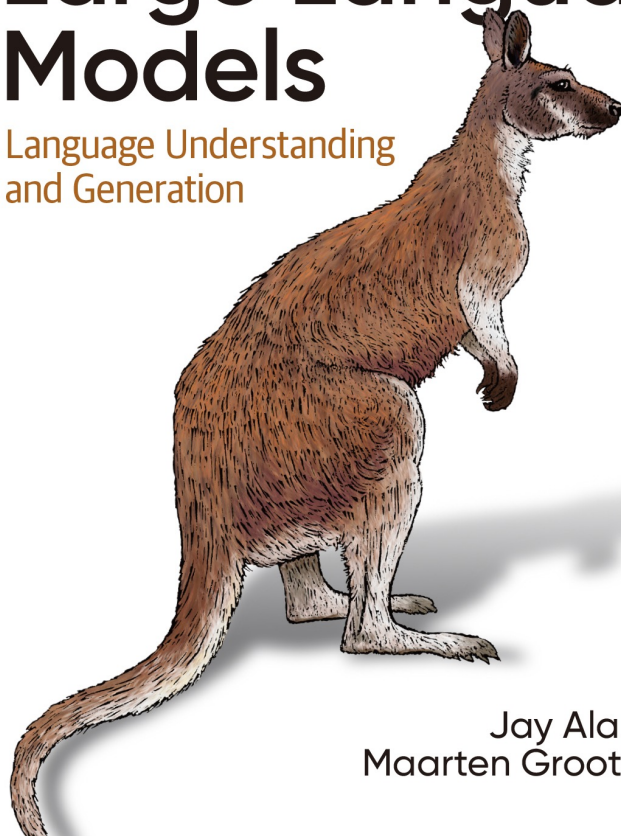
[Xipeng Qiu](#), Tianxiang Sun, Yige Xu, Yunfan Shao, Ning Dai, Xuanjing Huang

Recently, the emergence of pre-trained models (PTMs) has brought natural language processing (NLP) to a new era. In this survey, we provide a comprehensive review of PTMs for NLP. We first briefly introduce language representation learning and its research progress. Then we systematically categorize existing PTMs based on a taxonomy with four perspectives. Next, we describe how to adapt the knowledge of PTMs to the downstream tasks. Finally, we outline some potential directions of PTMs for future research. This survey is purposed to be a hands-on guide for understanding, using, and developing PTMs for various NLP tasks.

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